

ME 204 A : MECHANICS OF SOLIDS

2L, 1T 3 Hours, 100 Marks

Direct Stresses in Machine Parts : Analysis of stress and strain. -Hooke's law, stress-strain diagram, Poisson's ratio, Modulus of elasticity, Relation between elastic constants

Simple stresses : Tensile, compressive, shear, crushing and bearings. criteria. Hoop stress in thin cylindrical and spherical vessels subjected to internal pressure

Machine Members in Bending : Theory of bending of straight and curved beams. Bending and shear stress distribution in beams. Shear force and bending moment diagrams. Moment of inertia of different machine element sections

Deflection and stiffness : Mecauly's method, area moment method, numerical integration method and energy method Numerical problems of line shafts under simple loading with pulleys, ears, flywheel etc. Brief introduction to analysis of shafts with three Supports.

Machine Members in Torsion : Introduction to torsion of circular solid and hollow sections. Angular deformation and strain energy.

Stress in Slender Machines Members under axial Compressive Loads : Analysis of long and short columns. Euler's theory and effect of end conditions on stresses. Emperical formulas

Machine Members subjected to Combined Loading: Combination of axial, bending, and torsional stresses. Concept of principal stresses and planes. Mohr's circle of stresses. 2D state of strain, Plane stress and Plane strain conditions. Mohr's circle for strain. Theories of failure. Simple problems of line shaft.

SN Name of Authors /Books /Publisher

Year of

Pub.

1 Timoshenko, S.P., and Gere, J.M., "Mechanics of Materials", 2nd Ed., CBS Publishers

2 Crandall, S.H., Dahl, N.C., and Lardner, T.J., "An Introduction to the Mechanics of Solids", TMGH

3 Strength of materials , Ramamurtham

4 Punmia, Jain and Jain, "Mechanics of Materials", Laxmi Publication

5 Popov, E.P., Nagarajan, S., and Lu, Z. A., "Mechanics of Materials", Prentice-Hall of India

INTRODUCTION

In all engineering construction, the component parts of a structure or a machine must be assigned definite physical sizes. Such parts must be properly proportioned to resist the actual or probable forces that may be imposed upon them.

- Thus for example the walls of a pressure vessel (Such as boiler) must be of adequate strength to withstand the internal pressure.
- The floors of a building must be sufficiently strong for their intended purpose.
- The shaft of a Machine must be of adequate size to carry the required torque.
- A wing of an airplane must safely withstand the aerodynamic loads that may come upon it during take off, flight and landing.
- Likewise the parts of a composite structure must be rigid enough so as not to deflect excessively when in operation under the imposed loads.

The ability to determine the minimum load that an element of a structure can carry before failure happens is of great importance. In engineering practice such requirements must be met with the minimum expenditure of a given material.

- The subject strength of materials' involves analytical methods for determining the strength, stiffness and stability of various load carrying materials.
- This subject is very important to be studied by many braches of engineering. It is important for *civil engineers* in the design of bridges and buildings, for *nuclear engineers* in the design of reactor components, for *mechanical engineers* for the design of machineries and pressure vessels, for *metallurgical engineers*, who need fundamental concepts of this subject in order to understand how to improve existing materials further.

1. Simple Stresses And Strains

1.1 Introduction

Whenever a load is applied to a body, its shape changes i.e the body gets deformed and the process is called deformation. The amount by which body is deformed depends upon the amount of load and the nature as well as the geometrical properties. It has been experimentally found that the cohesive forces between molecules of a body resist the deformation and force of resistance increases with increment in deformation. It has also been observed that the process of deformation stops when the force of resistance is equal to the external load. If the force of resistance is less than the external load, the material gets failed.

1.2 Stress: When an external force or load is applied on a body internal resistance is set up. Let us consider a bar acted upon by a Force P along its axis as in Fig 1.1(a). If the bar AC is cut in two sections at B, it results in Fig 1.1(b). The externally applied forces to one side of the cut part must be balanced by internal forces developed at the cut.

If the internal resistance offered by the section against the deformation be assumed to be uniform across the section, the intensity of resistance per unit of area is called the intensity of stress or unit stress. In general usage the word stress is used to mean the intensity of stress.

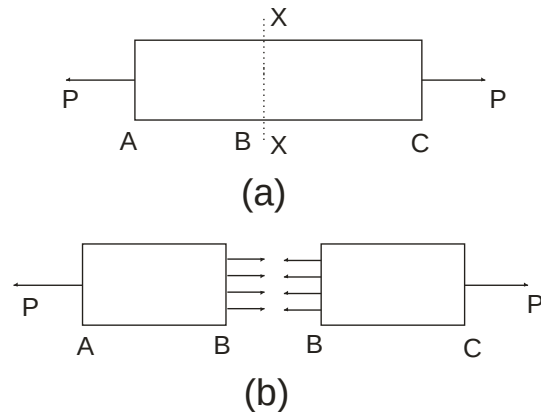


Fig. 1.1

$$\text{Intensity of stress } \sigma = \frac{F}{A} \quad \dots\dots(1.1)$$

$$\sigma = \text{Stress N/m}^2$$

$$A = \text{Cross Section area in m}^2$$

$$F = \text{Total force acting through the centroidal axis of the body in N.}$$

1.3 Strain: Strain is defined as the ratio of deformation of a body due to external load to the original length. It is a dimensionless parameter. If a load P is applied to a body the body deforms by δ and the original length be L the strain (ϵ) is give by

$$\epsilon = \frac{\delta}{L} \quad \dots(1.2)$$

1.4 Simple Stress & Strain:

(a) Tensile Stress (σ_t): When a body is subjected to two equal and opposite pulls and the length of the body increases, the stress induced is called tensile stress (Fig 1.2).

The stress in (σ_t) simple tension in a body is equal to the external force P divided by the resisting area and in this case the resisting area is cross-sectional area A of the member in m^2 . Thus



Fig. 1.2

$$\sigma_t = \frac{P}{A} \text{ N/m}^2 \quad \dots(1.3)$$

Corresponding strain is called tensile strain

$$\begin{aligned} \text{Tensile strain} &= \frac{\text{Increase in length}}{\text{Original length}} \\ &= e = \frac{dl}{l} \end{aligned} \quad \dots(1.4)$$

(b) Compressive Stress (σ_c): When a body is subjected to two equal and opposite pushes and length of the body reduces, the stress induced is called compressive stress.

(Fig 1.3). For a straight short compression member in which the resultant load P acts through centre of gravity of every cross section the stress σ_c



Fig. 1.3

$$\sigma_c = \frac{P}{A} \quad \dots(1.5)$$

Corresponding strain is called compressive strain.

$$\begin{aligned} \text{Compressive strain} &= \frac{\text{Decrease in length}}{\text{Original length}} \\ e &= \frac{dl}{l} \end{aligned} \quad \dots(1.6)$$

(C) Shear Stress (τ): When equal and opposite forces are applied tangentially to any cross sectional plane of a body, tending to slide its one part over other, the body is said to be in state of shear. The corresponding stress induced is shear stress. In Fig.1.4 riveted

joint tends to shear off at section XX due to externally applied load P. Shear stress is produced at the cross section of rivet. If the cross section area of rivet be A shear stress

$$\tau = \frac{P}{A} \quad \dots(1.7)$$

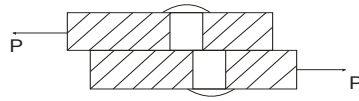


Fig. 1.4

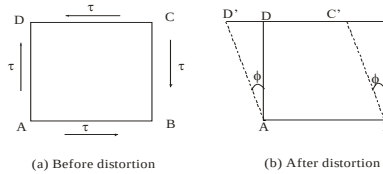


Fig. 1.5

Shear Strain: It is a measure of the angle through which a body is deformed by the applied load. Let a rectangular block ABCD, fixed at bottom is distorted through an angle ϕ by the externally applied tangential load P to face CD as in Fig. 1.5. The face CD moves to new position C'D'. Let the movement of end C is very small.

$$\text{Then shear strain} = \frac{CC'}{CB} = \tan \phi = \phi \text{ radian} \quad [\because \phi \text{ small}]$$

1.5 Elasticity: A material is composed of small particles and molecules, between which forces are acting. These molecular forces resist the change in the shape of the body which external forces tend to produce. Under the action of external forces the particles of body are displaced and the displacement continues till the equilibrium is established between the external and internal forces.

As in Fig. 1.6 a bar is loaded at the end. Under the action of this load a certain elongation of the bar will take place. When the load is diminished the elongation of the bar diminishes and the loaded end of the bar moves upward. The property by which a body returns to its original shape after removal of the load is called **elasticity**. The body is *perfectly elastic* if it recovers its original shape completely after unloading, *partially elastic* if the deformation disappear partially after unloading and *plastic* if the deformation is permanent.

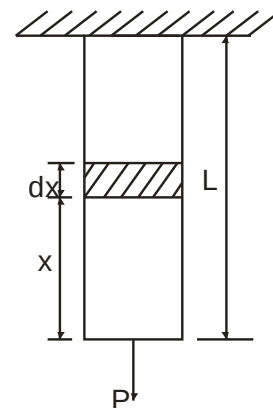


Fig. 1.6

1.6 Hooke's Law

It states that within an elastic limit stress is proportional to strain.

Stress $\propto$ Strain

or
$$\frac{\text{Stress}}{\text{Strain}} = E \quad [C \text{ is a constant}]$$

‘E’ is known as coefficient of elasticity or modulus of elasticity. The ratio of tensile stress and tensile strain or compressive stress and compressive strain is called young’s modulus of elasticity and is denoted by-.

$$\frac{\sigma_t}{\epsilon_t} \text{ or } \frac{\sigma_c}{\epsilon_c} = E \quad \dots(1.8)$$

The ratio of shear stress and shear strain is called shear modulus or modulus of rigidity. It is denoted by C or G.

$$\frac{\tau}{\epsilon_s} = C \quad \dots(1.9)$$

Let us consider a body subjected to a tensile force P, having length l and cross section area A, the stress induced

$$\sigma = \frac{P}{A}$$

And if the δ is the deformation and strain

$$\epsilon_t = \frac{\delta}{l}$$

As per Hooke’s law $\frac{\sigma}{\epsilon_t} = E$

$$\frac{Pl}{A\delta} = E$$

or deformation
$$\delta = \frac{Pl}{AE} \quad \dots(1.10)$$

1.7 Stress Strain Curve For Mild Steel: Curve (Fig. 1.7) for a round mild steel bar of which ordinate represents the stress intensities and the abscissa the corresponding strains. The points worth noting are:

- ❖ The line OA being straight. OA is called the proportional limit upto which Hooke's law is applicable i.e. stress is proportional to strain up to A.

- ❖ Mild steel remains elastic a little beyond A i.e. upto point A, so the proportional limit and elastic limit are not same for mild steel but same for rest of the materials.
- ❖ Upon loading beyond this point the elongation increases more rapidly and the diagram becomes curved. At B a sudden elongation of the bar takes place without an appreciable increase in the tensile force. The phenomenon is called 'Yielding' of the metal. The stress corresponding to this 'B' point is called yield point stress.

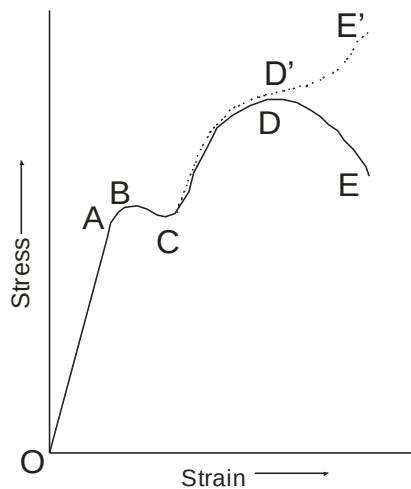


Fig. 1.7

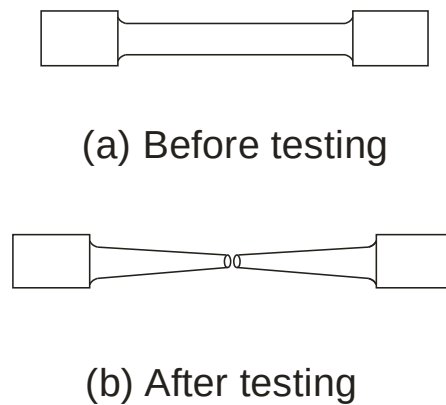


Fig. 1.8

- ❖ Upon further stretching of the bar, the material recovers its resistance and the tensile force increases with the elongation upto point 'D', where the force attains its maximum value and the process of necking begins.

The corresponding stress is called the 'Ultimate Stress'. Necking is displayed by contraction of the specimen. Fig. 1.8.

- ❖ Beyond the point D elongation of the bar takes place with a diminution of the load and fracture finally occurs at a load corresponding to point E of diagram. The corresponding stress is called the 'Breaking stress'.

The stress-strain curve is drawn experimentally and it is unique for every metal. Hence the points observed on the curve are the properties of the metal and unique for every metal. So the point A corresponds to **limit of elasticity**, point B corresponds to **Yield strength** and point D correspond to **Ultimate strength** of the metal.

Stretching of this bar is accompanied by lateral contraction but it is established for calculating the yield & ultimate point stress, hence initial cross-sectional area is used. If the load is divided by the original area of cross section it results in nominal stress which is less at breaking load than at the maximum load as indicated by E&D respectively. If actual area of cross section is used to find stress intensity, rupture or breaking stress is greater than at the maximum load. The dashed portion of the curve shows relation between actual stress intensity and strain. This is called true stress strain curve.

Fig. 1.9 represents a tensile test diagram for cast iron. The material has a very low proportional limit and has no definite yield point. For brittle materials stress-strain curve is as in Fig. 1.9. There is no yield point in case of brittle material.

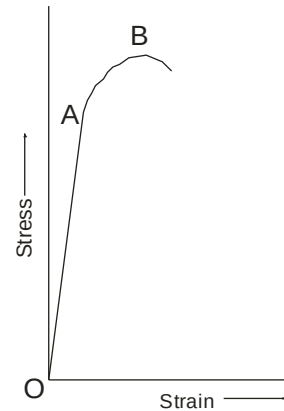


Fig. 1.9

1.8 Factor Of Safety And Working Stresses:

Working stress: The greatest calculated stress to which a part of machine or structure is ever subjected is called the working stress or allowable stress or permissible stress or design stress and the ratio

$$\frac{\text{Ultimate Strength}}{\text{Working Stress}} \text{ or } \frac{\text{Yield Stress}}{\text{Working Stress}} \quad \dots(1.11)$$

is called Factor of Safety. The ultimate or yield strength is the property of the metal and it is known while selecting the metal for design purpose. For design calculations the factor of safety is to be assumed and design stress is to be calculated. Hence from the equation (1.11) working stress is the ratio

$$\frac{\text{Ultimate Strength}}{\text{Factor of safety}} \text{ or } \frac{\text{Yield Stress}}{\text{Factor of safety}} \quad \dots(1.12)$$

Factor of safety depends upon:

1. **Nature of loading:** The factor of safety varies very greatly according to the nature of the stresses, whether constant, variable or alternating simple or compound.
2. **Homogeneity of materials used:** The factor of safety will be higher if the quality of material used is not of uniform properties.

3. **Degree of safety required:** If safety required is more than factor of safety will be higher. For example the areas where human hazards are involved we would require more safety like boiler applications.
4. **Degree of economy desired:** If we go for higher factor of safety the cost of the component will be more. Hence economical constraints are equally important while considering factor of safety.
5. **The accuracy of problem formulations:** If the estimation of stresses and load is not accurate we have to take higher factor of safety.
6. **Manufacturing processes:** If the quality of manufacturing methods used is good factor of safety required is lesser.

1.9 Elongation of Bars of Varying Cross Section

Fig 1.10 Shows a bar consisting of three lengths l_1 , l_2 , l_3 with cross section areas A_1 , A_2 , and A_3 , Subjected to an axial load P . Though the force on each section is same stress induced in each section is different.

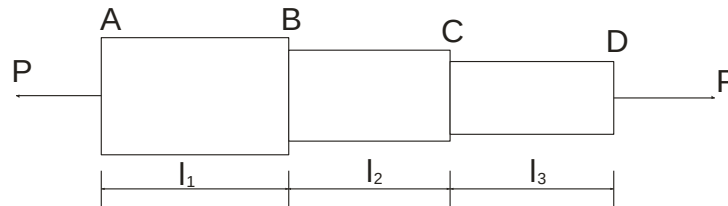


Fig. 1.10

Intensity of stress for section AB = $\sigma_1 = P/A_1$

Similarly intensity of stress for section BC = $\sigma_2 = P/A_2$

and intensity of stress for section CD = $\sigma_3 = P/A_3$

$$\therefore \text{Strain of section AB} = \epsilon_1 = \frac{\sigma_1}{E}$$

Where E – Modulus of elasticity

$$\therefore \text{Strain of section BC} = \epsilon_2 = \frac{\sigma_2}{E}$$

$$\therefore \text{Strain of section CD} = \epsilon_3 = \frac{\sigma_3}{E}$$

$$\text{Change in length of section AB} = \delta l_1 = \epsilon_1 l_1 = \frac{Pl_1}{A_1 E}$$

$$\text{Change in length of section BC} = \delta l_2 = \epsilon_2 l_2 = \frac{Pl_2}{A_2 E}$$

$$\text{Change in length of section CD} = \delta l_3 = \epsilon_3 l_3 = \frac{Pl_3}{A_3 E}$$

$$\begin{aligned} \therefore \text{Total change in length of bar} &= \delta l_1 + \delta l_2 + \delta l_3 \\ &= \frac{P}{E} \left[\frac{l_1}{A_1} + \frac{l_2}{A_2} + \frac{l_3}{A_3} \right] \end{aligned} \quad \text{.....(1.13)}$$

1.10 Elongation Of Bar Of Tapering Section

Let us consider a bar of length L , whose diameter increases uniformly from d_1 at one end to d_2 at the other end. The bar is subjected to an axial pull P as shown in Fig. 1.11. Considering an elementary length dx of the bar at a distance x from the smaller end denoting the diameter at this distance as d' ,

$$d' = d_1 + \frac{d_2 - d_1}{L} x$$

Extension of short length

$$\begin{aligned} \delta &= \frac{P dx}{\frac{\pi}{4} d'^2 E} \\ &= \frac{P dx}{\frac{\pi}{4} \left(d_1 + \frac{d_2 - d_1}{L} x \right)^2 E} \end{aligned}$$

Total elongation of bar

$$\begin{aligned} \delta &= \int_0^L \frac{4P dx}{\pi (d_1 + Kx)^2 E} \quad \left[\text{where } K = \frac{d_2 - d_1}{L} \right] \\ \delta &= - \frac{4P}{\pi E} \cdot \frac{1}{K} \left\{ \frac{1}{d_1 + Kx} \right\}_0^L \\ &= - \frac{4PL}{\pi E (d_2 - d_1)} \left\{ \frac{1}{d_1 + d_2 - d_1} - \frac{1}{d_1} \right\} \\ \delta &= \frac{4PL}{\pi E d_1 d_2} \end{aligned} \quad \text{.....(1.14)}$$

When $d_1 = d_2$ i.e. bar of constant diameter

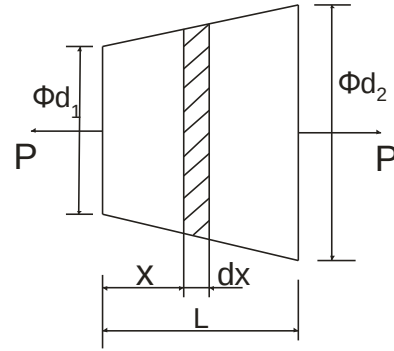


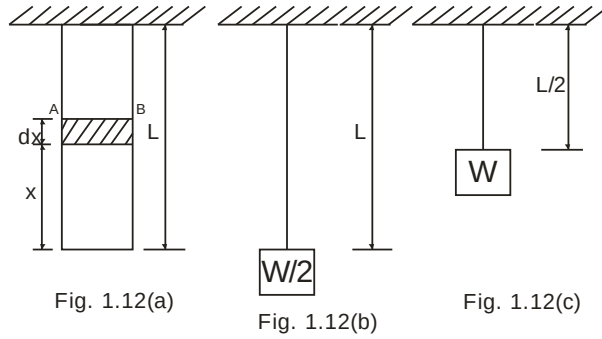
Fig. 1.11

$$\delta = \frac{4PL}{\pi d^2 E} = \frac{PL}{AE}$$

1.11 Elongation Due To Self Weight

1.11.1 Bar Of Uniform Section

Let us consider a bar of uniform cross section area A and length L . Considering an elementary strip AB of this bar of length dx at a distance x from bottom of the bar as in Fig. 1.12(a) .



The downward force acting on this strip is due to the weight of the bar that lies below this element $w_x = A \cdot x \cdot \lambda$

where λ is the density of material of bar. Deformation of length dx caused by this weight

$$\delta = \frac{w_x dx}{AE}$$

$$\delta = \frac{(A \cdot x \cdot \lambda) dx}{AE}$$

$$\delta = \frac{x \lambda dx}{E}$$

Deformation of the whole bar

$$\delta = \int_0^L \frac{x \lambda}{E} dx$$

$$\delta = \frac{\lambda L^2}{2E}$$

If W = total weight of bar we have

$$\lambda = \frac{W}{AL}$$

$$\therefore \delta = \frac{WL}{2AE} \quad \dots(1.15)$$

The above equation is applicable for following cases also

- i) When we have a weight of $W/2$ hanging at the end of a wire of length 'L' weight of wire is negligible Fig. 1.12(b).
- ii) When we have a weight of W hanging at the end of a wire of length $L/2$, weight of wire is negligible, Fig. 1.12(c).

1.12 Principal Of Superposition

Concept of free body diagram

If an elastic body is subjected to a no. of direct stresses at different sections along the length of the body, the deformation of individual sections can be very easily found if free body diagrams are drawn for the individual sections. According to principle of superposition the total deformation of the body will then be equal to the algebraic sum of the deformations of the individual sections. For example let us consider a bar subjected to

different forces at different sections as in Fig. 1.14 (a). To draw the free body diagram for AC all the forces to the left of the normal sections A to be considered. The net force is 20N, tensile considering the net force to right of C is $18 + 6 - 4 = 20\text{N}$. Thus the section AC of length l_1 , is subjected to tensile force of 20 N, as in Fig. 1.14 (b). Similarly, for

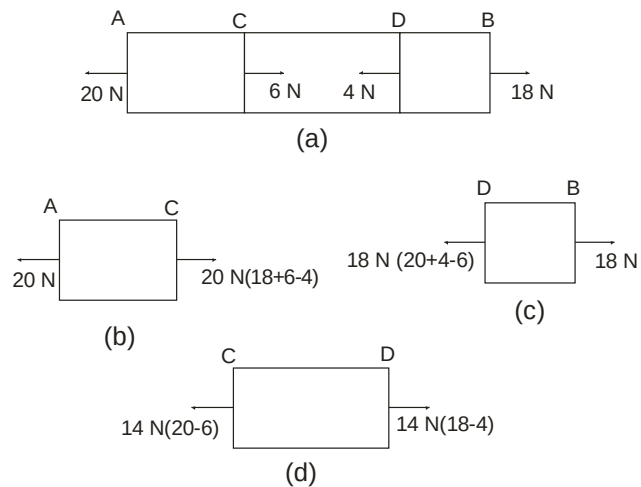


Fig. 1.14

the section CD of length l_2 consider the algebraic sum of the forces to the left of normal section at C or the sum of the forces to the right of the normal section at D. Net force on CD = $20 - 6 = 14$ or $18 - 4 = 14$. (Fig. 1.14 (d))

Similarly section DB is subjected to 18 N (Fig. 1.14 (c))

The extension of AC $\Delta_1 = \frac{P l_1}{AE} = \frac{20 l_1}{A_1 E}$

$$\text{CD } \Delta_2 = \frac{14 l_2}{A_2 E}$$

$$\text{DB } \Delta_3 = \frac{18 l_3}{A_3 E}$$

Total extension of the bar AB will be given by

$$\Delta = \Delta_1 + \Delta_2 + \Delta_3 \quad \dots(1.17)$$

1.13 Stresses In The Bars Of Composite Sections

When a bar consists of two different materials, it is said to be composite. The bars are joined in such a manner that the composite bar extends or contracts as one unit when subject to tension or compression. To find stresses and strain in a composite bar the following points are worth noting.

- (a) Extension or contraction of all the members is equal i.e. strain is also equal.
- (b) The total exit load is shared (not necessarily equally) by all the members.

Let one solid bar enclosed in a hollow tube and subjected to a load P as in Fig. 1.15.

From the condition of equilibrium

$$P_1 + P_2 = P \quad \dots(1.18)$$

Where P_1, P_2 are load shared by bar and tube.

Since the assembly is composite, the deformation of bar is equal to the deformation

of the tube:

$$\Delta_1 = \Delta_2$$

$$\frac{P_1 L_1}{A_1 E_1} = \frac{P_2 L_2}{A_2 E_2} \quad \dots(1.19)$$

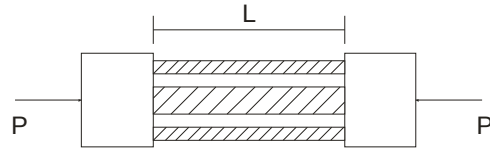


Fig. 1.15

From these two equations we can find the load shared by each member. if σ_1 , and σ_2 are the stresses in the two members respectively

$$\sigma_1 = \frac{P_1}{A_1} \quad \text{and} \quad \sigma_2 = \frac{P_2}{A_2}$$

1.15 Temperature Stresses And Strain

When a material is subjected to change in temperature, there will be corresponding change in dimensions. When a member is free to contract or extend no stresses are induced but, if natural change in dimension is prevented stresses are induced and are called thermal or temperature stresses.

Single Bar:

Let a bar of length L and coefficient of linear expansion α be heated through T° . If the expansion is allowed, the increase in length is $L\alpha T$ and there will be no stress. If

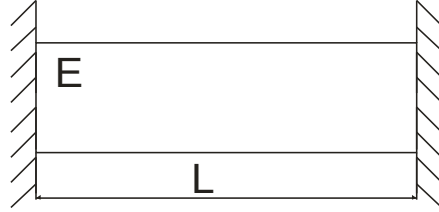


Fig. 1.17



Fig. 1.18

the ends of the bar are fixed as in Fig. 1.17 so that the expansion is prevented, then compressive strain (ϵ_c) will be produced in the bar.

$$\epsilon_c = \frac{L\alpha T}{L}$$

$$\epsilon_c = \alpha T \quad \dots(1.26)$$

and the corresponding stress

$$\sigma_c = E\epsilon_c$$

$$\sigma_c = E\alpha T \quad \dots(1.27)$$

Let us consider the case when the bar is permitted to expand by an amount a Fig. 1.18 temperature strain $\epsilon = \frac{\alpha TL - a}{L} = \alpha T - \frac{a}{L}$

$$\text{Compressive stress } \sigma = E\epsilon = \left(\frac{\alpha TL - a}{L} \right) E$$

(b) Composite Bar : A composite bar is made of two bars-steel and copper having same length rigidly connected at their free ends (Fig. 1.19)

Let L = Length of both the rods

A_s = Cross – sectional area of steel bar

A_c = Cross-sectional area of copper bar

α_s = Coefficient of linear expansion of steel bar

α_c = Coefficient of liner expansion of copper bar

t = Rise in temperature

E_s = Modulus of elasticity for steel bar

E_c = Modulus of elasticity for copper bar.

σ_c = Stress in bar

σ_s = Stress in Steel bar

As α_c is more than α_s , the copper bar will expand more than the steel bar. If both the bars are free to expand then let the copper bar expand to D and the steel bar to C. As the ends of both the bars are rigidly connected together so the composite bar will expand up to F. In attaining the position F the copper bar pulls forward the steel bar with a force P and brings it from C to F, whereas the steel bar pushes back the copper bar with a force P from D to F.

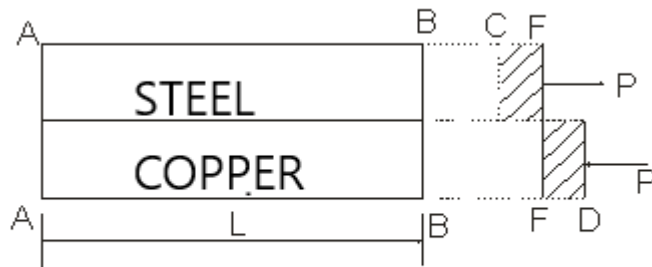


Fig. 1.19

The steel bar thus shall be subjected to a tensile force P and the copper bar to a compressive force P.

The final deformation of the composite bar = BF

$$BF = BC + CF = BD - DF \quad \dots(1.28)$$

$$BC = \alpha_s TL \text{ and } BD = \alpha_c TL$$

$$CF = \frac{PL}{A_s \cdot E_s}, \quad DF = \frac{PL}{A_c \cdot E_c}$$

Putting the values into equation 1.28

$$\alpha_s TL + \frac{PL}{A_s \cdot E_s} = \alpha_c TL - \frac{PL}{A_c \cdot E_c}$$

$$\text{Or } \alpha_s T + \frac{\sigma_s}{E_s} = \alpha_c T - \frac{\sigma_c}{E_c} \quad \dots\dots\dots(1.29)$$

Problem 1.1 An elastic rod 25 mm in diameter, 200 mm long extends by 0.25 mm under a tensile load of 80 kN. Find the intensity of stress, the strain and the elastic modulus for the material of the rod.

Solution: Diameter of rod $d = 25 \text{ mm}$

Length of the rod $l = 200 \text{ mm}$

Tension $P = 80 \text{ kN}$

Area of the section $A = \frac{\pi d^2}{4} = \frac{\pi \times 25^2}{4} = 490.87 \text{ mm}^2$

Intensity of stress $\sigma = \frac{P}{A} = \frac{80 \times 1000}{490.87} \text{ N/mm}^2$

$$= 162.97 \text{ N/mm}^2$$

Strain $= e = \frac{\delta}{L} = \frac{0.25}{200} = 0.00125$

Elastic Modulus $= E = \frac{\sigma}{e} = \frac{162.97}{0.00125}$

$$= 1.3 \times 10^5 \text{ N/mm}^2$$

Problem 1.2 A hollow steel column of external diameter 250 mm has to support an axial load of 4000 kN. If the ultimate stress for the steel column is 480 N/mm². Find the internal diameter of the column allowing a factor of safety of 4.

Solution: External diameter $D = 250 \text{ mm}$

Internal diameter $d \text{ mm}$

Load on the column $P = 4000 \times 1000 \text{ N}$

Ultimate stress $= 480 \text{ N/mm}^2$

Factor of safety $= 4$

$\therefore$ Safe stress $\sigma = \frac{480}{4} = 120 \text{ N/mm}^2$

$\therefore$ Sectional area required $A = \frac{P}{\sigma} = \frac{4000 \times 1000}{120} = 33333.33 \text{ mm}^2$

$\therefore \frac{\pi}{4}(250^2 - d^2) = 33333.33$

$\therefore 62500 - d^2 = 21221$

$\therefore d^2 = 29167$

$\therefore d = 170 \text{ mm}$

Problem 1.3 A short hollow cast iron cylinder of wall thickness 10 mm is to carry a compressive load of 600 kN. Determine its outside diameter if the ultimate crushing stress for the material is 540 N/mm². Use a factor of safety of 5.

Solution:

Let D_0 = Outside diameter of the cylinder in mm.

The wall thickness of the cylinder = 10 mm

$$\therefore \text{Inside diameter of the cylinder} = \frac{\pi}{4} \{ D_0^2 - (D_0 - 20)^2 \}$$

$$\text{Area of cross-section} = \frac{\pi}{4} \{ D_0^2 - (D_0 - 20)^2 \}$$

$$= \frac{\pi}{4} (10D_0 - 100) \text{ mm}^2$$

$$\text{Crushing load for the column} = \frac{\pi}{4} (10D_0 - 100) \times 540 \text{ N}$$

$$\text{Factor of safety} = 5 = \frac{\text{Crushing load}}{\text{Safe load}} = \frac{\text{Crushing load}}{600 \times 10^3}$$

$$\therefore \text{Safe load} = \frac{600 \times 10^3}{5} = \frac{\pi (10D_0 - 100) \times 540}{4}$$

$$600 \times 10^3 = \frac{\pi (10D_0 - 100) \times 540}{4}$$

$$600 \times 10^3 \times 4 = \pi (10D_0 - 100) \times 540$$

$$2400 \times 10^3 = \pi (10D_0 - 100) \times 540$$

$$\therefore D_0 = \frac{2400 \times 10^3}{\pi \times 540} + 100 = 186.93 \text{ mm}$$

Problem 1.4 Calculate the force required for punching a hole of 20 mm diameter through a mild steel plate 4 mm thick. Take maximum shear strength of mild steel as 320 N/mm². Also find the compressive stress on the punch.

Solution: Let d be the diameter of the hole to be punched, t be the thickness of the plate and τ be the maximum shear stress.

Force required for punching the hole = Shear stress $\times$ Area

$$= \tau \times \pi dt = 320 \times \pi \times 20 \times 4$$

$$= 80384 \text{ N}$$

Area of the punch in contact with the plate surface, $A = \frac{\pi}{4} d^2$

$$\therefore \text{Compressive stress on the punch, } \sigma_c = \frac{P}{A} = \frac{80384}{\frac{\pi}{4} \times (20)^2} = 256 \text{ N/mm}^2.$$

Problem 1.5 The following data refer to a tensile test conducted on a mild steel bar.

(i) Diameter of the steel bar = 30 mm

(ii) Gauge length = 200 mm

- (iii) Extension at a load of 100 kN = 0.139 mm
- (iv) Load at elastic limit = 250 kN
- (v) Maximum Load = 400 kN
- (vi) Total = 56 mm
- (vii) Diameter of the rod at failure = 22.25 mm

Calculate (a) the young's modulus, (b) The stress at elastic limit, (c) the percentage elongation and (d) The percentage decrease in area.

Solution: Area of the rod $= A = \frac{\pi}{4} (30)^2 \text{ mm}^2 = 706.86 \text{ mm}^2$

(a) Young's Modulus
 Stress $= \sigma = \frac{P}{A} = \frac{100 \times 1000}{706.86} \text{ N/mm}^2 = 141.47 \text{ N/mm}^2$

Strain $= e = \frac{\frac{dl}{l}}{\frac{\sigma}{E}} = \frac{0.139}{200} = 0.000695$

$\therefore$ Young's Modulus $E = \frac{\sigma}{e} = \frac{141.47}{0.000695} \text{ N/mm}^2 = 2.0355 \times 10^5 \text{ N/mm}^2$

(b) Stress at elastic limit $= \frac{250 \times 10^3}{706.86} = 35.68 \text{ N/mm}^2$

(c) Percentage elongation $= \frac{\text{Increase in Length}}{\text{Original Length}} \times 100 = \frac{56}{200} \times 100 = 28\%$

(d) Percentage decrease in area $= \frac{\frac{\pi}{4}(d^2 - d'^2)}{\frac{\pi d^2}{4}} \times 100 = \frac{d^2 - d'^2}{d^2} \times 100 = 44.99\%$

$= \frac{30^2 - 22.25^2}{30^2} \times 100 = 44.99\%$

Problem 1.6 The following data refer to a mild steel specimen tested in a laboratory:

- (i) Diameter of the specimen = 25 mm
- (ii) Length of the specimen = 300 mm
- (iii) Extension under a load of 15 kN = 0.045 mm
- (iv) Load at yield point = 130 kN
- (v) Maximum load = 210 kN

- (vi) Length of the specimen after failure = 375 mm
 (vii) Neck diameter = 17.75 mm

Determine.

- (i) Young's Modulus (ii) Yield point (iii) Ultimate stress (iv) Percentage of elongation (v) Percentage reduction in area (vi) Safe stress adopting a factor of safety of 2.

Solution: Area of the specimen $= A = \frac{\pi \times 25^2}{4} = 490.87 \text{ mm}^2$

At a load of 15 kN

Stress intensity $= \sigma = \frac{15 \times 10^3}{490.87} = 30.56 \text{ N/mm}^2$

Strain corresponding to this load $= \frac{0.045}{30.56} = 1.5 \times 10^{-4}$

Young's Modulus $E = \frac{1.5 \times 10^{-4}}{1.5 \times 10^{-4}} = 2.036 \times 10^5 \text{ N/mm}^2$

Stress at yield point $= \frac{130 \times 10^3}{490.87} = 264.84 \text{ N/mm}^2$

Ultimate stress $= \frac{210 \times 10^3}{490.87} = 427.81 \text{ N/mm}^2$

Percentage elongation $= \frac{375 - 300}{300} \times 100 = 25\%$

Percentage reduction in area $= \frac{d^2 - d'^2}{d^2} \times 100 = \frac{25^2 - 17.75^2}{25^2} \times 100 = 49.59\%$

Safe stress $= \frac{\text{Yield stress}}{\text{factor of safety}} = \frac{264.84}{2} = 132.42 \text{ N/mm}^2$

Problem 1.7 Find the total elongation of a copper bar as shown in Fig. 1.20 subjected to an axial load of 200 kN. Take $E = 100 \text{ GPa}$.

Solution:

Let δL_1 = Elongation for the portion AB of the bar.

δL_2 = Elongation for the portion BC of the bar.

δL_3 = Elongation for the portion CD of the bar.

δL = Total elongation of the copper bar.

$\therefore L_1, L_2$ and L_3 be the length of the portions AB, BC and CD respectively.

The load $P = 200 \text{ kN} = 200 \times 10^3 \text{ N}$.

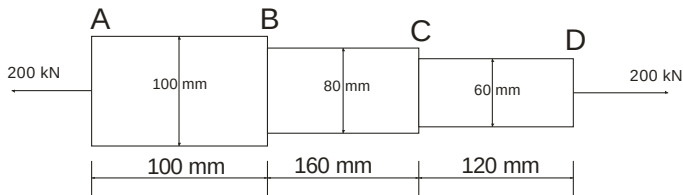


Fig. 1.20

Now $\delta_{L_1} = \frac{PL_1}{A_1 E} = \frac{200 \times 10^3 \times 0.10}{\frac{\pi}{4} \times (0.1)^2 \times 100 \times 10^9} = 0.0254 \text{ mm}.$

$$\delta_{L_2} = \frac{PL_2}{A_2 E} = \frac{200 \times 10^3 \times 0.16}{\frac{\pi}{4} \times (0.08)^2 \times 100 \times 10^9} = 0.0636 \text{ mm}$$

$$\delta_{L_3} = \frac{PL_3}{A_3 E} = \frac{200 \times 10^3 \times 0.12}{\frac{\pi}{4} \times (0.06)^2 \times 100 \times 10^9} = 0.0848 \text{ mm}$$

$\therefore$ Total elongation, $\delta_L = \delta_{L_1} + \delta_{L_2} + \delta_{L_3} = 0.0254 + 0.0636 + 0.0848 = 0.1738 \text{ mm}$

Problem 1.8 A steel bar of 30 mm diameter is loaded as shown in Fig 1.21. Determine the stresses in each part and the total elongation. Take $E = 210 \text{ GPa}$.

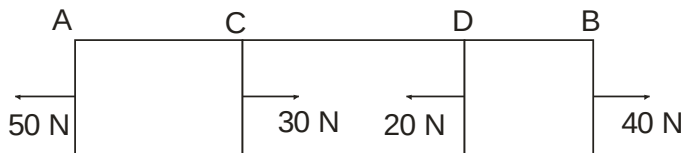


Fig. 1.21

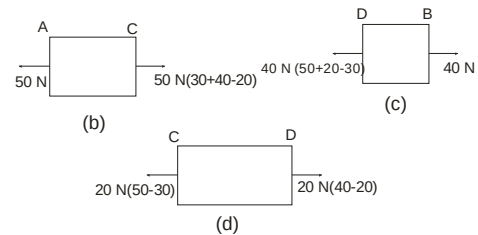


Fig. 1.22

Solution: The free body diagram for each portion is shown in Fig. 1.22.

Area of cross-section, $A = \frac{\pi}{4} \times (0.03)^2 = \frac{\pi \times 2.25 \times 10^{-4}}{50 \times 1000} \text{ m}^2$

$$\sigma_{AB} = \frac{\pi \times 10^{-4} \times 2.25}{20 \times 1000} = 70.77 \times 10^6 \text{ N/m}^2$$

$$\sigma_{BC} = \frac{\pi \times 10^{-4} \times 2.25}{40 \times 1000} = 28.31 \times 10^6 \text{ N/m}^2$$

$$\sigma_{CD} = \frac{\pi \times 10^{-4} \times 2.25}{50 \times 1000} = 56.61 \times 10^6 \text{ N/m}^2$$

$$\begin{aligned}
 \text{Total elongation, } \delta L &= \frac{P_1 L_1 + P_2 L_2 + P_3 L_3}{AE} \\
 &= \frac{50 \times 10^3 \times 0.4 + 20 \times 10^3 \times 0.3 + 40 \times 10^3 \times 0.2}{\pi \times 2.25 \times 10^{-4} \times 210 \times 10^9} \\
 &= 0.229 \text{ mm}
 \end{aligned}$$

Problem 1.9 If a tension bar is found to taper uniformly from $(D - a)$ mm diameter to $(D + a)$ mm, prove that the error involved in using the mean diameter to calculate young's Modulus is $\left(\frac{10a}{D}\right)^2$ percent.

Solution: we know that the elongation of a tapering rod is

$$\delta L = \frac{4PL}{\pi E D d}$$

Here $D_1 = (D + a)$ and $D_2 = (D - a)$

$$\therefore \delta L = \frac{4PL}{\pi E (D + a)(D - a)} = \frac{4PL}{\pi E (D^2 - a^2)}$$

or

$$E = \frac{4PL}{\pi \delta L (D^2 - a^2)}$$

It is the actual value of young's modulus.

Let E_1 be the young's modulus calculated by taking mean diameter.

$$\begin{aligned}
 \text{Mean diameter} &= \frac{D_1 + D_2}{2} = D \\
 \therefore E_1 &= \frac{4PL}{\pi \delta L D^2} \quad \left[\because d = D \right] \\
 \therefore \text{Percentage error} &= \frac{E - E_1}{E} \times 100 = \frac{\frac{4PL}{\pi \delta L (D^2 - a^2)} - \frac{4PL}{\pi \delta L D^2}}{\frac{4PL}{\pi \delta L (D^2 - a^2)}} \times 100 \\
 &= \frac{\frac{1}{D^2 - a^2} - \frac{1}{D^2}}{\frac{1}{D^2 - a^2}} \times 100 = \left(1 - \frac{D^2 - a^2}{D^2} \right) \times 100 \\
 &= \frac{100a^2}{D^2} = \left(\frac{10a}{D} \right)^2 \text{ Proved.}
 \end{aligned}$$

Problem 1.10 A rod of square section of side D at one end tapers to a square section of side d at the other end. The rod is of length l and is subjected to an axial pull P . Find the increase in length of the rod.

Solution: Consider a section AB at a distance x from the smaller end (Fig. 1.23)

$$AB = d + \frac{D-d}{l}x$$

$$\text{Cross-sectional area at } AB = \left[d + \frac{D-d}{l}x \right]^2$$

$$\text{The elongation in length} = \frac{P \cdot dx}{\left[d + \frac{D-d}{l}x \right]^2} \times E$$

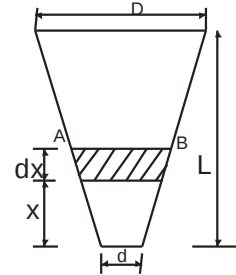


Fig. 1.23

$$\text{The increase in length of the rod} = \int_0^L \frac{P \left(\frac{-l}{D-d} \right) \left[\frac{1}{d + \frac{D-d}{l}x} \right]}{E} dx$$

$$= \frac{P}{E} \left(\frac{-l}{D-d} \right) \left(\frac{1}{D} - \frac{1}{d} \right) = \frac{Pl}{EDd}$$

Problem 1.11 A reinforced concrete column has a square cross-section. It is reinforced with 4 steel rods of diameter 20 mm at each corner. The column carries an axial compressive load of 1000 kN. Find the required size of the column so that the stress in the concrete does not exceed 4 N/mm^2 . Find also the load carried by steel and the stress induced in it. Assume modular ratio $\frac{E_s}{E_c} = 15$.

Solution: Given

Safe load, $P = 1000 \text{ kN}$, steel rod diameter, $d = 20 \text{ mm}$.

Number of steel rods = 4, stress in the concrete, $\sigma_c = 4 \text{ N/mm}^2$

Area of steel rods $A_s = 4 \times \frac{\pi}{4} \times (20)^2 = 1256 \text{ mm}^2$

Let the side of the column be $= x$

Area of the concrete A_c

$$\text{Now } \frac{\sigma_s}{E_c} = \frac{\sigma_c}{E_c} \text{ or } \frac{\sigma_s}{\sigma_c} = 15 \quad \therefore \sigma_s = 15 \sigma_c$$

Here $\sigma_c = 4 \text{ N/mm}^2$
 $\therefore \sigma_s = 15 \times 4 = 60 \text{ N/mm}^2$
 $\therefore$ Safe load, $P = P_s + P_c = \sigma_s \cdot A_s + \sigma_c \cdot A_c$
 $1000 \times 10^3 = 60 \times 1256 + 4 \times A_c$
 $\therefore A_c = 231160 \text{ mm}^2$
 Now, Total cross-sectional area,
 $x^2 = A_s + A_c = 1256 + 231160$
 $= 232416 \text{ mm}^2$
 $\therefore x = 482.1 \text{ mm}$
 Load carried by steel $= A_s \times \sigma_s = 1256 \times 60$
 $= 75360 \text{ N} = 75.36 \text{ kN}$
 Size of the column $= 485 \times 485 \text{ mm}$.

Problem 1.12 A solid steel bar 500 mm long and 70 mm in diameter is placed inside an aluminium tube having 75 mm inside diameter and 100 mm outside diameter. The aluminium tube is 0.15 mm longer than the steel bar. An axial load of 600 kN is applied to the bar and tube through rigid cover plates. Find the stresses developed in the steel bar and aluminium tube. E for steel $= 2.2 \times 10^5 \text{ N/mm}^2$, E for aluminium $= 0.7 \times 10^5 \text{ N/mm}^2$.

Solution:

$$A_s = \frac{\pi}{4} (70)^2 = 3846.5 \text{ mm}^2$$

$$A_{al} = \frac{\pi}{4} [(100)^2 - (75)^2] = 3434.4 \text{ mm}^2$$

Let the aluminium tube be compressed by δL_{al} and steel bar by δL_s

$$\delta L_{al} = \frac{\sigma_{al}}{E_{al}} \times L_{al}, \delta L_s = \frac{\sigma_s}{E_s} \times L_s$$

$$\frac{\sigma_{al}}{E_{al}} \times L_{al} - \frac{\sigma_s}{E_s} \times L_s = 0.15$$

or
$$\frac{\sigma_{al}}{0.70 \times 10^5} \times 500 - \frac{\sigma_s}{2.20 \times 10^5} \times 500 = 0.15$$

or
$$\frac{\sigma_{al}}{70} - \frac{\sigma_s}{220} = 300000$$

$$\sigma_{al} \cdot A_{al} + \sigma_s \cdot A_s = 600 \times 10^3$$

or
$$\sigma_{al} \times 3434.4 + \sigma_s \times 3846.5 = 600 \times 10^3$$

or
$$\sigma_{al} = 174.7 - 1.12 \sigma_s$$

Substituting the value of σ_{al} in Equation (i)

$$\frac{174.7 - 1.1199\sigma_s}{70} - \frac{\sigma_s}{220} = 300000$$

$$\therefore \sigma_s = 106.82 \text{ N/mm}^2$$

$$\sigma_{al} = 54.99 \text{ N/mm}^2$$

Problem 1.13 Two copper rods and one steel rod together support a load 500 kN as shown in Fig. 1.24. Find the stresses in the rods. Take $E_s = 2 \times 10^5 \text{ N/mm}^2$ and $E_c = 1 \times 10^5 \text{ N/mm}^2$.

Solution: Each rod will be compressed by the same amount.

Let the decrease in length of each rod be δ mm

Let the strain in copper and steel be e_c and e_s respectively

$$\therefore e_c l_c = e_s l_s = \delta$$

$$\therefore e_s = \frac{l_c}{l_s} e_c = e_c = \frac{140}{210} e_c$$

$$\therefore e_s = \frac{2}{3} e_c$$

Let the stress in steel and copper be σ_s and σ_c respectively.

$$\therefore \frac{\sigma_s}{E_s} = \frac{\sigma_c}{E_c} \text{ and } \sigma_c = e_c E_c$$

$$\therefore \frac{\sigma_s}{E_s} = \frac{\sigma_c}{E_c}$$

$$\therefore \frac{\sigma_s}{\sigma_c} = \frac{2}{3} \times 2 = \frac{4}{3}$$

$$\therefore \sigma_s = \frac{4}{3} \sigma_c$$

Fig. 1.24

.....(1.36)

Load on steel + Load on copper = Total load applied

$$\text{i.e., } \sigma_s A_s + \sigma_c A_c = P$$

$$\therefore \frac{4}{3} \sigma_c \times 3200 + \sigma_c \times (2 \times 2000)$$

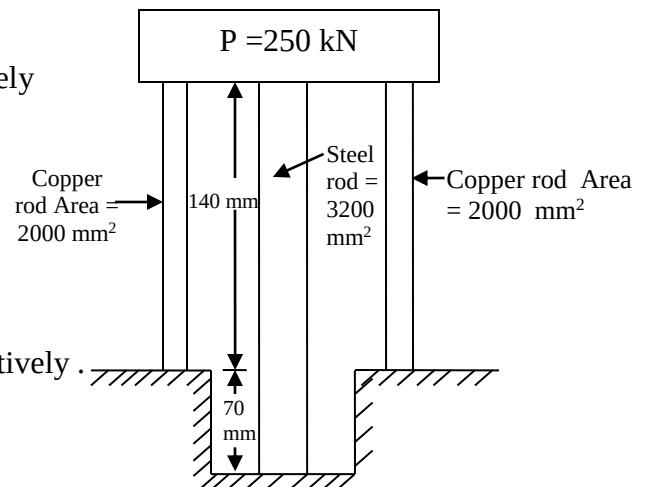
$$= 250000$$

$$\therefore \frac{128}{3} \sigma_c + 40 \sigma_c = 2500$$

$$\therefore \frac{248}{3} \sigma_c = 2500$$

$$\therefore \sigma_c = \frac{30.24 \text{ N/mm}^2}{3}$$

$$\text{And } \sigma_s = \frac{4}{3} \times 30.24 = 40.32 \text{ N/mm}^2$$



Problem 1.14 A steel rod 18 mm in diameter passes centrally through a steel tube 25 mm in internal diameter and 30 mm in external diameter. The tube is 700 mm long and is closed by rigid washers of negligible thickness which are fastened by nuts threaded on the rod. The nuts are tightened until the compressive load on the tube is 15 kN. Calculate the stresses in the tube and the rod.

Find the increase in these stresses when one nut is tightened by one quarter of a turn relative to the other. There are 4 threads per 10 mm. Take $E = 2 \times 10^5 \text{ kN/mm}^2$

Solution: When the nuts are tightened the tube will be compressed and the rod will be elongated. Since no external forces have been applied, the compressive load on the tube must be equal to the tensile load on the rod.

Area of the tube

$$A_t = \frac{\pi}{4} (30^2 - 25^2) \text{ mm}^2 = \frac{275\pi}{4} \text{ mm}^2$$

Area of the rod

$$A_r = \frac{\pi}{4} \times 18^2 = 81\pi \text{ mm}^2$$

Let the stresses in the rod and tube be p_r and p_t N/mm² respectively.

Tensile load on the rod = compressive load on the tube

$$\begin{aligned} \therefore \sigma_r A_r &= \sigma_t \cdot A_t \\ \therefore \sigma_r &= \frac{A_t}{A_r} \cdot \sigma_t \\ \therefore \sigma_r &= \frac{275\pi}{4} \cdot \frac{1}{81\pi} \sigma_t \\ \therefore \sigma_r &= \frac{275}{324} \sigma_t \end{aligned}$$

(I) when the compressive load on the tube is 15 kN

$$\therefore \text{Stress in the tube } \sigma_t = \left(\frac{15000}{\frac{275\pi}{4}} \right) = 69.5 \text{ N/mm}^2 \text{ (Compressive)}$$

$$\therefore \text{Stress in the rod } \sigma_r = \frac{275}{324} \times 69.6 \text{ N/mm}^2 = 58.99 \text{ N/mm}^2 \text{ (tensile)}$$

(ii) when one nut is tightened by one quarter of a turn.

Let σ_t and σ_r be the stresses due to tightening of the nut by one-quarter of a turn.

Obviously σ_t is a compressive stress and σ_r is tensile, and

$$\sigma_r = \frac{275}{324} \sigma_t$$

$\therefore$ Reduction in the length of the tube

$$= \frac{\sigma_t \times 700}{E} = \frac{\sigma_t \times 700}{200000} \text{ mm} = 0.00350 \sigma_t \text{ mm}$$

Extension of the rod

$$\begin{aligned} \frac{\sigma_r \times 700}{E} &= \frac{\sigma_r \times 700}{200000} \text{ mm} \\ &= \frac{275}{324} \cdot \frac{\sigma_t \times 700}{200000} \text{ mm} \\ &= 0.002971 \sigma_t \text{ mm} \end{aligned}$$

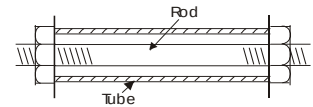


Fig. 1.25

But contraction of the tube + extension of the rod = axial advance of the nut.

$$0.00350 \sigma_t + 0.002971 \sigma_t = \frac{1}{4} \times \frac{1}{4} (10) \text{ mm} = 0.625 \text{ mm}$$

$$\therefore 0.006471 \sigma_t = 0.625$$

$$\therefore \sigma_t = \frac{96.6}{2.75} \text{ N/mm}^2 \text{ (compressive)}$$

$$\sigma_t = \frac{2.75}{324} \times 96.6 = 81.98 \text{ N/mm}^2 \text{ (tensile)}$$

Problem 1.15 A rod is 2 meters long at 15°C . Find the expansion of the rod when the temperature is raised to 85°C . If this expansion is prevented, find the stress in the material. Take $E = 1 \times 10^5 \text{ N/mm}^2$ and $\alpha = 0.000012 \text{ per } ^\circ\text{C}$.

Solution: Rise in temperature = $T = 85 - 15 = 70^\circ \text{C}$

$$\text{Free expansion} = \alpha Tl = 0.000012 \times 70 \times 2 \times 1000 = 1.68 \text{ mm}$$

$$\text{Temperature stress} = \alpha TE = 0.000012 \times 70 \times 1 \times 10^5 = 84 \text{ N/mm}^2$$

Problem 1.19 A rod of steel is 20 meters long at a temperature of 30°C . Find the free expansion of the rod when the temperature is raised to 75°C . Find the temperature stress produced (i) When the expansion of the rod is prevented (ii) when the rod is permitted to expand by 6.8 mm. Take $\alpha = 12 \times 10^{-6} \text{ per } ^\circ\text{C}$ and $E = 2 \times 10^5 \text{ N/mm}^2$

Solution:

$$\text{Free expansion of the rod} = \alpha Tl = 12 \times 10^{-6} (75 - 30) 20 \times 2 \times 1000 \text{ mm} = 10.80 \text{ mm}$$

(i) When the expansion is fully prevented

$$\begin{aligned} \text{Temperature stress} &= \alpha TE = 12 \times 10^{-6} \times (75 - 30) 2 \times 10^5 \text{ N/mm}^2 \\ &= 108 \text{ N/mm}^2 \end{aligned}$$

(ii) when the rod is permitted to expand by 6.8 mm

In this case, expansion prevented = 10.8 - 6.8 = 4 mm

$$\therefore \text{Strain} = \frac{\text{Expansion prevented}}{\text{Original Length}} = \frac{4}{20 \times 1000} = \frac{3}{5000}$$

$$\therefore \text{Temperature stress} = \text{Strain} \times E = \frac{1}{5000} \times 2 \times 10^5 \text{ N/mm}^2 = 40 \text{ N/mm}^2$$

Problem 1.16 A 15 mm diameter steel rod passes centrally through a copper tube 60 mm external diameter and 50 mm internal diameter. The tube is closed at each end by rigid plates of negligible thickness. The nuts are tightened lightly on the projecting parts of the rod. If the temperature of the assembly is raised by 60°C, calculate the stresses developed in copper and steel.

Take $E_s = 2.10 \times 10^5 \text{ N/mm}^2$; $E_c = 1.05 \times 10^5 \text{ N/mm}^2$;

$\alpha_s = 12 \times 10^{-6} \text{ per } ^\circ\text{C}$; $\alpha_c = 17.5 \times 10^{-6} \text{ per } ^\circ\text{C}$

Solution: Area of the steel rod $= A_s = \frac{\pi}{4} (15)^2 \text{ mm}^2 = 56.25 \pi \text{ mm}^2$

Area of the copper tube $= A_c = \frac{\pi}{4} (60^2 - 50^2) \text{ mm}^2 = 275 \pi \text{ mm}^2$

Free expansion of steel $= \alpha_s Tl$

Free expansion of copper $= \alpha_c Tl$

Let the actual expansion of each component be δ

$$\alpha_c Tl > \delta > \alpha_s Tl$$

$\therefore$ Steel is in tension and copper is in compression

Let σ_s and σ_c be the stresses in steel and copper.

For the equilibrium of the system

Tension in steel = Compression in copper

$$\sigma_s A_s = \left[\frac{A_c}{A_s} \right] \sigma_c A_c$$

$$\sigma_s = \left[\frac{275\pi}{26.25\pi} \right] \sigma_c$$

$\therefore$

$$\therefore \sigma_s = 4.89 \sigma_c$$

Actual expansion of steel = Actual expansion of copper

$$\alpha_s Tl + \frac{\sigma_s l}{E} = \alpha_c Tl - \frac{\sigma_c l}{E_c}$$

$$\therefore \alpha_s T + \frac{\sigma_s}{E_s} = \alpha_c T - \frac{\sigma_c}{E_c}$$

But $\sigma_s = 4.89 \sigma_c$ and substituting for α_s , α_c , E_s and E_c , we get

$$12 \times 10^{-6} \times 60 + \frac{4.89 \sigma_c}{2.10 \times 10^5} = 17.5 \times 10^{-6} \times 60 - \frac{\sigma_c}{1.05 \times 10^5}$$

$$72 + 2.32 \sigma_c = 105 - 0.952 \sigma_c$$

$$3.28 \sigma_c = 33$$

$$\sigma_c = 10.06 \text{ N/mm}^2 \text{ (compressive)}$$

$$\sigma_s = 4 \times 11.55 = 46.20 \text{ N/mm}^2 \text{ (tensile)}$$

Problem 1.17 A copper rod 22 mm diameter screwed at the ends passes through a steel tube 25 mm and 30 mm internal and external diameters. The temperature of the whole assembly is raised to 116°C and the nuts on the rod are then screwed lightly on the ends of the tube. Find the intensity of stress in the rod and the tube when the common temperature has fallen to 16°C .

$$\text{Coefficient of expansion for steel} = 12 \times 10^{-6} \text{ per } ^\circ \text{C}$$

$$\text{Coefficient of expansion for copper} = 17.5 \times 10^{-6} \text{ per } ^\circ \text{C}$$

$$\text{Modulus of Elasticity for steel} = 2.1 \times 10^5 \text{ N/mm}^2$$

$$\text{Modulus of Elasticity for copper} = \frac{1.05 \times 10^5 \text{ N/mm}^2}{4}$$

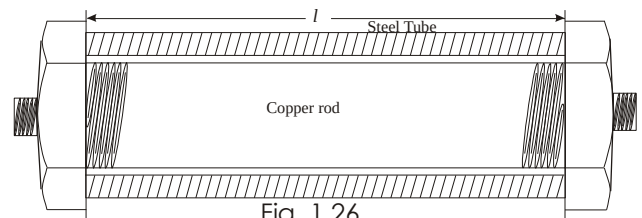
$$\text{Area of the steel tube } A = \frac{\pi}{4} [30^2 - 25^2] = \frac{275}{4} \pi \text{ mm}^2$$

Solution:

$$\text{Area of the gun metal rod } A_s = \frac{\pi}{4} \times 22^2 = 121\pi \text{ mm}^2$$

Let the length of the rod and the tube between the nuts be l mm. If the two members had been free to contract, free contraction of the copper rod $= \alpha_c Tl$, free contraction of the steel tube $= \alpha_s Tl$

Since α_c is greater than α_s the free contraction of the copper rod is greater than the free contraction of the steel tube. But, since the ends of the rod have been provided with nuts the two members are not free to contract fully, each of the member will contract by the same amount. Let δ mm be the final contraction of the each rod. The free



contraction of the copper rod is greater than δ , while the free contraction of the steel tube is less than δ . Hence the steel tube will be subjected to compressive stress while the copper rod will be subjected to tensile stress. Let σ_s and σ_g be the stress in steel and copper.

For the equilibrium of the whole system:

Total compressive force in steel = Total tension in gun metal.

$$\therefore \sigma_s A_s = \sigma_c A_g = \left(\frac{25\pi}{4} \right) \sigma_c$$

$$\sigma_s =$$

$$\therefore \sigma_s = 1.76 \sigma_c$$

Final contraction of steel = final contraction of gun metal

$$\therefore \alpha_s T l + \frac{\sigma_s}{E_s} l = \alpha_g T l - \frac{\sigma_c}{E_g} l$$

$$\therefore \alpha_s T + \frac{\sigma_s}{E_s} = \alpha_g T - \frac{\sigma_c}{E_g}$$

$$\text{Since } T = 126^\circ - 16^\circ = 110^\circ \text{ C}$$

$$\therefore 12 \times 10^{-6} \times 110 + \frac{1.76 \sigma_c}{2.1 \times 10^5} = 17.5 \times 10^{-6} \times 110 - \frac{\sigma_c}{1.05 \times 10^5}$$

$$132 + 0.838 \sigma_c = 192.5 - 0.952 \sigma_c$$

$$1.79 \sigma_c = 60$$

$$\sigma_c = 33.8 \text{ N/mm}^2 \text{ (Tensile)}$$

$$\sigma_s = 1.76 \times 33.8 = 59.5 \text{ N/mm}^2 \text{ (Compressive)}$$

Problem 1.18 A steel bar is placed between two copper bars each having the same area and length as the steel bar at 20° C . At this stage they are rigidly connected together at

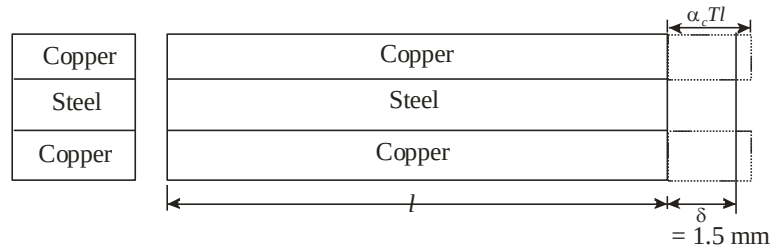


Fig. 1.27

both the ends. When the temperature is raised to 320° C , the length of the bars increases by 1.50 mm. Determine the original length and the final stresses in the bars.

Take $E_s = 2.1 \times 10^5 \text{ N/mm}^2$; $E_c = 1 \times 10^5 \text{ N/mm}^2$

$$\alpha_s = 0.000012 \text{ per } ^\circ\text{C}$$

$$\alpha_c = 0.1111175 \text{ per } ^\circ\text{C}$$

Solution: Let the sectional area of the steel component be $A \text{ mm}^2$

$\therefore$ Sectional area of the copper component = $2A \text{ mm}^2$

Free expansion of steel component = $\alpha_s Tl$

Free expansion of copper component = $\alpha_s Tl$

Let δ be the actual expansion of each bar.

$$\alpha_s Tl < \delta < \alpha_c Tl$$

$\therefore$ Steel is in tension and copper is in compression.

Let σ_s and σ_c be the stresses in steel and copper respectively.

For the equilibrium of the system.

Tension in steel = Compression in copper.

$$\therefore \sigma_s A = \sigma_c (2A)$$

$$\therefore \sigma_s = 2 \sigma_c$$

Actual expansion of steel = Actual expansion of copper.

$$\alpha_s Tl + \frac{\sigma_s l}{E_s} = \alpha_c Tl - \frac{\sigma_c l}{E_c}$$

$$\alpha_s T + \frac{\sigma_s}{E_s} = \alpha_c T - \frac{\sigma_c}{E_c}$$

$$T = 320 - \frac{2\sigma_c}{2.1 \times 10^5} \text{ } ^\circ\text{C}$$

$$\therefore 0.000012 \times 300 + \frac{\sigma_s}{2.1 \times 10^5} = 0.0000175 \times 300 - \frac{\sigma_c}{1 \times 10^5}$$

$$360 + 0.952 \sigma_c = 525 \sigma_c$$

$$1.952 \sigma_c = 165$$

$$\therefore \sigma_c = 84.52 \text{ N/mm}^2 \text{ (compressive)}$$

$$\therefore \sigma_s = 2 \times 84.52 = 169.04 \text{ N/mm}^2 \text{ (tensile)}$$

Actual expansion of the steel bar = Actual expansion of the copper bar

$$= \alpha_s Tl + \frac{\sigma_s l}{E_s} = 1.50$$

$$0.000012 \times 300l + \frac{169.04}{2.1 \times 10^5} l = 1.50$$

$$360l + 80.495l = 150000$$

$$440.495l = 150000$$

$$l = 340.526 \text{ mm}$$

Problem 1.19 A steel bar is placed between two copper bars each having the same area and length as the steel bar at 15°C . At this stage they are rigidly connected together at both the ends. When the temperature is raised to 315°C , the length of the bars increases by 1.5 mm. Determine the original length and the final stresses in the bars.

Take $E_s = 2.1 \times 10^5 \text{ N/mm}^2$ $\alpha_s = 12 \times 10^{-6}/^{\circ}\text{C}$

$E_c = 1 \times 10^5 \text{ N/mm}^2$ $\alpha_c = 17.2 \times 10^{-6}/^{\circ}\text{C}$

Solution:

Let a = Area of steel bar
 $2a$ = Area of copper bar

Free expansion of steel bar = $L \alpha_s T$

Free expansion of copper bar = $L \alpha_c T$

Actual expansion of each bar = δL

$\therefore$ $L \alpha_s T < \delta L < L \alpha_c T$

$\therefore$ Steel bar is in tension and copper bar is in compression.

For equilibrium of the system

Tension in steel bar = Compression in copper bar

$\sigma_s a = \sigma_c (2a)$

$\therefore$ $\sigma_s = 2 \sigma_c$

Actual expansion of steel = Actual expansion of copper

$$L \alpha_s T + \frac{\sigma_s}{E_s} \times L = L \alpha_c T - \frac{\sigma_c}{E_c} \times L$$

$$L \alpha_s T + \frac{\sigma_s}{E_s} = L \alpha_c T - \frac{\sigma_c}{E_c}$$

Hear $T = 315 - 15 = 300^{\circ}\text{C}$

$$12 \times 10^{-6} \times 300 + \frac{\sigma_s}{2.1 \times 10^5} = 17.2 \times 10^{-6} \times 300 - \frac{\sigma_c}{1 \times 10^5}$$

$\therefore$ $\sigma_c = 84.511 \text{ N/mm}^2$

$\sigma_s = 169.022 \text{ N/mm}^2$

Actual expansion of steel bar = $L \alpha_s T + \frac{\sigma_s}{E_s} \times L = 1.5$

$$L \left(12 \times 10^{-6} \times 300 + \frac{169.022}{2.1 \times 10^5} \right) = 1.5$$

$L \times 4.405 = 1.5$

$\therefore$ $L = 340.5 \text{ mm}$

Problem 1.20 The composite bar consisting of steel and aluminium compound shown in fig is connected to two grips at the ends at a temperature of 60°C . Find the stress in the two rods when the temperature falls to 20°C (i) if the ends do not yield, (ii) if the ends

yield by 0.25 mm. Take $E_s = 2 \times 10^5 \text{ N/mm}^2$, $E_a = 0.07 \times 10^5 \text{ N/mm}^2$, $\alpha_s = 1.17 \times 10^{-5} \text{ per } ^\circ\text{C}$ and $\alpha_a = 2.34 \times 10^{-5} \text{ per } ^\circ\text{C}$. Area of the steel and aluminium bars are 250 mm^2 and 400 mm^2 respectively.

Solution: $\frac{A_a}{A_s} = \frac{400}{250} = 1.6$

Free contraction of the composite bar

$$= \alpha_s T l_s + \alpha_a T l_a$$

$$= 1.17 \times 10^{-5} (60 - 20) 800 + 2.34 \times 10^{-5}$$

$$(60 - 20) 400 = 0.7488 \text{ mm}$$

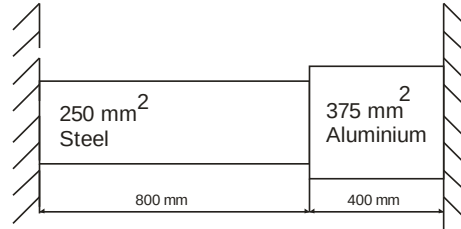


Fig. 1.28

When the contraction is prevented completely or partially tensile stress are induced in the rods. Let σ_s and σ_a be the stress in the steel and aluminium rods.

Since the same force exists in the two rods. $A_s \sigma_s = A_a \sigma_a \quad \therefore \sigma_s = \frac{A_a}{A_s} \sigma_a = \frac{400}{250} \sigma_a$

$$\therefore \sigma_s = 1.6 \sigma_a$$

Case (i) When the ends do not yield

Contraction prevented = 0.7488 mm

Contraction prevented in steel = $\frac{P}{E_s} l_s$ Contraction prevented in aluminium = 0.7488 mm.

$$\frac{P}{E_s} l_s + \frac{P}{E_a} l_a = 0.7488$$

$$\frac{1.6 \sigma_a}{2 \times 10^5} \times 800 + \frac{\sigma_a}{0.70 \times 10^5} \times 400 = 0.7488 \quad \therefore 12.1143 \times 10^{-3} \sigma_a = 0.7488$$

$$\sigma_a = 61.8 \text{ N/mm}^2$$

$$\sigma_s = 1.6 \times 61.8 = 98.88 \text{ N/mm}^2$$

Case (ii) When the ends yield by 0.25 mm.

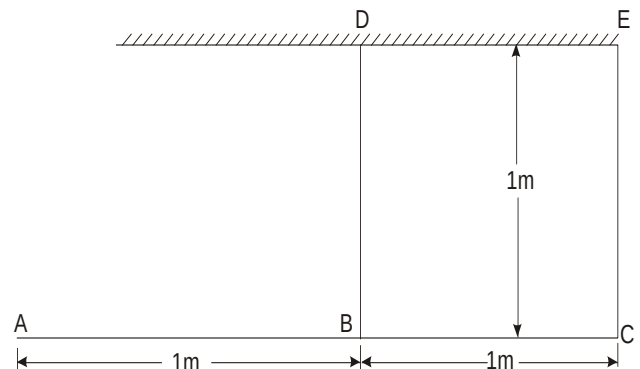
Contraction prevented = 0.7488 – 0.25 = 0.4988 mm

$$12.1143 \times 10^{-3} \sigma_a = 0.4988 \text{ mm} \quad \therefore \sigma_a = 41.17 \text{ N/mm}^2$$

$$\therefore \sigma_s = 1.6 \times 41.17 = 65.88 \text{ N/mm}^2.$$

Problem 1.21 Fig. 1.29 shows a rigid bar ABC hinged at A and suspended at two point B

and C by two bars BD and CE made of aluminium and steel respectively. The bar carries a load of 20 kN midway between B and C. The cross-sectional area of the aluminium bar BD is 4 sq. mm and that of the steel bar CE



is 3 sq. mm. Determine the load taken by the two bars BD and CE.

Modulus of Elasticity for aluminium $E_a = 7 \times 10^4 \text{ N/mm}^2$

Modulus of Elasticity for steel $E_s = 2 \times 10^5 \text{ N/mm}^2$

Solution: A rigid rod will not bend. It will remain straight. But the aluminium and steel rods will extend due to their elasticity. Fig shows the position of the rigid bar after the aluminium and steel rods have undergone their extensions.

Let the tensions in the aluminium and steel rods be P Newton and Q Newton respectively.

Since the rigid bar remains straight, the extensions of CE and BD are proportional to their distances from A.

Let the extension of BD be $BB_1 = \Delta$

$$\therefore \text{Extension of CE} = CC_1 = \frac{2}{1} \times \Delta = 2\Delta$$

$$\text{But extension of BD} = \Delta = \frac{P}{A_a E_a}$$

$$\text{And extension of CE} = 2\Delta = \frac{Q}{A_s E_s}$$

$$\therefore \Delta = \frac{Pl}{A_a E_a} = \frac{Ql}{2A_s E_s}$$

$$\therefore \frac{P}{Q} = \frac{1}{2} \cdot \frac{A_a}{A_s} \cdot \frac{E_a}{E_s}$$

$$\therefore \frac{P}{Q} = \frac{1}{2} \cdot \frac{4}{3} \cdot \frac{70000}{200000}$$

$$\therefore \frac{P}{Q} = \frac{28}{120}$$

$$\therefore P = \frac{28}{120} Q$$

.....(1.37)

Now consider the equilibrium of the rigid bar.

Taking moments of the forces of the rigid bar, about the end A, we have.

$$P \times 1 + Q \times 2 = 20,000 \times 1.5$$

$$\therefore P + 2Q = 30000$$

$$\text{But } P = \frac{28}{120} Q$$

.....(1.39)

$$\therefore \frac{28}{120} Q + 2Q = 30000$$

$$\therefore \frac{268}{120} Q = 30,000$$

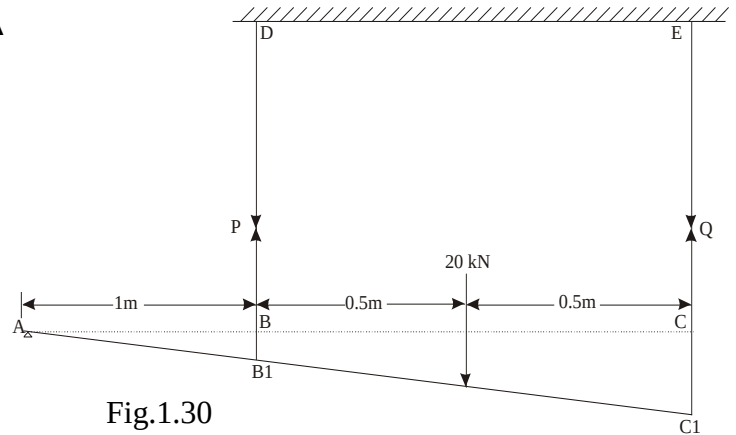
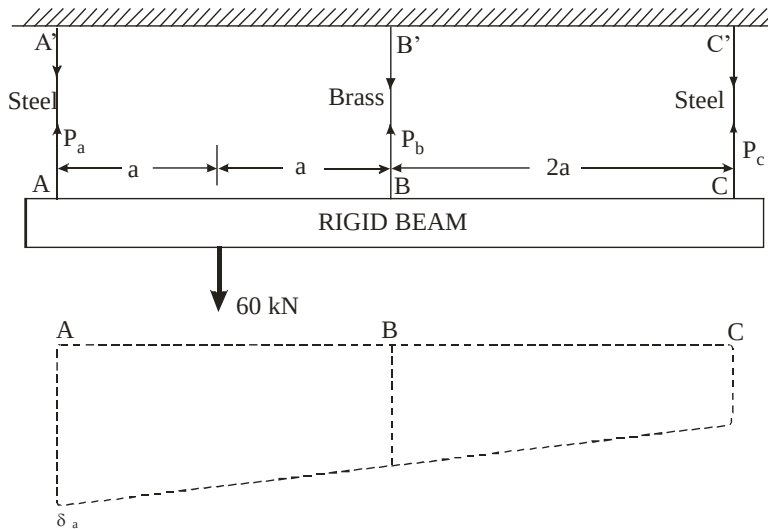


Fig.1.30

$$\therefore Q = \frac{30000 \times 120}{268} = 13432.8 \text{ N (tensile)}$$

$$\therefore P = \frac{28}{120} \times 13260 = 3134 \text{ N (tensile)}$$

Problem 1.22 A rigid beam carrying a load 60 kN is supported by three bars as shown in fig. All the bars are of the same area A and same length l . Find the load carried by each bar. Neglect the weight of the beam. Take $E_s = 2E_b$.



Solution: Let P_a , P_b and P_c be the tensions in the bars AA' , BB' and CC' respectively. Consider the equilibrium of the beam.

Resolving moments about C,

$$P_a \times 4a + P_b \times 2a = 60 \times 3a$$

$$\therefore 2P_a + P_b = 90$$

Let extensions of AA' , BB' and CC' be Δ_a , Δ_b and Δ_c respectively. Obviously Δ_a , Δ_b and Δ_c are the deflections of the beam at A, B and C.

$$\Delta_a = \frac{P_a l}{AE_s}, \Delta_b = \frac{P_b l}{AE_b} \text{ and } \Delta_c = \frac{P_c l}{AE_s}$$

Since the beam is rigid, it remains straight

$$\therefore \frac{\Delta_b}{\Delta_c} = \frac{2}{1} \left[\frac{P_a l}{AE_s} + \frac{P_c l}{AE_s} \right] = \frac{l}{2AE_s} [P_a + P_c] = \frac{l}{4AE_b} [P_a + P_c]$$

$$\therefore P_b = \frac{P_a + P_c}{4}$$

$$\therefore P_a + P_c = 4P_b \quad \dots(1.40)$$

Solving equations (i), (ii) and (iii), we get,

$$P_a = 39 \text{ kN}, P_b = 12 \text{ kN} \text{ and } P_c = 9 \text{ kN}$$

Exercise

What is the procedure of finding thermal stresses in the composite bars?

1.1 Find an expression for the total elongation of a uniformly tapering rectangular bar when it is subjected to an axial load P .

1.2 Find the Young's modulus of a rod of diameter 30 mm and of length 300 mm which is subjected to a tensile load of 60 kN and the extension of the rod is 0.4 mm.

(Ans. 63.5 GN/m²)

1.4 An axial pull of 40 kN is acting on a bar consisting of three sections of length 300 mm, 250 mm and 200 mm and of diameters 20 mm, 40 mm and 50 mm respectively. If the Young's modulus is 2×10^5 N/mm² determine (a) stresses in each section (b) total extension of the bar.

(Ans. 127.32, 31.8, 20.37 N/mm², 0.25 mm)

1.5 A rod which tapers uniformly from 50 mm diameter to 30 mm diameter in a length of 500 mm, is subjected to an axial load of 6 kN. If the Young's modulus is 2×10^5 N/mm² find the extension of the rod.

(Ans. 0.0127 mm)

1.6 A rectangular bar of made of steel is 3 m long and 10 mm thick. The bar is subjected to an axial tensile pull of 50 kN. The width of the bar varies from 70 mm at one end to 28 mm at the other. If the Young's modulus is 2×10^5 N/mm² find the extension of the rod.

(Ans. 1.636 mm)

1.7 Find the Young's modulus for a rod which tapers uniformly from 40 mm diameter to 25 mm diameter in a length of 400 mm. The rod is subjected to a load of 6 kN and extension of the rod is 0.04 mm.

(Ans. 76.39 kN/mm²)

1.8 A reinforced concrete column 250 mm x 250 mm in section is reinforced with 8 steel bars. The total area of steel bars is 1608.5 mm². The column carries a load of 270 kN. If the Young's modulus for steel is 18 times that of concrete, find the stresses in concrete and steel. If the stress in concrete shall not exceed 4 N/mm² find the area of steel required so that the column may support a load of 400 kN.

(Ans. $\sigma_c = 3$ N/mm², $\sigma_s = 54$ N/mm², $A_s = 2206$ mm²)

1.9 A vertical bar fixed at upper end and of uniform strength carries a tensile load of 500 kN. The bar is 18 m long and having weight per unit volume as 0.00008 N/mm^2 . If the area of the bar at the lower end is 500 mm^2 find the area of the bar at the upper end.

(Ans. 500.72 mm^2)

1.10 A rod is 3 m long at a temperature of 15°C . Find the expansion of the rod when the temperature is raised to 95°C . If this expansion is prevented find the stress induced in the material of the rod. If the Young's modulus is $1 \times 10^5 \text{ N/mm}^2$ and $\alpha = 0.000012 / ^\circ\text{C}$

(Ans. 2.88 mm, 96 N/mm^2)

3 Shear Force and Bending Moment

3.1 Introduction

A beam is a structural member subjected to a system of external forces at right angles to its axis. The process of deformation of a bar because of the forces transverse to its axis is called bending. The algebraic sum of the vertical forces at any section of a beam to the right or left of the section is known as shear force. It is briefly written as S.F. The algebraic sum of the moments of all the forces acting to the right or left of the section is known as bending moment. It is written as B.M. idered.

3.2 Shear Force And Bending Moment Diagrams

A shear force diagram is one which shows the variation of the shear force along the length of the beam. A bending moment diagram is one which shows the variation of the bending moment along the length of the beam.

3.3 Types Of Loads

A beam is normally horizontal and the loads acting on the beams are generally vertical. The following are the important types of load acting on a beam:

3.3.1. Concentrated Or Point Load. A concentrated load is one which is considered to act at a point. Although in practice it must really be distributed over a small area because such knife edge contacts are neither possible nor desirable. In Fig. 3.1(a) W shows the point load acting on a beam.

3.3.2. Uniformly Distributed Load. A uniformly distributed load is one which is spread over a beam in such a manner that intensity of loading w is uniform along the length (*i.e.* each unit length is loaded to the same rate) as shown in Fig. 3.1(b). The intensity of loading is expressed as w N/m run.

3.3.3. Uniformly Varying Load. A uniformly varying load is one which is spread over a beam in such a manner that intensity loading varies from point to point along the beam as shown

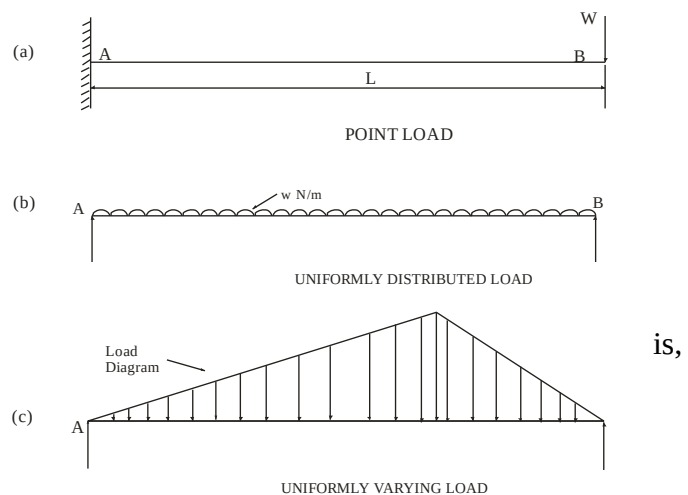


Fig. 3.1

in Fig. 3.1(c) in which load is zero at one end and increases uniformly to one point and then decreases to zero to the other end.

3.4 Types Of Beams

The following are the important types of beams:

3.4.1. Cantilever Beam. A beam which is fixed at one end and free at other end is known as cantilever beam. Such beam is shown in Fig. 3.2(a).

3.4.2. Simply Supported Beam. A beam supported or resting on the supports at its both ends is known as simply supported beam. beam is shown in Fig. 3.2(b).

3.4.3 Overhanging Beam. If end portion of a beam is ex-tended beyond the support, such beam is known as overhanging beam. Overhanging beam is shown in Fig. 3.2(c).

3.4.4. Fixed Beams. A beam, whose both ends are fixed or built in walls, is known as fixed beam. Such beam is shown in Fig. 3.2(d). A fixed beam is also known as a *built-in*, or encastred beam.

3.4.5. Continuous Beam. A beam which is provided more than two supports or more than one span as shown in Fig. 3.2(e) is known as continuous beam.

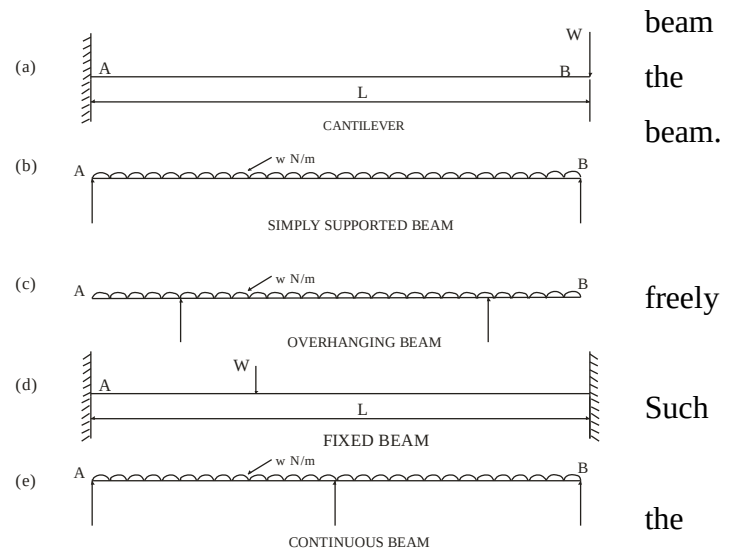


Fig. 3.2

3.5 Shear Force and Bending Moment

Fig. 3.3(a) shows a simply supported of span L. A single point load Pacts at a distance of L_1 from left support. The two reactions at left and right supports may be calculated from equations of static equilibrium. These equations state that the algebraic sum of all forces acting along two mutually perpendicular directions is zero. In case of beams one direction is taken along the beam axis and second direction is transverse to the beam axis. The equations of equilibrium further state that algebraic sum of moments of all forces acting on a body taken about any point in the plane of forces will vanish.

Taking moment about right support:

$$R_1 L = W (L - L_1) \text{ or } R_1 = \frac{W(L - L_1)}{L}$$

Summing up the forces transverse to the beam axis

$$R_2 - W + R_1 = 0 \text{ or } R_1 = \frac{WL_1}{L}$$

Where R_1 & R_2 are the support reactions.

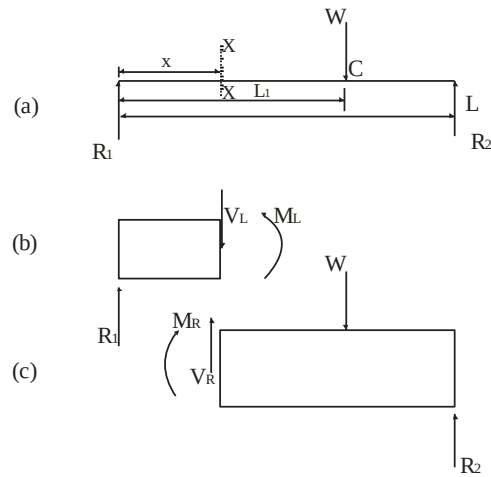


Fig. 3.3 of

Fig. 3.3(b) shows the free body diagram left hand portion of the beam cut by a transverse plane XX at a distance of x from left support. It can be seen that to balance R_1 on the right side of the cut portion a downward internal force V_L equal and opposite to R_1 ($V_L = -R_1$) will act on the section at a distance x from the left support. The force V_L is called shear force. R_1 and V_L will form a clockwise couple $R_1 x$ and hence an internally developed moment i.e. moment of resistance ($M_L = -R_1 x$) will have to balance the moment $R_1 x$ and will act anticlockwise. (We are taking clockwise moment positive.)

Thus it now becomes clear that in a loaded beam at any section the static equilibrium is achieved by two internal reactions viz a shear force V_L and a moment of resistance M_L . By drawing a free body diagram of right hand portion of the beam as in Fig. 3.3(c) it can be shown that to balance $(-W + R_2)$ on the left side of the cut portion a upward internal force V_R equal and opposite to $(-W + R_2)$ will act on the section.

$$V_R = -(-W + R_2) = +R_1$$

The force V_R is called shear force. $(-W + R_2)$ and V_R will form a anticlockwise couple:

$$M = W(L_1 - x) - \frac{WL_1}{L}(L - x) = -\frac{WL_2}{L}x = -R_1 x$$

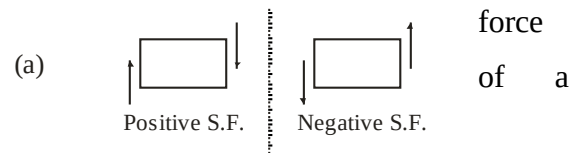
Hence an internally developed moment i.e. moment of resistance ($M_R = R_1 x$) will have to balance the above moment $R_1 x$ and will act clockwise. (We are taking clockwise moment positive). We can observe that V_L and V_R are same in magnitude but opposite in signs and acting

on opposite sides of the sectioning plane. Similarly M_L and M_R are same in magnitude but opposite in directions.

‘The **shear force** at a section of a loaded beam is the algebraic sum of all the forces acting either left or right side of the section. Similarly **bending moment** at a section is the algebraic sum of moments of all the forces either to the right or to the left of the section taken about the section.’

3.6 Sign Conventions:

(a) **Shear Force:** shear force having a downward direction to the right hand side of a section will be taken as positive. Similarly shear force having an upward direction to the right hand side of a section will be taken as negative. (Fig. 3.4(a))



(b) **Bending Moment:** A bending moment causing convexity upwards will be taken as negative and will be called a hogging. Similarly a bending moment causing concavity upwards will be taken as positive and will be called a sagging. (Fig. 3.4(b))

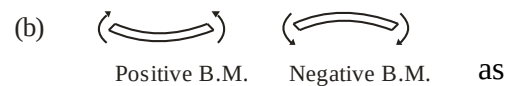


Fig. 3.4

3.7 Important Points For Drawing Shear Force And Bending Moment Diagrams

The shear force diagram is one which shows the variation of the shear force along the length of the beam. A bending moment diagram is one which shows the variation of the bending moment along the length of beam. In these diagrams, the shear force or bending moment are represented by ordinates whereas the length of the beam represents abscissa.

The following are the important points for drawing shear force and bending moment diagrams:

1. Consider the left or the right portion of the section.
2. Add the forces (including reaction) normal to the beam on one of the portion. If right portion of the section is chosen, a force on the right portion acting downwards is positive while a force on the right portion acting upwards is negative.

If the left portion of the section is chosen, a force on the left portion acting upwards is positive while a force on the left portion acting downwards is negative.

3. The positive values of shear force and bending moments are plotted above the base line, and negative values below the base line.
4. The shear force diagram will increase or decrease suddenly i.e. by a vertical straight line at a section where there is a vertical point load.
5. The shear force between any two vertical loads will be constant and hence the shear force diagram between two vertical loads will be horizontal.
6. The bending moment at the two supports of a simply supported beam and at the free end of a cantilever will be zero.

3.8 Shear Force And Bending Moment Diagrams For A Cantilever

3.8.1 A Cantilever With A Point Load At The Free End

Fig. 3.5 (a) shows a cantilever AB of length L fixed at A and free at B and carrying a point load W at the free end B. Let us take a section X at a distance x from the free end.

Let F_x = Shear force at X, and

M_x = Bending moment at X.

S.F.D: Considering the right portion of the section the shear force at this section is equal to the resultant force acting on the right portion at the given section. The resultant force acting on the right portion at the section X is W which is acting in the downward direction. Since a force on the right portion acting downwards is considered positive, hence shear force at X is positive.

$$\therefore F_x = +W$$

The shear force will be constant at all sections of the cantilever between A and B as there is no other load between A and B. The shear force diagram is shown in Fig. 3.5 (b).

B.M.D.

The bending moment section X is given by

$$M_x = -W \times x \quad \dots(i)$$

(Bending moment will be negative as B.M. is of hogging nature, bending the

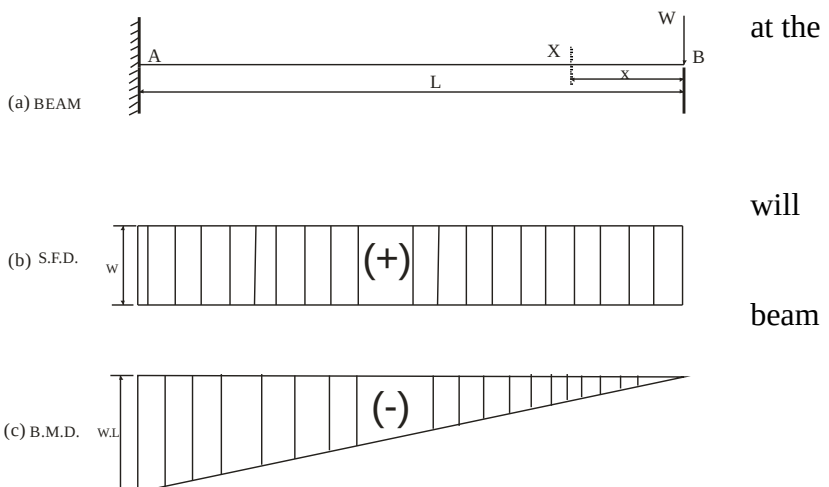


Fig. 3.5

convex upwards.)

From equation (i), it is clear that B.M. at any section is proportional to the distance of the section from the free end.

At $x=0$ i.e. at B, B.M. = 0

At $x=L$ i.e. at A, B.M. = $W \times L$

Hence B.M. follows the straight line law. The B.M. diagram is shown in Fig. 3.5 (c).

3.8.2 A Cantilever With Several Point Loads

S.F.D.: Between B and D, $F_x = +4$ (constant)

Between D and C, $F_x = +4 + 4 = 8$ (constant)

Between C and A, $F_x = +4 + 4 + 2 = 10$ (constant)

Therefore the S.F.D. consists of rectangles of different sizes as in Fig. 3.6 (b).

B.M.D.: Between B and D, $M_x = -4x$

At B, $x = 0 \therefore M_B = 0$.

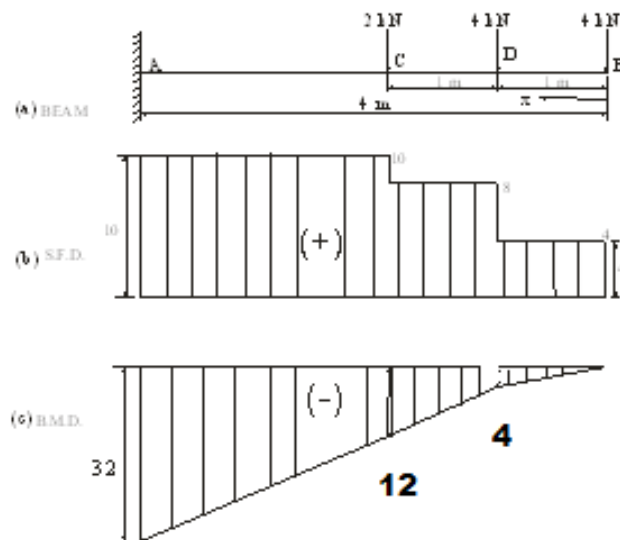


Fig. 3.6

At D, $x = 1 \text{ m} \therefore M_D = -4 \times 1 = -4 \text{ kN-m}$.

Between D and C, $M_x = -4x - 4(x - 1)$

At D, $x = 1 \text{ m} \therefore M_D = -4 \text{ kN-m}$.

At C, $x = 2 \text{ m} \therefore M_C = -12 \text{ kN-m}$.

Between C and A, $M_x = -4x - 4(x - 1) - 2(x - 2)$

At C, $x = 2 \text{ m} \therefore M_C = -12 \text{ kN-m}$.

At A, $x = 4 \text{ m} \therefore M_A = -32 \text{ kN-m}$.

Therefore the B.M.D. consists of lines of various slopes as in Fig. 3.6 (c).

3.8.3 A Cantilever With A Uniformly Distributed Load

Fig. 3.7 shows a cantilever of length L fixed at A and carrying a uniformly distributed load of w per unit length over the entire length of the cantilever.

S.F.D: The shear force at any section X at a distance x from the free end B will be equal to the resultant force acting on the right portion of the section. This resultant force is acting downwards. Resultant force on the right portion acting downwards is considered positive. Hence shear force at X is positive.

The resultant force on the right portion = $w \times \text{Length of right portion} = +w.x$.

At B, $x = 0 \therefore F_B = 0$.

At A, $x = L \therefore F_A = +wL$.

Therefore the S.F.D. consists of triangle as shown in Fig. 3.7(b).

B.M.D.:

To find the effect of the uniformly distributed load over a it is converted into point load at the C.G. of the uniformly distributed load.

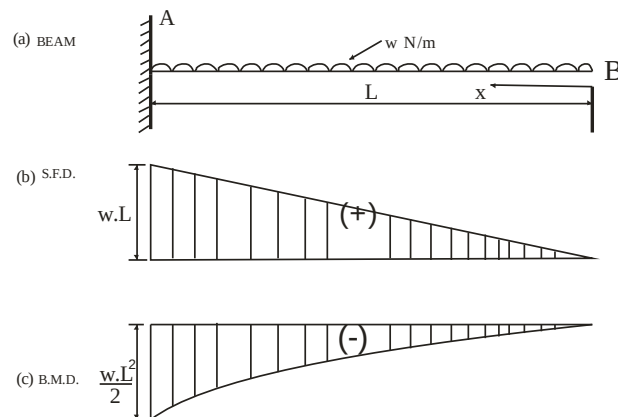


Fig. 3.7

section
acting

The bending moment at the section X is given by

$M_x = -(Total \text{ load on right portion}) \times \text{Distance of section from the C.G. udl}$

$$= - (w \cdot x) \cdot \frac{x}{2}$$

$$= - w \cdot x \cdot \frac{x}{2} = - w \cdot \frac{x^2}{2}$$

.... (i)

(The bending moment will be negative as for the right portion of the section, the moment of the load at x is clockwise. Also the bending of cantilever will take place in such a manner that convexity will be at the top of the cantilever).

From equation (i), it is clear that B.M. at any section follows a parabolic law.

At B, $x = 0$ hence $M_B = 0$

At A, $x = L$ hence $M_A = -w \cdot \frac{L^2}{2}$

The bending moment diagram is drawn as shown in Fig. 3.7(c).

3.8.4 A Cantilever With A U.D.L. On Part Span From The Free End

S.F.D:

For section BC, $F_x = +w \cdot x$

At B, $x = 0 \therefore F_B = 0$.

At A, $x = a \therefore F_C = +w \cdot a$.

For section CA, $F_x = +w \cdot a$

(Constant)

At A, $x = L \therefore F_A = F_C =$

The S.F.D. is triangular from B to C and rectangular from C to A as shown in Fig. 3.8 (b).

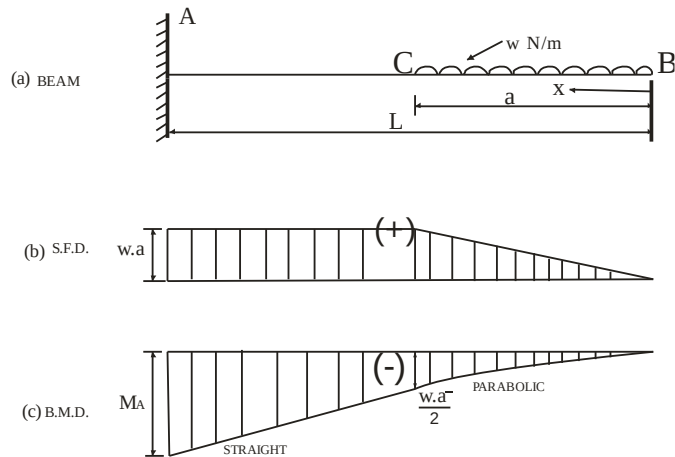


Fig. 3.8

+w.a.
B to C
as

B.M.D.: For section BC, $M_x = -w \cdot \frac{x^2}{2}$ (Parabolic)

At B, $x = 0 \therefore M_B = 0$.

At C, $x = a \therefore M_C = -w \cdot \frac{a^2}{2}$.

For section CA, $M_x = -w \cdot a \cdot \left(x - \frac{a}{2}\right)$ (linear)

At C, $x = a \therefore M_C = -w \cdot \frac{a^2}{2}$.

$$\text{At A, } x = L \therefore M_A = -w.a \left(L - \frac{a}{2} \right).$$

The B.M.D. is parabolic from B to C and linear from C to A as shown in Fig. 3.8 (c).

3.8.5 A Cantilever With Combination of Loads (U.D.L. and Point loads)

S.F.D:

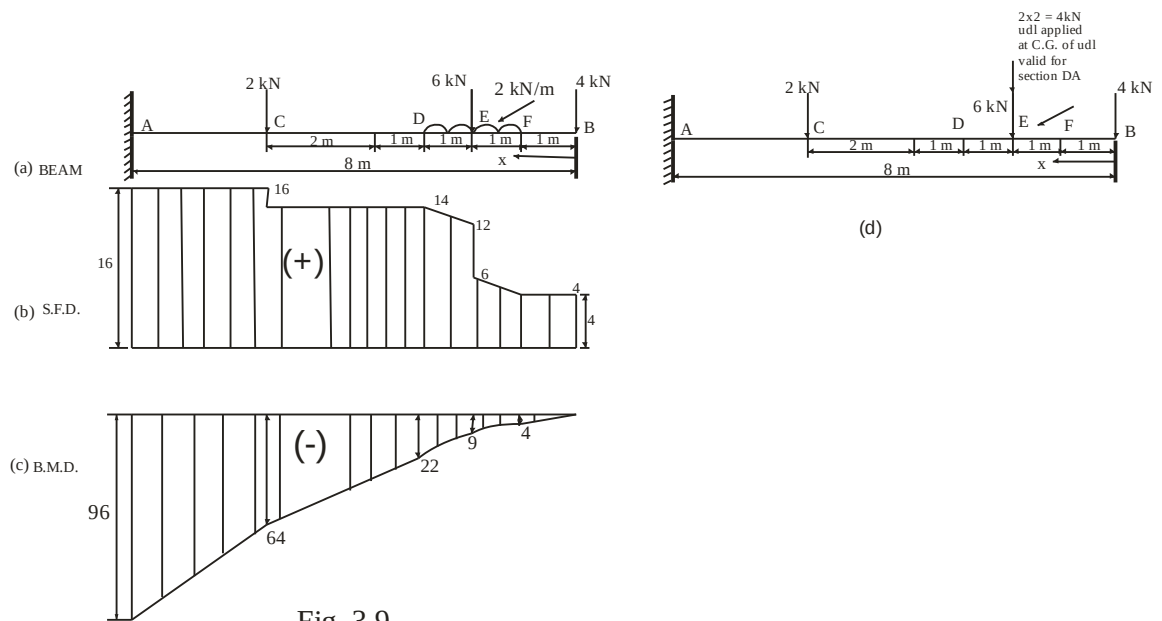


Fig. 3.9

For section BF, $F_x = +4$ kN (Constant)

At B, $x = 0 \therefore F_B = +4$ kN

At F, $x = 1 \therefore F_F = +4$ kN

For section FE, $F_x = +4 + 2(x-1)$

At F, $x = 1 \therefore F_F = +4$ kN

At E, $x = 2 \therefore F_E = +6$ kN

For section ED, $F_x = +4 + 2(x-1) + 6$

At E, $x = 2 \therefore F_E = +1$ kN. (Sudden change in S.F. due to point load)

At D, $x = 3 \therefore F_D = +14$ kN

For section DC, $F_x = +14$ kN (Constant)

At D, $x = 3 \therefore F_D = +14$ kN

At C, $x = 6 \therefore F_C = +14 \text{ kN}$

For section CA, $F_x = +14 + 2 = 16 \text{ kN}$ (Constant)

At C, $x = 6 \therefore F_C = +16 \text{ kN}$ (Sudden change in S.F. due to point load)

At A, $x = 8 \therefore F_A = +16 \text{ kN}$

The S.F.D. is shown in Fig. 3.9 (b).

B.M.D.: For section BF, $M_x = 4.x$

At B, $x = 0 \therefore M_B = 0$.

At F, $x = 1 \therefore M_F = -4 \text{ kN-m}$

For section FE, $M_x = -4.x - \frac{2}{2}(x-1)^2$ (Parabolic)

At F, $x = 1 \therefore M_F = -4 \text{ kN-m}$

At E, $x = 2 \therefore M_E = -9 \text{ kN-m}$

For section ED, $M_x = -4.x - \frac{2}{2}(x-1)^2 - 6(x-2)$ (Parabolic)

At E, $x = 2 \therefore M_E = -9 \text{ kN-m}$

At D, $x = 3 \therefore M_D = -22 \text{ kN-m}$

For section DC, This section is out of the udl. Hence the udl is to be converted into equivalent point load. The total udl (2×2) is assumed to be applied at C.G. of the udl i.e. 2 m away from the free end as shown in Fig. 3.9 (d). To draw BMD for section DA Fig. 3.9 (d) is to be considered.

$M_x = -4.x - 4.(x-2) - 6(x-2)$ (Linear)

At D, $x = 3 \therefore M_D = -22 \text{ kN-m}$

At C, $x = 6 \therefore M_C = -64 \text{ kN-m}$

For section CA

$M_x = -4.x - 4.(x-2) - 6.(x-2) - 2.(x-6)$ (Parabolic)

At C, $x = 6 \therefore M_C = -64 \text{ kN-m}$

At A, $x = 8 \therefore M_A = -96 \text{ kN-m}$

The B.M.D. is shown in Fig. 3.8 (c).

3.9 Shear Force And Bending Moment Diagrams For A Simply Supported Beam

3.9.1 A Simply Supported Beam With A Point Load At Mid-Point

Fig. 3.10 (a) shows a beam AB of length L simply supported at the ends A and B and carrying a point load W at its middle point C. Since the beam is loaded symmetrically the reactions at the support will be equal to $\frac{W}{2}$.

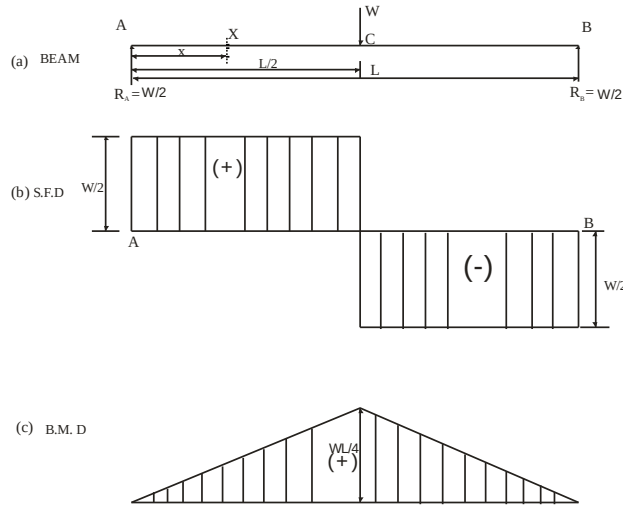


Fig. 3.10

Hence $R_A = R_B = \frac{W}{2}$.

S.F.D.: For the section AC let us take a section X at a distance x from the end A between A and C. Here we have considered the left portion of the section. The shear force at X will be equal to the resultant force acting on the left portion of the section. But the resultant force on the left portion is $\frac{W}{2}$ acting upwards. According to the sign convention, the resultant force on the left portion acting upwards is considered positive. Hence shear force at X is positive and its magnitude is $\frac{W}{2}$.

$$\therefore F_x = + \frac{W}{2} \text{ (Constant)}$$

For the section CB let us take a section X at a distance x from the end A. The resultant

force on the left portion will be

$$\left(\frac{W}{2} - W\right) = -\frac{W}{2}$$

This force will also remain constant between C and B. Hence shear force between C and B is equal to $-\frac{W}{2}$.

At the section C the shear force changes from $+\frac{W}{2}$ to $-\frac{W}{2}$.

The shear force diagram is shown in Fig. 3.10 (b).

B.M. D:

Section AC, $M_x = R_A \cdot x$

$$\text{or } M_x = +\frac{W}{2} \cdot x \text{ (Linear)} \quad \dots(i)$$

B.M. will be positive as for the left portion of the section, the moment of all forces at X is clockwise. Moreover, the bending of beam takes place in such a manner that concavity is at the top of the beam.

$$\text{At A, } x = 0 \quad \text{Hence } M_A = \frac{W}{2} \times 0 = 0$$

$$\text{At C, } x = \frac{L}{2} \quad \text{hence } M_C = \frac{W}{2} \times \frac{L}{2} = \frac{W \times L}{4}$$

$$\text{Section CB, } M_x = R_A \cdot x - W \times \left(x - \frac{L}{2}\right)$$

$$= \frac{W}{2} \cdot x - Wx + W \times \frac{L}{2} = \frac{WL}{2} - \frac{Wx}{2} \text{ (Linear)}$$

$$\text{At C, } x = \frac{L}{2} \quad \therefore M_C = \frac{WL}{2} - \frac{W}{2} \times \frac{L}{2} = \frac{W \times L}{4}$$

$$\text{At B, } x = L \quad \therefore M_B = \frac{WL}{2} - \frac{W}{2} \times L = 0$$

The B.M. diagram can be completed as shown in Fig. 3.10 (c).

It is to be worth noted the bending moment is maximum at the middle point C, where the shear force changes its sign.

3.9.2 A Simply Supported Beam With An Eccentric Point Load

Let us first calculate the reactions, by taking moments about A or about B.

Taking moments of the forces on the beam about A, we get

$$R_B \times L = W \times a$$

$$\therefore R_B = \frac{W.a}{L}$$

and

$$R_A = W - R_B = W - \frac{W.a}{L}$$

$$= W \left(1 - \frac{a}{L} \right) = W \left(\frac{L-a}{L} \right) = \frac{W \times b}{L} \quad (\because L = a + b)$$

S.F.D.: For the section AC let us consider a section X at distance x from the end A. The shear force F_x at the section is given by,

$$F_x = +R_A = + \frac{W.b}{L} \quad (\text{Constant})$$

The shear force will be positive as the resultant force on the left portion of the section is acting upwards.

For the section CB let us consider a section X at a distance x from the end A. The resultant force on the left portion will be $R_A - W$.

or

$$\frac{W.a}{L} - W = W \left(\frac{b-L}{L} \right) = -W \left(\frac{L-b}{L} \right) = - \frac{W.a}{L} \quad (\text{Constant})$$

At the section C, the shear force changes from $\frac{W.b}{L}$ to $-\frac{W.a}{L}$. The shear force diagram is shown in Fig 3.11 (b).

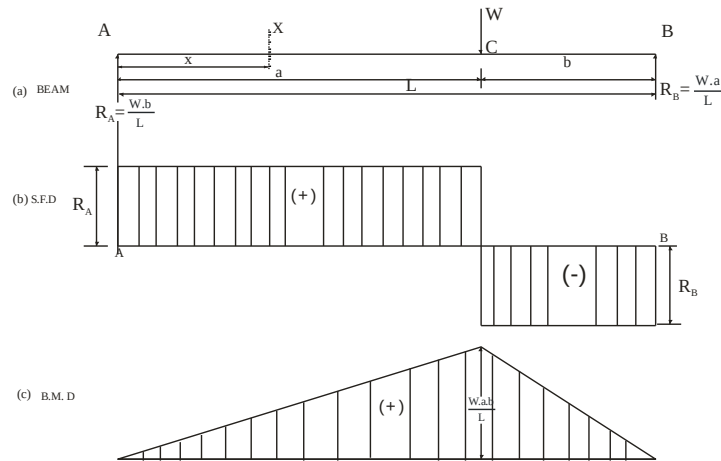


Fig. 3.11

B.M.D.

Section AC, $M_x = R_A \cdot x$

$$= + \frac{W.b}{L} \cdot x \text{ (Linear) (Plus sign due to sagging)}$$

$$\text{At A, } x = 0 \quad \text{Hence } M_A = \frac{W.b}{L} \times 0 = 0$$

$$\text{At C, } x = a \quad \text{Hence } M_C = \frac{W.b}{L} \cdot a = \frac{W \cdot a \cdot b}{L}$$

The B.M. is zero at B. Hence B.M. will decrease from $\frac{W \cdot a \cdot b}{L}$ at C to zero at B following a straight line law. The B.M. diagram is drawn in Fig. 3.11 (c).

It is to be worth noted from the shear force and bending moment diagrams, the B.M. is maximum at C where the S.F. changes its sign.

3.9.3 A Simply Supported Beam Carrying A Uniformly Distributed Load

Fig. 3.12 shows a beam AB of length L simply supported at the ends A and B and carrying a uniformly distributed load of w per unit length over the entire length. The reactions at the supports will be equal and their magnitude will be half the total load on the entire length.

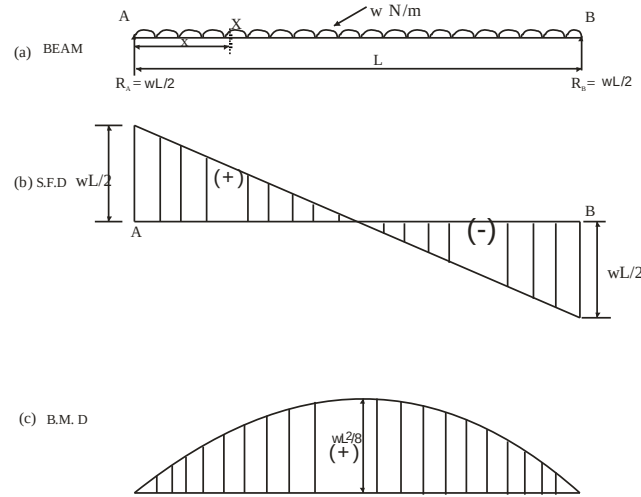


Fig. 3.12

$$\therefore R_A = R_B = \frac{wL}{2}$$

S.F.D.: For the section AC let us consider a section X at distance x from the end A. The shear force F_x at the section is given by,

$$F_x = + R_A - w \cdot x = + \frac{wL}{2} - w \cdot x \text{ (Linear)} \quad \dots(i)$$

The values of shear force at different points are :

$$\text{At A, } x = 0 \quad \therefore F_A = + \frac{wL}{2} - \frac{w \cdot 0}{2} = + \frac{wL}{2}$$

$$\text{At B, } x = L \quad \therefore F_B = + \frac{wL}{2} - w \cdot L = - \frac{wL}{2}$$

$$\text{At C, } x = \frac{L}{2} \quad \therefore F_C = + \frac{wL}{2} - w \cdot \frac{L}{2} = 0$$

The shear force diagram is drawn as shown in Fig. 3.12 (b).

B.M.D.

Section AC, $M_x = + R_A \cdot x - w \cdot x \cdot \frac{x}{2}$

$$= \frac{w.L}{2} \cdot x - \frac{w.x^2}{2} \quad \left(\because R_A = \frac{w.L}{2} \right) \quad \text{.....(ii)}$$

(Parabolic)

The values of B.M. at different points are :

$$\text{At A, } x = 0 \quad \therefore M_A = \frac{w.L}{2} \cdot 0 - \frac{w.0}{2} = 0$$

$$\text{At B, } x = L \quad \therefore M_B = \frac{w.L}{2} \cdot L - \frac{w}{2} \cdot L^2 = 0$$

$$\begin{aligned} \text{At C, } x = \frac{L}{2} \quad \therefore M_C &= \frac{w.L}{2} \cdot \frac{L}{2} - \frac{w}{2} \cdot \left(\frac{L}{2} \right)^2 \\ &= \frac{w.L^2}{4} - \frac{w.L^2}{8} = + \frac{w.L^2}{8} \end{aligned}$$

Thus the B.M. increases according to parabolic law from zero at A to $+\frac{w.L^2}{8}$ at the middle point of the beam and from this value the B.M. decreases to zero at B according to the parabolic law.

The B.M. diagram is drawn as shown in Fig. 3.12(c).

3.9.4 A Simply Supported Beam With Combination of Loads (U.D.L. And Point loads)

Let us first calculate the reactions, by taking moments about A or about B.

Let us take moments of the forces on the beam about A.

$$R_B \times 8 = 6 \times 6 + 2 \times 2 \times 2 \times 6$$

($2 \times 2 \times 6$ is the moment of udl about support A. The udl is converted into point load (Fig. 3.12 (d)) and applied at C.G. of the udl.)

$$\therefore R_B = 8 \text{ kN}$$

Similarly taking moments of the forces on the beam about B.

$$R_A \times 8 = 2 \times 6 + 6 \times 2 - 2 \times 2 \times 2$$

($2 \times 2 \times 2$ is the moment of udl about support B. The udl is converted into point load (Fig. 3.13 (d)) and applied at C.G. of the udl.)

$$\therefore R_A = 4 \text{ kN}$$

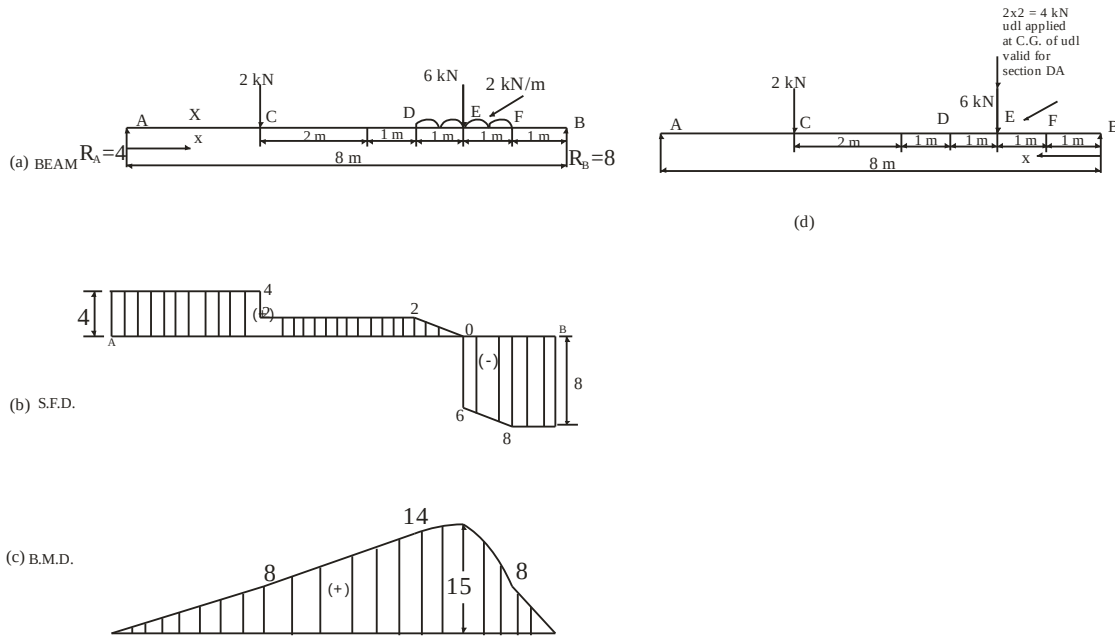


Fig. 3.13

S.F.D.: For the section AC let us consider a section X at distance x from the end A. The shear force F_x at the section is given by,

$$F_x = +R_A = +4 \text{ kN (Constant)}$$

The shear force will be positive as the resultant force on the left portion of the section is acting upwards.

Section CD, $F_x = +4 - 2 = 2 \text{ kN (Constant)}$

Section DE, $F_x = +4 - 2 - 2(x-5) \text{ (Linear)}$

At D, $x = 5 \therefore F_D = 2 \text{ kN}$

At E, $x = 6 \therefore F_E = 0 \text{ kN}$

Section EF, $F_x = +4 - 2 - 2(x-5) - 6 \text{ (Linear)}$

At E, $x = 6 \therefore F_E = -6 \text{ kN}$ (Sudden change due to point load)

At F, $x = 7 \therefore F_F = 8 \text{ kN}$ (constant)

Section FB, $F_x = +4 - 2 - 2x - 6$
 $= 8 \text{ kN (Constant)}$

At F, $x = 7 \therefore F_F = -8 \text{ kN}$

At B, $x = 8 \therefore F_B = -8 \text{ kN}$

The shear force diagram is shown in Fig 3.13 (b).

B.M.D.

Section AC, $M_x = R_A \cdot x$ (Linear)

$= +4 \cdot x$ (Linear) (Plus sign due to sagging)

At A, $x = 0 \therefore M_A = 4 \times 0 = 0$

At C, $x = 2 \therefore M_C = 4 \times 2 = 8 \text{ kN-m}$

Section CD, $M_x = +4 \cdot x - 2 \cdot (x-2)$ (Linear)

At C, $x = 2 \therefore M_C = 8 \text{ kN-m}$

At D, $x = 5 \therefore M_D = 14 \text{ kN-m}$

Section DE, $M_x = +4 \cdot x - 2 \cdot (x-2) - \frac{2}{2}(x-5)^2$ (Parabolic)

At D, $x = 5 \therefore M_D = 14 \text{ kN-m}$

At E, $x = 6 \therefore M_E = 15 \text{ kN-m}$

Section EF, $M_x = +4 \cdot x - 2 \cdot (x-2) - \frac{2}{2}(x-5)^2 - 6(x-6)$ (Parabolic)

At E, $x = 6 \therefore M_E = 15 \text{ kN-m}$

At F, $x = 7 \therefore M_F = 8 \text{ kN-m}$

For section FB, This section is out of the udl. Hence the udl is to be converted into equivalent point load. The total udl (2×2) is assumed to be applied at C.G. of the udl i.e. 2 m away from the

support B as shown in Fig. 3.13 (d). To draw BMD for section FB Fig. 3.13 (d) is to be considered.

$$\text{Section FB, } M_x = +4 \cdot x - 2 \cdot (x-2) - 4 \cdot (x-6) - 6 \cdot (x-6) \text{ (Linear)}$$

$$\text{At F, } x = 7 \quad \therefore M_F = 8 \text{ kN-m}$$

$$\text{At B, } x = 8 \quad \therefore M_B = 0$$

The B.M. is zero at B. The B.M. diagram is drawn in Fig. 3.13 (c).

It is to be worth noted from the shear force and bending moment diagrams, the B.M. is maximum at E where the S.F. changes its sign.

3.10 Shear Force And Bending Moment Diagrams For Over-Hanging Beams

If the end portion of a beam is extended beyond the support such beam is known as overhanging beam. In case of overhanging beams, the B.M. is positive between the two supports, whereas the B.M. is negative for the over-hanging portion. Hence at some point, the B.M. is zero after changing its sign from positive to negative or vice-versa. That point is known as the point of contraflexure or point of inflexion.

3.10.1 Beam With Both Sides Overhang: Point Loads

Let us first calculate the reactions, by taking moments about A or about B.

Let us take moments of the forces on the beam about B.

$$8 \times 5 = R_A \times 4 + 10 \times 2$$

$$\therefore R_A = 5 \text{ kN}$$

$$\therefore R_B = 8 + 10 - 5 = 13 \text{ kN}$$

S.F.D.: For the section CA,

$$F_x = - 8 \text{ kN (Constant)}$$

The shear force will be negative as the resultant force on the left portion of the section is acting downwards.

$$\text{Section AB,} \quad F_x = - 8 + 5 = - 3 \text{ kN (Constant)}$$

$$\text{Section BD,} \quad F_x = - 8 + 5 + 13 = +10 \text{ kN}$$

Thus S.F. changes sign at B.

The shear force diagram is shown in Fig 3.14 (b).

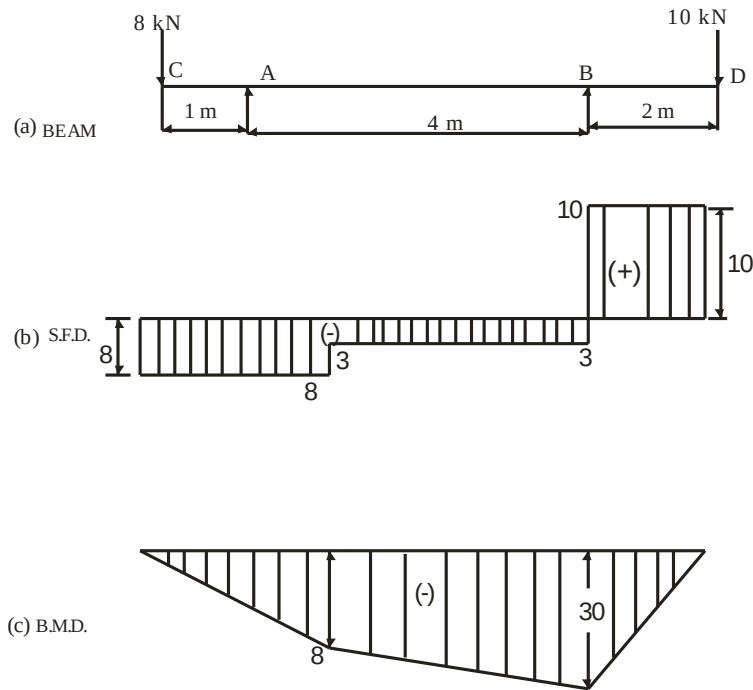


Fig. 3.14

B.M.D.

Section CA, $M_x = -8 \cdot x$ (Linear)

$$\text{At C, } x = 0 \quad \therefore M_C = -8 \times 0 = 0$$

$$\text{At A, } x = 1 \quad \therefore M_C = -8 \times 1 = -8 \text{ kN-m}$$

Section AB, $M_x = -8 \cdot x + 5 \cdot (x-1)$ (Linear)

$$\text{At A, } x = 1 \quad \therefore M_A = -8 \text{ kN-m}$$

$$\text{At B, } x = 5 \quad \therefore M_B = -20 \text{ kN-m}$$

Section BD, $M_x = -8 \cdot x + 5 \cdot (x-1) + 13 \cdot (x-5)$ (Linear)

$$\text{At B, } x = 5 \quad \therefore M_B = -20 \text{ kN-m}$$

$$\text{At D, } x = 7 \quad \therefore M_D = 0$$

The B.M. diagram is drawn in Fig. 3.14 (c). Maximum bending moment is where S.F. changes sign.

3.10.2 Beam With Both Sides Overhang: Uniformly Distributed Load

Fig. 3.15 (a) shows a beam EABD of length $(l+2a)$ with supports at A and B. The beam carries a udl w N/m over entire length.

Since the beam is symmetrically loaded the reactions R_A and R_B are

$$R_A = R_B = \frac{w}{2}(l + 2a)$$

S.F.D.: For the section EA,

$$F_x = -wx \text{ (Linear)}$$

The shear force will be negative as the resultant force on the left portion of the section is acting downwards.

$$\text{S.F. at E, } x = 0 \therefore F_E = 0$$

$$\text{S.F. at A, } x = a \therefore F_A = -w.a$$

For the section AB,

$$F_x = R_A - wx = \frac{w}{2}(l + 2a) - wx \text{ (Linear)}$$

$$\text{S.F. at A, } x = a \therefore F_A = + \frac{wl}{2} \text{ (Sudden change in S.F. due to point load } R_A)$$

$$F_x = R_A - wx = \frac{w}{2}(l + 2a) - wx \text{ (Linear)}$$

$$\text{S.F. at B, } x = (a+l) \therefore F_B = - \frac{wl}{2}$$

Hence from A to B the S.F. changes uniformly from $+\frac{wl}{2}$ to $-\frac{wl}{2}$. The S.F. at C the middle point of the beam is zero.

For the section BD, the S.F. will change uniformly from $+wa$ at B to zero at D.

The shear force diagram is shown in Fig 3.15 (b).

B.M.D.

Section EA, $M_x = -\frac{wx^2}{2}$ (Hogging and parabolic)

$$\text{At E, } x = 0 \quad \therefore M_E = 0$$

$$\text{At A, } x = a \quad \therefore M_A = -\frac{wa^2}{2}$$

Section AB, $M_x = R_a(x-a) - \frac{wx^2}{2} = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2}$

$$\text{At A, } x = a \quad \therefore M_A = -\frac{wa^2}{2}$$

$$\text{At C, } x = a + \frac{l}{2} \quad \therefore M_C = \frac{w(l^2 - 4a^2)}{8}$$

Case (A): when $l > 2a$

For this case M_c is positive.

For this case there will be two points of contraflexure C_1 and C_2 between A and B. The positions of these points can be determined by equating the general expression for B.M. for any section in AB to zero.

$$M_x = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2}$$

For the points of contraflexure

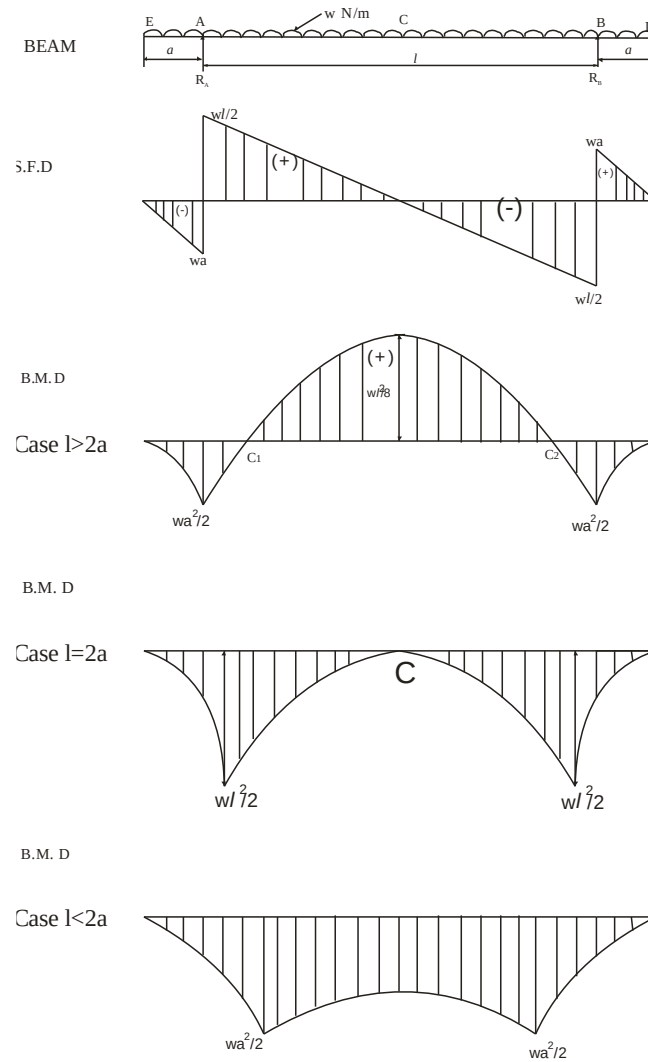


Fig. 3.15

$$M_x = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2} = 0$$

$$\text{Or } \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2} = 0$$

$$\text{Or } x^2 - (2a+l)x + a(2a+l) = 0$$

$$x = \frac{(2a + l) \pm \sqrt{(2a + l)^2 - 4a(2a + l)}}{2}$$

$$\text{or } x = a + \frac{l}{2} \pm \frac{\sqrt{(l)^2 - 4a^2}}{2}$$

$$\text{Here EC} = a + \frac{l}{2}$$

Hence the points of contraflexure C_1 and C_2 are at a distance $\frac{\sqrt{(l)^2 - 4a^2}}{2}$ from the middle point C of the beam. The BMD is as shown in Fig. 3.15 (c).

Case (B): when $l=2a$ B.M. at C = 0

The points of contraflexure will coincide with C and the BMD between A and B just touches the span. The BMD is as shown in Fig. 3.15 (d).

Case (C): when $l < 2a$

$$M_C = \frac{w(l^2 - 4a^2)}{8} \quad \text{since } 4a^2 > l^2 \therefore M_C \text{ is negative.}$$

The BMD is as shown in Fig. 3.15 (e).

Chapter 3

Problem 3.1 A cantilever AB of length l is subjected to a downward load P at its free end and an upward load P at a distance a from the free end. Draw S.F. and B.M. diagrams for the cantilever.

Solution:

S.F.D.

Section in BC = $+W$

Section in CA = 0

B.M. at any section in BC = $-W.x$ (Linear);

At B $x = 0 \therefore M_B = 0$

At C $x = a \therefore M_C = -Wa$

B.M. at any section in CA = $-W.x + W.(x -$

a)

$= -Wa$ (Constant)

At C $x = a \therefore M_C = -Wa$

At A $x = l \therefore M_A = -Wa$

From C to D the

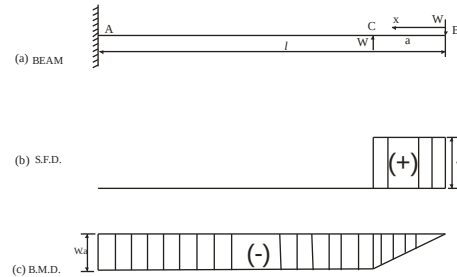


Fig. 3.16

Problem 3.2 A cantilever beam 5 m long carries concentrated loads of 30, 30 and 30 kN at distance of 1.5, 3 and 4.5 meters from the fixed end. In addition to this the beam carries a uniformly distributed load of 10 kN/m over the entire length of the beam. Draw the S.F. and B.M. diagrams.

Solution :

The loaded beam is shown in Fig. 3.17

S.F. at B = 0

S.F. at C = $(10 \times 0.5 + 30) = 35$ kN

S.F. at D = $(10 \times 2 + 30 + 30) = 80$ kN

S.F. at E = $(10 \times 3.5 + 30 + 30 + 30) = 125$ kN

S.F. at A = $(10 \times 5 + 30 + 30 + 30) = 140$ kN

The S.F. diagram is shown in Fig. 3.17(b)

B.M. at B = 0

B.M. at C = $-10 \times 0.5 \times \frac{0.5}{2} = -1.25$ kN.m

B.M. at D = $-10 \times 2 \times \frac{2}{2} - 30 \times 1.5 = -65$

kN.m

B.M. at E = $-10 \times 3.5 \times \frac{3.5}{2} - 30 \times 3 - 30 \times 1.5 = -196.25$ kN.m

B.M. at A = $-10 \times 5 \times \frac{5}{2} - 30 \times 4.5 - 30 \times 3 - 30 \times 1.5$

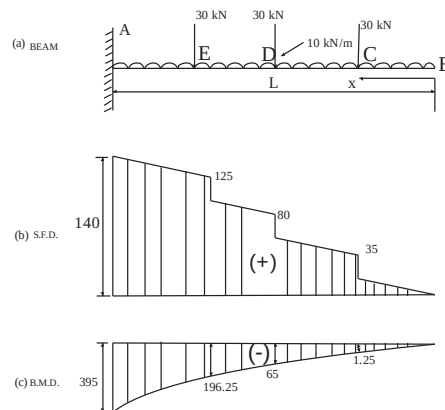


Fig. 3.17

$$= -395 \text{ kN.m}$$

The B.M. diagram is shown in Fig. 3.17(c)

Problem 3.3 Draw the shear force and bending moment diagrams for the beam shown in Fig. 3.18. Also find the position and magnitude of the maximum bending moment.

Solution: Let the salient point of the beam be named as shown in the Fig. 3.18. Let R_a and R_b be the vertical reactions at the left and right supports respectively.

Taking moments about the left support, we have

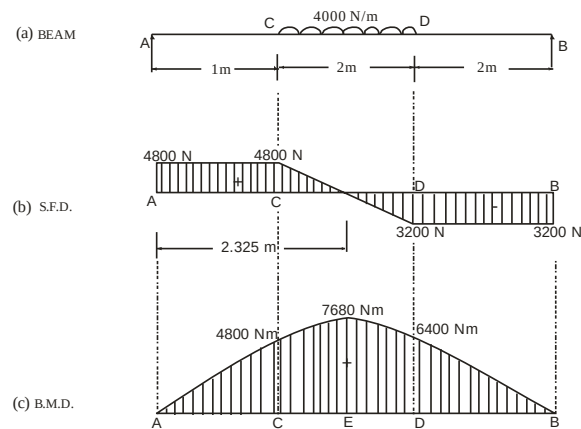


Fig. 3.18

$$R_b \times 5 = 4000 \times 2(1+1)$$

$$\therefore R_b = 3200 \text{ N}$$

$$\therefore R_a = (4000 \times 2) - 3200$$

$$= 4800 \text{ N}$$

$$\text{S.F. between A and C} = +4800 \text{ N}$$

$$\text{S.F. between D and B} = -3200 \text{ N}$$

Let the S.F. be zero at a section E, x meters from A

Equating the S.F. at E to zero, we have

$$4800 - 4000(x - 1) = 0$$

$$\therefore x = 2.2 \text{ m}$$

B.M. calculations

$$\text{B.M. at A} = M_a = 0$$

$$\text{B.M. at C} = M_c = 4800 \times 1 = 4800 \text{ Nm}$$

$$\text{B.M. at D} = M_d = 3200 \times 2 = 6400 \text{ Nm}$$

B.M. at 2.2 m from A

$$= M_E = 4800 \times 2.2 - 4000 \frac{(2.2 - 1)^2}{2} \text{ Nm}$$

$$= 7680 \text{ Nm}$$

Problem 3.4 Draw S.F. and B.M. diagrams for the simply supported beam shown in Fig. 3.19.

Solution: Let R_a and R_b be the vertical reactions at the supports A and B respectively. For the equilibrium of the beam taking moments of the forces on the beam about the left support, we have,

$$R_b \times 8 = 4 \times 1.5 + 10 \times 4 + 3 \times 6 = 64 \text{ kN.m}$$

$$\therefore R_b = \frac{64}{8} = 8 \text{ kN}$$

$\therefore R_a = \text{total load on the beam} - R_b = 17 - 8 = 9 \text{ kN}$
 S.F. between A and C = 9 kN
 S.F. between C and D = $+9 - 4 = +5 \text{ kN}$
 S.F. between D and E = $+9 - 4 - 10 = -5 \text{ kN}$
 or alternatively = $-8 + 3 = -5 \text{ kN}$
 S.F. between E and b = $+9 - 4 - 10 - 3 = -8 \text{ kN}$
 or alternatively = $-R_b = -8 \text{ kN}$

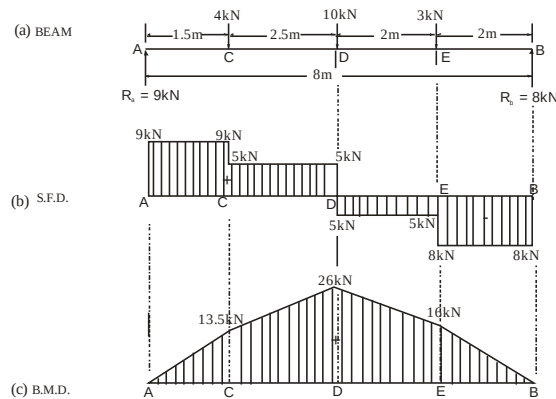


Fig. 3.19

B.M. at A = 0
 B.M. at C = $+9 \times 1.5 = +13.5 \text{ kNm}$ (sagging)
 B.M. at D = $+9 \times 4 - 4 \times 2.5 = +26 \text{ kNm}$ (Sagging)
 B.M. at E = $+8 \times 2 = +16 \text{ kNm}$ (Sagging)

It may be observed from the S.F. and B.M. diagrams that the maximum B.M. occurs at D where the S.F. change its sign. SFD and BMD are shown in Fig. 3.19.

Problem 3.5 A beam AB 10 meters long has supports at its ends A and B. It carries a point load of 10 kN at 3 meters from A and a point load of 5 kN at 7 metres from A and a uniformly distributed load of 1 kN per meter between the point loads. Draw SF and BM diagrams for the beam.

Solution:

Reactions :

Since the loading is symmetrical reaction at each support equals half the total load

$$\therefore R_a = R_b = \frac{10 + 10 + 2 \times 4}{2} = 14 \text{ kN}$$

S.F. analysis

S.F. at any section in AD = + 14 kN
 S.F. just on RHS of D = $+14 - 10 = +4 \text{ kN}$
 S.F. just on LHS of E = $-14 + 10 = -4 \text{ kN}$
 S.F. at any section in EB = - 14 kN
 S.F. at centre C = 0

B.M. analysis

B.M. at A = BM at B = 0

$$\text{B.M. at D} = \text{B.M. at E} = + 14 \times \frac{2}{2} \times 3 = + 42 \text{ kNm}$$

$$\text{B.M. at C} = 14 \times 5 - 10 \times 2 - 2 \times \frac{2}{2} = 70 - 20 - 4 = + 46 \text{ kNm}$$

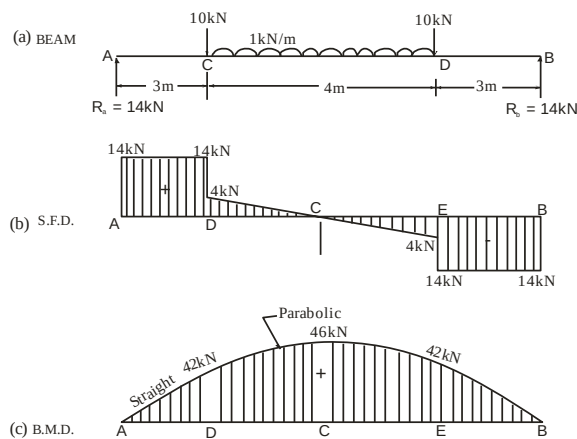


Fig. 3.20

(Maximum bending moment)

The B.M. diagram is linear for the parts AD and BE and is parabolic for the part DE. SFD and BMD are shown in Fig. 3.20.

Problem 3.6 A simply supported beam is loaded as shown in Fig. 3.21. Draw the S.F. and B.M. diagrams and find the value of maximum bending moment.

Solution: Let R_A and R_B be the reactions at the supports A and B respectively.

$$R_A + R_B = 30 \times 2 + 120 + 50 \times 4 = 380 \text{ kN} \quad \dots(i)$$

Taking moment about A

$$R_B \times 8 = 30 \times 2 \times \left(\frac{2}{2} + 6 \right) + 120 \times 6 + 50 \times 4 \times \frac{4}{2} = 1540$$

$$\therefore R_B = \frac{1540}{8} = 192.5 \text{ kN}$$

Putting the value of R_B in Equation (i) we get $R_A = 380 - 192.5 = 187.5 \text{ kN}$

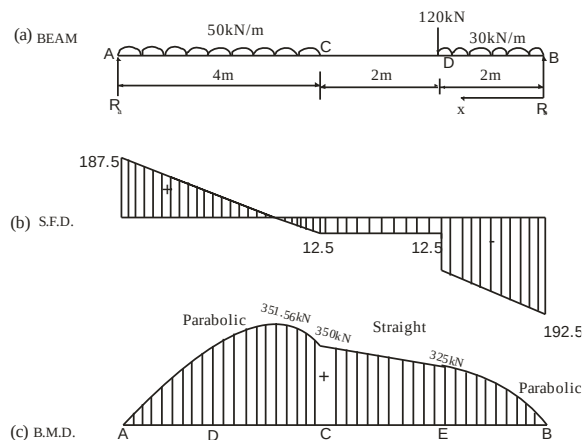


Fig. 3.21

$$\text{S.F. at B} = 192.5 \text{ kN (-) ve}$$

$$\text{S.F. at C} = 192.5 - 30 \times 2 - 120 = 12.5 \text{ kN (-) ve}$$

$$\text{S.F. at D} = 12.5 \text{ kN (-) ve}$$

S.F. at A = 187.5 kN (+) ve

The S.F. diagram is shown in Fig. 3.21 There is a point between A and D where S.F. is zero. Let the point be E and BE = x

Now S.F. at E = $-192.5 + 30 \times 2 + 120 + 50(x - 4)$

$$0 = -12.5 + 50x + 200$$

or $50x = 212.5$

$\therefore x = 4.25 \text{ m}$

B.M. at A = 0

B.M. at B = 0

$$\text{B.M. at C} = 192.5 \times 2 - 30 \times 2 \times \frac{2}{2} = 325 \text{ kN.m}$$

$$\text{B.M. at D} = 192.5 \times 4 - 30 \times 2 \times \left(\frac{2}{2} + 2\right) - 120 \times 2 = 350 \text{ kN.m}$$

$$\begin{aligned} \text{B.M. at E} &= 192.5 \times 4.25 - \frac{30 \times 2}{2} \times (4.25 - 1) - 120 \times 2.25 \\ &\quad - 50 \times 0.25 \times \frac{2}{2} = 351.56 \text{ kN.m} \end{aligned}$$

Therefore the maximum bending moment is at E and is equal to 351.56 kN.m.

The B.M. diagram is shown in Fig. 3.21.

Problem 3.7 A simply supported beam ABC with supports at A and B, 6 meters apart and with an overhang BC 2 meters long carries a uniformly distributed load of 30 kN per meter over the whole length as shown in Fig. 3.22 Draw S.F. and B.M. diagrams.

Solution:

Reactions. Taking moments about the end A, $\sum$

$$R_b \times 6 = 30 \times 8 \times \frac{8}{2}$$

$\therefore R_b = 160 \text{ kN}$

$\therefore R_a = \text{Total load} - R_b = (30 \times 8) - 160 = 80 \text{ kN}$

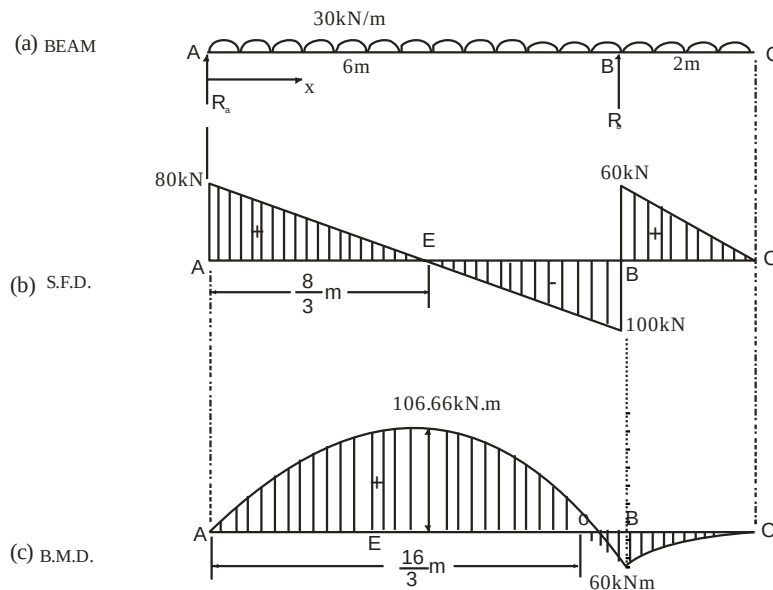


Fig. 3.22

S.F.D.: At any section AB distant x from A,

Shear force $S = 80 - 30x$

At $x = 0$, $S = +80$ kN

At $x = 6$ m $S = 80 - 30 \times 6 = -100$ kN

Section of zero shear: Equating the general expression for shear force to zero.

$$80 - 30x = 0$$

$$\therefore x = \frac{8}{3} \text{ m} = 2.67 \text{ m}$$

At any section in CB, distant x from C,

Shear force $S = +30x$

At $x = 0$ $S = 0$

At $x = 2$ m, $S = 30 \times 2 = +60$ kN

B.M.D.: At any section in AB, distant x from A, $\frac{x^2}{2}$

Bending moment $= M = 80x - 30 \times \frac{x^2}{2}$

$$\therefore M = 80x - 15x^2$$

This is a parabolic law.

At $x = 0$, $M = 0$

At $x = 6$ m, $M = 80 \times 6 - 15 \times 6^2 = -60$ kNm

Maximum bending moment which occurs at the section of zero shear

$$\begin{aligned} &= M_e = 80 \times \frac{8}{3} - 15 \left(\frac{8}{3} \right)^2 \\ &= + \frac{320}{3} \text{ kNm} = +106.66 \text{ kNm} \end{aligned}$$

At any section in CB, distant x and C, $\frac{x^2}{2}$

Bending moment $= M = -30 \frac{x^2}{2}$

$$\therefore M = -15x^2 \text{ . this is a parabolic law.}$$

At $x = 0$, $M = 0$

At $x = 2$ m, $M = -15 \times 2^2 = -60$ kNm

Problem 3.8 Draw shear force and bending Moment diagrams for the beam shown in

Fig. 3.23.

Solution : Taking moments about A, $R_b \times 10 = 20 \times 14(7-3)$

$$\therefore R_b = 112 \text{ KN} \uparrow \quad \therefore R_a = (20 \times 14) - 112 = 168 \text{ KN} \uparrow$$

S.F.D.: S.F. at C = 0,

S.F. on LHS of A = $-20 \times 3 = -60$ kN

S.F. on RHS of A = $-60 + 168 = +108$ kN;

S.F. on LHS of B = $168 - 20 \times 13 = -82$ kN

S.F. on RHS of B = $-82 + 112 = +20$ kN ;

S.F. at D = 0

Section of zero shear. Let the shear force be zero at a section distant x from A.

Equating the shear force to zero,

$$168 - 20(3+x) = 0 \quad \therefore x+3 = 8.4 \quad \therefore x = 5.4 \text{ m}$$

B.M.D. BM at C = 0

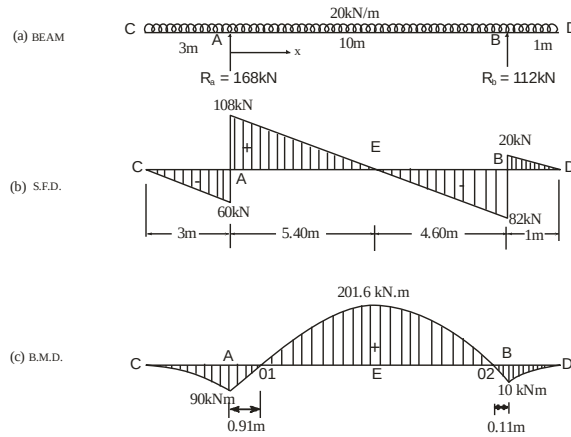


Fig. 3.23

$$\text{B.M. at A} = -\frac{20 \times 3^2}{2} = -90 \text{ kNm};$$

$$\text{B.M. at B} = -\frac{20 \times 1^2}{2} = -10 \text{ kNm}$$

B.M. at the section of zero shear, i.e. at a distance of 5.40 m from A,

$$M_{\max} = 168 \times 5.4 - \frac{20 \times 5.4^2}{2} = 201.6 \text{ kNm}$$

Points of contra flexure between A and B. Let the B.M. be zero at a distance x from A.

$$\text{A. Equating the general expression for B.M. to zero, } 168x - \frac{20(x+3)^2}{2} = 0$$

$\therefore x^2 - 10x + 9 = 0$ solving we get,

$$x = 0.91 \text{ m and } x = 9.89 \text{ m}$$

Distance of point of contra flexure O_1 from A = 0.91 m

Distance of point of contra flexure O_2 from B = $10 - 9.89 = 0.11 \text{ m}$

Problem 3.9 Calculate the reactions at the supports A and B of the beam shown in Fig. 3.24. Draw bending moment and shearing force diagrams.

Solution: Let reactions at A and B be R_a and R_b respectively.

Taking moments about A, We have, $R_b \times 7 + 1000 \times 2$

$$= \frac{2500 \times 40}{2} + 1500 \times 10$$

$$R_b = \frac{23000}{7} = 3285.71$$

$$\therefore R_a = \text{Total load} - R_b = 5000 - 3285.71 = +1714.29$$

S.F. Calculations

S.F. between D and A = - 1000 N

S.F. between A and C = - 1000 + 1714.29 = + 714.29

S.F. between C and B = + 1500 - 3285.71 = - 1785.71

S.F. between B and E = + 1500 N

B.M. Calculations

B.M. at D = $M_d = 0$

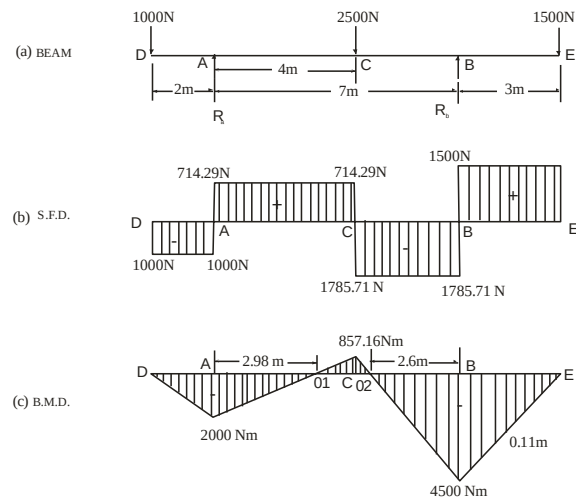


Fig. 3.24

$$\text{B.M. at A} = M_a = -1000 \times 2 = -2000 \text{ Nm}$$

$$\begin{aligned} \text{B.M. at C} = M_c &= +1714.29 \times 4 - 1000 \times 6 \text{ Nm} \\ &= -857.16 \end{aligned}$$

$$\text{B.M. at B} = M_b = -1500 \times 3 = -4500 \text{ Nm}$$

$$\text{B.M. at E} = M_e = 0$$

There will be two points of contra flexure O_1 and O_2 . One of them lies between A and C and the other lies between C and B.

Points of contra flexure between A and C: Let this point be x meters from A. Equating the general expression for B.M. to zero, $1714.29x - 1000(x+2) = 0$

Solving we get, $x = 2.8$ m from A.

Points of contra flexure between C and B: Let this point be x meters from B. Equating the general expression for B.M. to zero, $3285.71x - 1500(x+3) = 0$

Solving we get, $x = 2.52$ m from B.

Problem 3.10 A beam of uniform section 10 meters long carries a uniformly distributed load of 10 kN per meter over the whole length and a concentrated load of 10 kN at the right end. If the beam is freely supported at the left and, find the position of the second support so that maximum bending moment for the beam shall be as small as possible. Find also the maximum bending moment for this case.

Solution: Let the supports be at A and B so that the overhang BC = a meter
 $\therefore AB = (10 - a)$ meter

Taking moments about A, $R_b(10 - a) = \frac{10 \times 10 \times (10 - a)}{2} + 10 \times 10$

$$R_b = \left(\frac{600}{10 - a} \right) \text{ kN}$$

$$\therefore R_a = (10 \times 10) + 10 - \frac{600}{10 - a} = \frac{500 - 110a}{10 - a} \text{ kN}$$

$$\text{B.M. at B} = -10 \times a - 10 \times \frac{a^2}{2} = -5a(a + 2)$$

Maximum positive B.M. will occur at a section in AB.

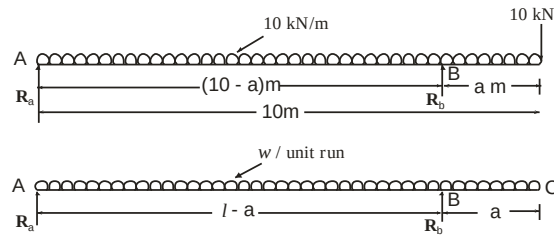


Fig. 3.24

Let this section be x meters from A.

$$\therefore \text{S.F. at this section} = 0$$

$$\therefore \frac{500 - 110a}{10 - a} - 10x = 0$$

$$\therefore x = \frac{500 - 110a}{(10 - a)100} = \frac{50 - 11a}{10 - a} m$$

$$\text{Maximum positive B.M.} = R_a \cdot x - \frac{wx^2}{2} = \frac{500 - 110a}{10 - a} \cdot \frac{50 - 11a}{10 - a} - \frac{10}{2} \left[\frac{50 - 11a}{10 - a} \right]^2$$

$$= 5 \frac{(50 - 11a)^2}{(10 - a)^2}$$

For the condition that the maximum bending moment shall be as possible, the maximum positive bending moment and the hogging moment over the support should be numerically equal.

$$\therefore 5 \frac{(50 - 11a)^2}{(10 - a)^2} = 5a(a + 2)$$

$$\therefore \left[\frac{50 - 11a}{10 - a} \right]^2 = a(a + 2)$$

Solving by trial and error, we get $a = 2.33$ m.

Maximum B.M. with the above value of a

$$= 5a(a + 2) = 5 \times 2.33(2.33 + 2) = 50.44 \text{ kNm}$$

Problem 3.11 A beam AB, 20 meters long supported on two intermediate props 12 meters apart carries a uniformly distributed load 6 kN/m per meter together with concentrated loads of 30 kN at the left end A and 45 kN at the right end B. The supports are so located that the reaction is the same at each support. Determine the position of the props and draw B.M. and S.F. diagrams. Mark the values of the maximum B.M. and S.F.

Solution: Let the left support be at C and the right support be at D

Let AC = a meters

$$\therefore \text{BD} = 20 - 12 - a = (8 - a) \text{ meters}$$

Let R_1 and R_2 be the reactions at the left and right supports respectively.

$$\text{Total load on the beam} = 35 + 45 + (6 \times 20) \text{ kN} = 200 \text{ kN}$$

Since the reactions at the supports are given to be equal, we have

$$R_1 = R_2 = 100 \text{ kN}$$

Taking moments about A, we have ,

$$45 \times 20 + 6 \times 2 \times 10 = 100a + 100(12 + a)$$

$$900 + 1200 = 100a + 1200 + 100a$$

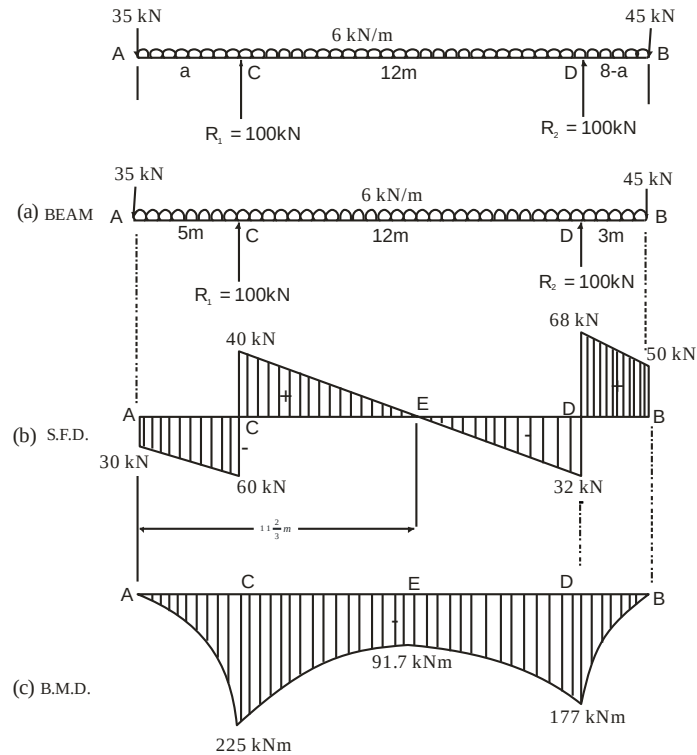


Fig. 3.25

$$200a = 900$$

$\therefore$

$$a = 4.5 \text{ meters}$$

$\therefore$ The left support is at 4.5 meters from A and the right support is at $8 - 4.5 = 3.5$ meters from B.

Shear force calculations

S.F. just on the right hand side of A	=	- 30 kN
S.F. just on the left hand side of C	=	- 30 - 6 $\times$ 4.5 = - 57 kN
S.F. just on the right hand side of C	=	- 57 + 100 = + 43 kN
S.F. just on the right hand side of D	=	+ 45 + 6 $\times$ 3.5 = 66
S.F. just on the left hand side of D	=	+ 66 - 100
	=	- 34 kN

Let the S.F. be zero at a distance of x meters from A (between the two supports)

Equating the S.F. to zero, we have,

$$100 - 35 - 6x = 0$$

$$\therefore 6x = 65$$

$$\therefore x = \frac{65}{6} = 10 \frac{5}{6} \text{ m}$$

Bending moment calculations

$$\text{B.M at A} = 0$$

$$\text{B.M. at C} = M_d = -35 \times 4.5 - 6 \times \frac{4.5^2}{2} \text{ kNm} = -218.25 \text{ kNm}$$

$$\text{B.M. at D} = M_d = -45 \times 3.5 - 6 \times \frac{3.5^2}{2} \text{ kNm} = -194.25 \text{ kNm}$$

$$\text{B.M. at distance of } 10\frac{5}{6} \text{ meter from A}$$

$$= M_e = 100 \times \frac{5}{6} - 30 \times \frac{10\frac{5}{6}}{6} - \frac{6\left(10\frac{5}{6}\right)^2}{2} \text{ kNm} = -93.75 \text{ kNm}$$

Problem 3.12 Calculate the reactions for the beam shown in Fig. 3.26 Construct the bending moment and shear force diagrams. Determine the location of the maximum bending moment and mark it clearly on each of the diagrams.

Solution: Let R_a and R_c be the reactions at the supports A and C respectively.
Taking moments about A, we have

$$R_c \times 5 = 80 \times \frac{3}{2} + 15 \times 7$$

$$\therefore R_c = 45 \text{ kN}$$

$$\therefore R_a = 80 + 15 - 45 = 50 \text{ kN}$$

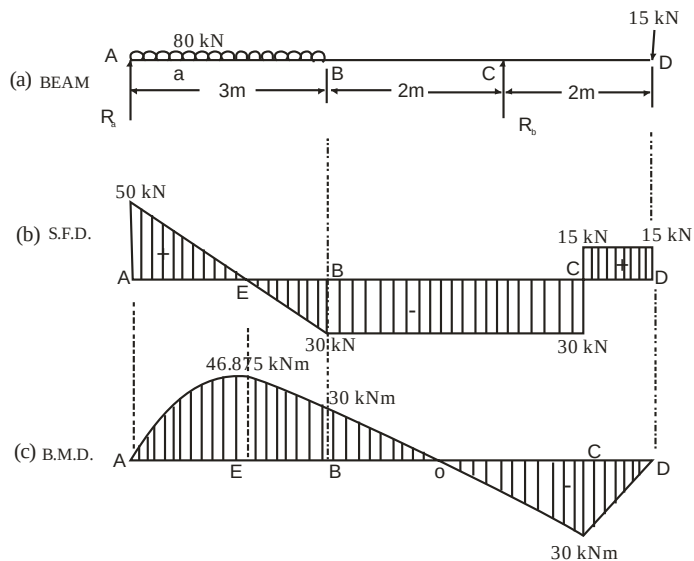


Fig. 3.26

S.F. Calculations

$$\text{S.F. at A} = +50 \text{ kN}$$

$$\text{S.F. at B} = 50 - 80 = -30 \text{ kN}$$

$$\text{S.F. between B and C} = -30 \text{ kN}$$

$$\text{S.F. between C and D} = +15 \text{ kN}$$

Let the S.F. be the zero at x meters from A.

Equating the S.F. to zero, we have,

$$50 - \frac{80}{3}x = 0$$

$$\therefore x = \frac{3}{80} \times 50 = 1.875 \text{ m from A}$$

B.M. Calculations

$$\text{B.M. at A} = M_a = 0$$

$$\text{B.M. at B} = M_b = 50 \times 3 - 80 \times \frac{3}{2} \text{ kNm} = + 30 \text{ kNm}$$

$$\text{B.M. at E i.e., 1.875 meters from A} = 50 \times 1.875 - \frac{80}{2} (1.875)^2 \text{ kNm} = + 46.875 \text{ kNm}$$

$$\text{B.M. at C} = M_c = - 15 \times 2 = - 30 \text{ kNm}$$

$$\text{B.M. at D} = M_d = 0$$

Point of contra flexure

There will be a point of contraflexure between B and C.

Let this point of contraflexure be at x meter from C

Equating the B.M. to zero, we get,

$$40x - 15(2 + x) = 0$$

$$\therefore x = 1.2 \text{ m}$$

Exercise

3.1 What are the different types of beams?

3.2 What are the sign conventions for shear force and bending moment?

3.3 What do you mean by point of contraflexure?

3.4 What are the different types of load acting on a beam?

3.5 Draw SFD and BMD for a cantilever subjected to udl.

3.6 A cantilever of length 2 m carries a udl of 3 kN/m over a length of 1 m from the fixed end. Draw SFD and BMD.

(Ans. $F_{\max} = 3\text{kN}$, $M_{\max} = -1.5\text{kNm}$)

3.7 A cantilever of length 4 m carries a udl of 1 kN/m over whole length and a point load of 2 kN at 1 m from the free end. Draw SFD and BMD.

(Ans. $F_{\max} = 14\text{kN}$, $M_{\max} = -14\text{kNm}$)

3.8 A simply supported beam of length 8 m carries point loads of 4, 10 and 7 kN at a distance of 1.5, 2.5 and 2 m respectively from left end. Draw the SFD and BMD.

(Ans. $M_{\max} = 90\text{kNm}$)

3.9 A simply supported beam of length 8 m carries point loads of 4 and 6 kN at a distance of 2 and 4 m respectively from left end. Draw the SFD and BMD.

(Ans. $M_{\max} = 20\text{kNm}$)

3.10 A simply supported beam of length 6 m carries point loads of 1, 2 and 3 kN at a distance of 1.5, 3 and 4.5 m respectively from left end. There is a udl of 1.5 kN/m over the entire length. Draw the SFD and BMD.

(Ans. $M_{\max} = 12.75\text{kNm}$)

4 Centre Of Gravity And Moment Of Inertia

4.1 Introduction:

Center Of Gravity And Centroid

The center of gravity of a body is the point at which the resultant gravitational force by the weights of all the particles of the body passes or the point at which the whole of its mass may be assumed to be concentrated. The centre of gravity of a plane figure which has area but no weight and where the whole area is assumed to be concentrated is known as the centroid of that area.

4.2 Location of Centroid

Analytical Method For The Location Of Centroid: Let us take a plane area as shown in

Fig. 4.1. Let $\bar{x}$ and $\bar{y}$ be the co-ordinates of centre of gravity of the area from OY and OX axis respectively. Divide the entire area A into small elementary areas $A_1, A_2, A_3, A_4 \dots$ etc. whose centroids are at distances of $x_1, x_2, x_3, x_4 \dots$ etc. respectively from the vertical axis OY. The algebraic sum of the moments of these elemental areas about Y axis = $A_1 x_1 + A_2 x_2 + A_3 x_3 + A_4 x_4 + \dots$

The moment of the total area about y – axis = $A \bar{x}$

The algebraic sum of the moments of the element areas about any axis is equal to the moment of the total area 'A' about the same axis.

$$\therefore \frac{A_1 x_1 + A_2 x_2 + A_3 x_3 + A_4 x_4 + \dots}{A_1 + A_2 + A_3 + A_4 + \dots} = \bar{x}$$

or

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3 + A_4 x_4 + \dots}{A_1 + A_2 + A_3 + A_4 + \dots} = \frac{\sum Ax}{\sum A}$$

Similarly by equating moments of the elements of the elemental areas about x – axis with that of the total area we have

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3 + A_4 y_4 + \dots}{A_1 + A_2 + A_3 + A_4 + \dots} = \frac{\sum Ay}{\sum A}$$

Knowing $\bar{x}$ and $\bar{y}$, the position of centroid of the given figure is fixed.

4.3 Centroid of Regular Areas

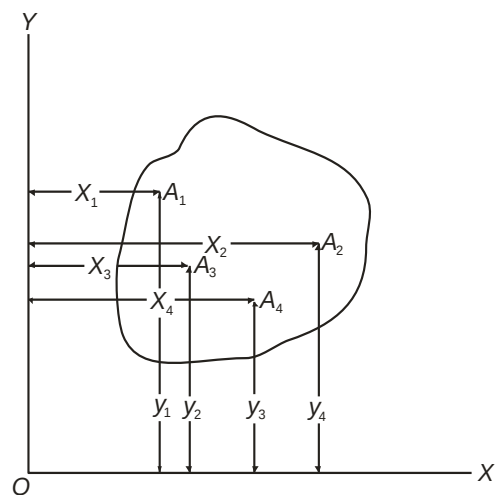


Fig. 4.1

(a) **Rectangle, square, parallelogram and rhombus:** These figures are having two axes of symmetry and their intersection point is the location of centroid.

(b) **Triangle:** the centroid is the point of intersection of medians. It is situated at $\frac{2}{3}$ of height from vertex or $\frac{1}{3}$ height from base.

(c) **Circle:** A circle is symmetrical about its diameters. The intersection of any two diameters, i.e. the centre of the circle is the centroid.

(d) **Semicircle:** A semicircle has one axis of symmetry. The centroid G lies at a distance of $\frac{4r}{3\pi}$ from its base, as shown in Fig. 4.2.

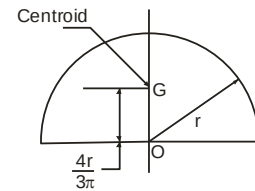


Fig. 4.2

4.4 Procedure for Locating the Centroid:

- (i) First observe whether the given area has any axis of symmetry or not.
- (ii) Divide the area into areas of regular shape whose centroids are known.
- (iii) Find out the areas of each broken up geometrical figure.
- (iv) Choose suitable axes of reference OX and OY (where there is one axis of symmetry, one of the axes chosen will coincide with the line of symmetry) with respect to which the position of centroid of the given figure is to be determined.
- (v) Find the distance of the centroids of the figure from the axes of reference.
- (vi) Take moments of the broken up areas about the axes of the reference.

First moment of area = area x perpendicular distance of centroid of respective area from the axes of reference.

$$= Ax \quad \dots(5.1)$$

(vii) The distance of the centroid from the axis of reference:

$$\bar{x} = \frac{\sum Ax}{A} \quad \text{and} \quad \bar{y} = \frac{\sum Ay}{A}$$

(viii) $\bar{x}$ and $\bar{y}$ fix the position of the centroid of the given area.

4.5 Moment of Inertia

As per Newton's first law of motion the property of matter of resisting any change of its state of rest or motion in a straight line is known as inertia. For bodies in linear motion, the force required to change its state of motion shall depend upon the mass of the body. For bodies in

rotation about an axis, the force required to change the state of motion shall depend on the mass of the body and its distance from the axis around which it is rotating.

4.6 Moment of Inertia of an area

Moment of inertia of an area about x – x axis is written as I_{xx} and about y – y axis as I_{yy} . The moment of inertia of an area about an axis passing through its centroid is written by I_G or simply I . If the moment of area given by equation (5.1) is again multiplied by the perpendicular distance between C.G. of the area and axis OY i.e. distance x , then the quantity $(Ax).x = Ax^2$ is known as moment of moment of the area or second moment of area or moment of inertia about the axis OY. Similarly the moment of inertia about the axis OX = Ay^2 .

Now let us find out the moment of inertia of an irregular area A (Fig. 4.1) about x – x and y – y axis. Assume the given area to be composed of several elemental areas $A_1, A_2, A_3, A_4, \dots$ etc. and their distances from the two axes as $x_1, x_2, x_3, x_4, \dots$ etc. and $y_1, y_2, y_3, y_4, \dots$ etc. Then

$$I_{xx} = A_1 y_1^2 + A_2 y_2^2 + A_3 y_3^2 + A_4 y_4^2 + \dots = \sum Ay^2 \quad \dots(5.2)$$

$$I_{yy} = A_1 x_1^2 + A_2 x_2^2 + A_3 x_3^2 + A_4 x_4^2 + \dots = \sum Ax^2$$

.....(5.3)

4.7 Radius of Gyration

The radius of gyration of a body rotating about an axis is the distance from the axis where the whole area of the body is assumed to be concentrated so that the moment of inertia about the axis is unaltered. Now let us assume that the area shown in Fig. 4.3 is squeezed into a uniform strip of area A as shown in Fig. 4.4. Each elemental area will be then at the same distance k from the axis xx . The moment of inertia of this area is

$$I_{xx} = Ak^2. \quad \dots(5.3)$$

If the value of k is such that the moment of inertia given by Equation (5.2) remains the same as given by equation (5.4) then k is known as the radius of gyration of the section about the axis xx .

$$I_{xx} = Ak^2 \text{ or } k = \sqrt{\frac{I_{xx}}{A}} \quad \dots(5.4)$$

4.8 Theorems Involving Moment of Inertia of Plane Figures

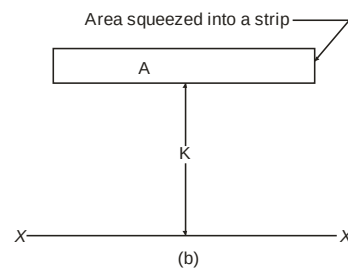


Fig. 4.3

4.8.1 Theorem of Parallel Axis

It states that “The moment of inertia of a figure about any axis in the plane of the figure is equal to the sum of the moment of inertia of that figure about the centroidal axis parallel to the given axis and the product of the area of the figure and square of the perpendicular distance between the two axes.”

Suppose the moment of inertia of a figure of area A as in Fig. 4.4 is to be determined about axis AB . Let G be the centroid of the figure. Draw a

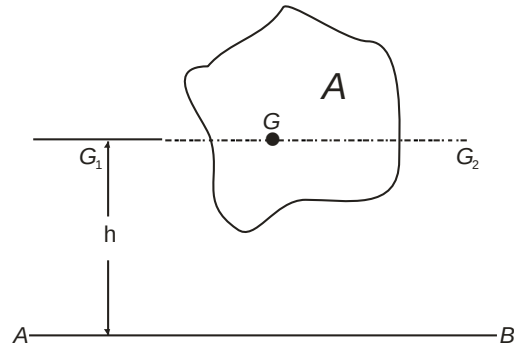


Fig. 4.4

centroidal line G_1G_2 parallel to AB . Then the moment of inertia of the figure about AB

$$I_{AB} = I_G + Ah^2 \quad \dots(5.5)$$

where h is the distance between the two axes and I_G is moment of inertia of the area about axis G_1G_2 .

4.8.2 Theorem of Mutually Perpendicular Axis

It states that, “The moment of inertia of a figure about an axis perpendicular to the figure passing through its centroid is equal to the sum of its moment of inertia about two mutually perpendicular centroidal axes lying in the plane of the figure”.

Consider a plane figure of area A . Let two mutually perpendicular axes OX and OY be in the plane of the figure and passing through the centroid O as shown in Fig. 4.5.

Then the moment of inertia of the area about an axis OZ passing through the centroid O and perpendicular to the plane of the figure will be the sum of the moments of inertia of the area about the x -axis and y -axis, i.e.

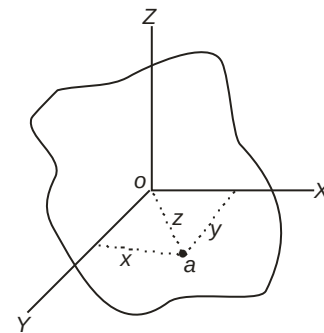


Fig. 4.5

$$I_{zz} = I_{xx} + I_{yy} \quad \dots(5.6)$$

All the three axes OX , OY and OZ are mutually perpendicular to one another.

4.9 Moment of Inertia of Regular Sections

4.9.1 Rectangle

Consider a rectangle of breadth B and depth D as shown in Fig 4.6. Let the axes $X-X$ and $Y-Y$ pass through the C.G. Consider an elementary strip of width B and thickness dy at a distance y from $X-X$ axis and parallel to it.

Area of the strip = $B \times dy$

Moment of inertia of this elementary strip about $X-X$ axis = $B \times dy \times y^2$

$\therefore$ Moment of inertia of the whole rectangle about $X-X$ axis (centroidal)

$$I_{xx} = \int_{-\frac{D}{2}}^{\frac{D}{2}} B \times dy \times y^2$$

$$I_{xx} = B \left[\frac{y^3}{3} \right]_{-\frac{D}{2}}^{\frac{D}{2}} = B \left[\frac{D^3}{24} + \frac{D^3}{24} \right] = \frac{BD^3}{12}$$

$$I_{xx} = \frac{DB^3}{12}$$

Similarly we can get $I_{yy} = \frac{BD^3}{12}$

Moment of inertia of the rectangle about its base AB will be obtained by applying the theorem of parallel axis.

$$I_{AB} = I_{xx} + B \times D \times \left(\frac{D}{2} \right)^2 = \frac{BD^3}{12} + \frac{BD^3}{4} = \frac{BD^3}{3}$$

4.9.2 Hollow Rectangle (Fig 4.7)

$$I_{xx} \text{ for the hollow rectangle} = \frac{BD^3}{12} - \frac{bd^3}{12}$$

Similarly

$$I_{yy} = \frac{DB^3}{12} - \frac{db^3}{12}$$

$\therefore$

$$I_{zz} = I_{xx} + I_{yy} = \frac{BD^3}{12} - \frac{bd^3}{12} + \frac{DB^3}{12} - \frac{db^3}{12}$$

$$= \frac{BD}{12} (B^2 + D^2) - \frac{bd}{12} (b^2 + d^2)$$

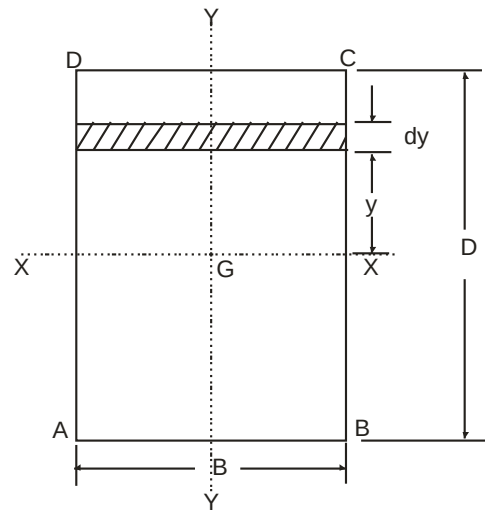


Fig. 4.6

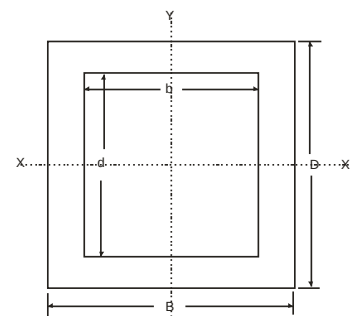


Fig. 4.7

4.9.3 Triangle

Consider a triangle ABC of base width B and height H . It is desired to find out the moment of inertia about the base BC and about the axis $x-x$ passing through the centroid and parallel to the base as shown in the Fig. 4.8.

Consider an elementary strip of thickness dy at a distance y below the vertex A . Triangle ABC and ADE are similar.

$$\therefore \frac{DE}{BC} = \frac{y}{H}$$

$$\text{or} \quad DE = \frac{BC \times y}{H} = \frac{by}{H}$$

$$\text{Area of the strip} = \frac{By}{H} \times dy$$

Moment of inertia of the strip about the base = area $\times$ (distance)²

$$= \frac{By}{H} \times y dy \times (H - y)^2 = \frac{B}{H} \cdot y (H^2 + y^2 - 2Hy) \cdot dy$$

$$= \frac{B}{H} (H^2 y + y^3 - 2Hy^2) dy$$

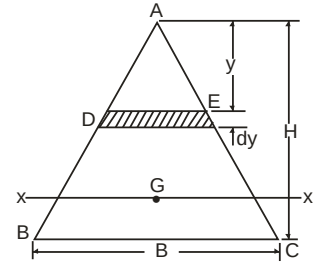


Fig. 4.8

Moment of inertia of the triangle about the base BC is given by

$$I_{BC} = \int_0^H \frac{B}{H} (H^2 y + y^3 - 2Hy^2) dy = \frac{B}{H} \left[\frac{H^2 y^2}{2} + \frac{y^4}{4} - \frac{2Hy^3}{3} \right]_0^H$$

$$= \frac{B}{H} \left[\frac{H^4}{2} + \frac{H^4}{4} - \frac{2H^4}{3} \right] = \frac{BH^3}{12}$$

Moment of inertia of the triangle about an axis x-x passing through the centroid and parallel to the base is found out as follows:

Now from the theorem of parallel axis we have

$$I_{BC} = I_G + \frac{BH}{2} \times \left(\frac{H}{3} \right)^2$$

$$I_G = I_{BC} - \frac{BH^3}{18} = \frac{BH^3}{12} - \frac{BH^3}{18} = \frac{BH^3}{36}$$

4.9.4 Circle

Consider a circle of radius R as shown in Fig 4.9. Consider an elementary ring of thickness dr at a radius r from o. Area of this ring = $2\pi r \cdot dr$.

Moment of inertia of this ring about z axis passing through o

and perpendicular to the plane of the given area = $2\pi \times r^2 dr$.

$\therefore$ Moment of inertia of the whole area

$$I_{zz} = \int_0^R 2\pi r^3 dr = 2\pi \int_0^R r^3 dr = 2\pi \left[\frac{r^4}{4} \right]_0^R = \frac{\pi R^4}{2}$$

$$\text{But} \quad I_{zz} = I_{xx} + I_{yy}$$

The given area is symmetrical about

xx as well as yy axis.

$$\therefore \quad I_{xx} = I_{yy}$$

$$I_{zz} = 2 I_{xx} = 2 I_{yy}$$

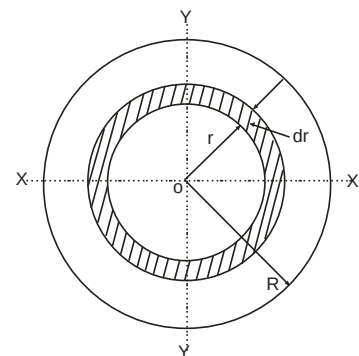


Fig. 4.9

$$\text{or } I_{xx} = I_{yy} = \frac{I_{zz}}{2} = \frac{\pi R^4}{4} = \frac{\pi D^4}{64}$$

For a hollow circular area of external diameter D and internal diameter d.

$$I_{xx} = I_{yy} = \frac{\pi D^4}{64} - \frac{\pi d^4}{64} = \frac{\pi}{64} (D^4 - d^4)$$

4.10 Section Modulus and Polar Moment of Inertia

It is defined as ratio of the moment of inertia of the section about the axis passing through the centroid and the distance of extreme fiber of the section from the centroidal axis. It is written as Z. Every section has got two section moduli.

For rectangle shown in Fig. 4.6

$$Z_{xx} = I_{xx} / \frac{D}{2}$$

$$\text{and } Z_{yy} = I_{yy} / \frac{B}{2}$$

Polar moment of inertia of a body is the moment of inertia of the body about its centroidal axis which is perpendicular to the plane of the body.

Problem 4.1 Find the centroid of the section shown in Fig. 4.10(a).

Solution: The section will be split up into two rectangular areas ABCD and EFGH as shown in Fig. 4.10(b) of areas,

$$20 \times 100 = 2000 \text{ mm}^2$$

$$\text{and } 100 \times 20 = 2000 \text{ mm}^2.$$

respectively

Centroidal distance of

ABCD from the axis 1 – 1 = 70 mm

Centroidal distance of EFGH from the axis 1 – 1 = 10 mm

Let $\bar{y}$ be the height of the centroid of the section from the axis 1 – 1

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{2000 \times 70 + 2000 \times 10}{2000 + 2000} \text{ mm}$$

$$= 40 \text{ mm above the axis 1 – 1}$$

The above computation may be conveniently worked out in a tabular form as shown below:

Component	Area A mm ²	Centroidal Distance From 1 – 1 y mm	Ay mm ²
ABCD	2000	70	140000
EFGH	2000	10	20000
Total	4000		60000

$$\therefore \bar{y} = \frac{\sum Ay}{\sum A} = \frac{160000}{4000} = 40 \text{ mm}$$

Problem 4.2 Find the centroid of the section in Fig. 4.11.

Solution: The given section will be split up into a number of components. The area of the various components and their centroidal distance from axis 1 – 1 and the moments of the individual components about the axis 1- 1 are shown in the following table:

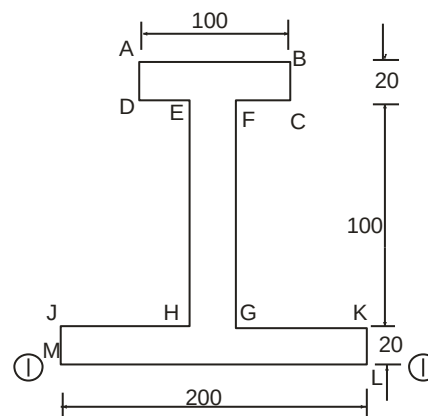


Fig. 4.11

Component	Area A mm ²	Centroidal Distance From y 1 – 1 mm	Ay mm ²
ABCD 100 × 20	2000	130	260000
EFGH 10 × 100	1000	70	70000
JKLM 200 × 20	4000	10	40000
Total	7000		370000

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{370000}{7000} \text{ mm} = 52.86 \text{ mm above the axis 1 – 1.}$$

Problem 4.3 Find the centroid of the section shown in Fig. 4.12.

Solution: The given section may be split up into two rectangles ABCD and EFGH as shown in

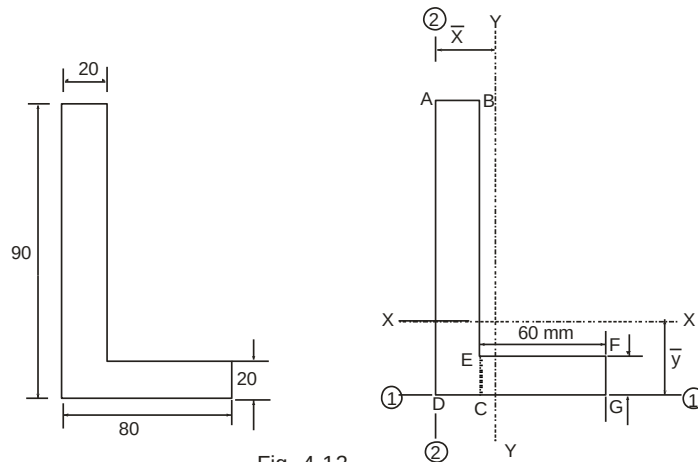


Fig. 4.12

Fig. 4.12. The position of the centroid of the section with respect to the axis 1 – 1 and 2 – 2 will now be worked out. The relevant computations are shown in the following table:

Component	Area A mm ²	Centroidal Distance y From 1 – 1 mm	Centroidal Distance x From 2 – 2 mm	Ay mm ³	Ax mm ³
ABCD 20 × 90	1800	45	10	81000	18000
EFGC 60 × 20	1200	10	50	12000	60000
Total	3000			93000	78000

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{93000}{3000} = 31 \text{ mm}$$

$$\bar{x} = \frac{\sum Ax}{\sum A} = \frac{78000}{3000} = 26 \text{ mm}$$

Problem 4.4 Find the moment of inertia of the section shown in Fig. 4.13 about the centroidal axis perpendicular to the web.

Solution: The given section will be split up into components ABCD, EFGH and JKLM. The areas of the individual components, their centroidal distances from the axis 1 – 1 through the bottom edge and the moments of inertia of the individual components about their respective centroidal axes parallel to the axis 1 – 1 are given in the following table:

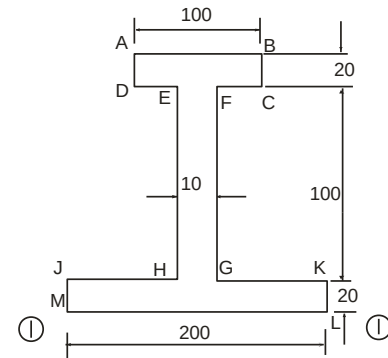


Fig. 4.13

Component	Area A mm ²	Centroidal distance y from 1 – 1 mm	Ay mm ³	Ay ² mm ⁴	I _{self} mm ⁴
ABCD	2000	130	260000	33800000	$\frac{100 \times 20^3}{12} = 66667$
EFGH	1000	70	70000	4900000	$\frac{10 \times 100^3}{12} = 833333.33$
JKLM	4000	10	40000	400000	$\frac{200 \times 20^3}{12} = 133333$
TOTAL	7000		370000	39100000	1033333.33

$$\text{Distance of the centroidal axis XX from the axis 1-1} \\ = \bar{y} = \frac{\sum Ay}{\sum A} = \frac{370000}{7000} \text{ mm} = 52.86 \text{ mm}$$

$$\text{Moment of inertia about the axis 1 – 1} \\ = I_{1-1} = \sum I_{self} + \sum Ay^2 \\ = 1033333.33 + 3910.0000 = 40133333.33 \text{ mm}^4$$

$$\text{But } I_{1-1} = I_{xx} + \sum A\bar{y}^2 \\ \therefore 40133333.33 = I_{xx} + 7000 \times 52.86^2 \\ \therefore I_{xx} = 20574076.13 \text{ mm}^4$$

Problem 4.5 Find the moment of inertia about the centroidal axes XX and YY of the section shown in Fig. 4.14.

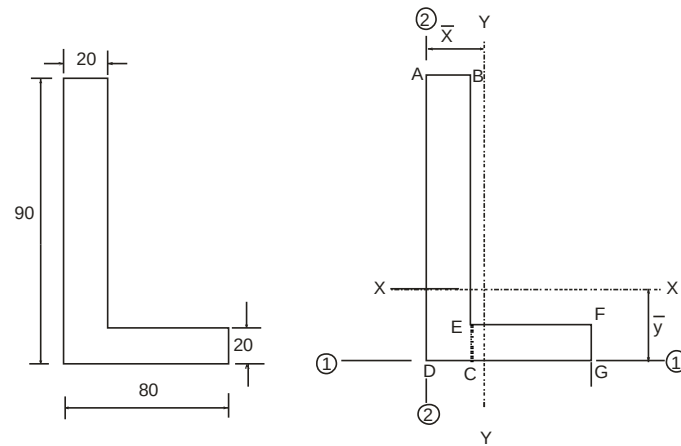


Fig. 4.14

Solution: To find the moment of inertia about the axis XX:

The following table shows the relevant computations:

Component	Area A mm ²	Centroidal distance y from 1 – 1 mm	Ay mm ³	Ay ² mm ⁴	I _{self} mm ⁴
ABCD	1800	45	81000	3645000	$\frac{20 \times 90^3}{12} = 1215000$
EFGH	1200	10	12000	120000	$\frac{60 \times 20^3}{12} = 40000$
TOTAL	3000		93000	3765000	1255000

$$\therefore \bar{y} = \frac{\sum Ay}{\sum A} = \frac{93000}{3000} = 31$$

$$\text{Moment of inertia about the axis 1 – 1} = I_{1-1} = \sum I_{self} + \sum Ay^2$$

$$\text{But } I_{1-1} = 1255000 + 3765000 = 5020000 \text{ mm}^4$$

$$5020000 = I_{xx} + 3000 \times 31^2$$

$$\therefore I_{xx} = 2137000 \text{ mm}^4$$

To find the moment of inertia about the axis YY:

The following table shows the relevant computations:

Component	Area A mm ²	Centroidal distance x from 2 – 2	Ax mm ³	Ax ² mm ⁴	I _{self} mm ⁴
-----------	------------------------------	---	-----------------------	------------------------------------	--------------------------------------

		mm			
ABCD	1800	10	18000	180000	$\frac{90 \times 20^3}{12} = 60000$
EFGH	1200	50	60000	3000000	$\frac{20 \times 60^3}{12} = 360000$
TOTAL	3000		78000	3180000	420000

$$\therefore \bar{x} = \frac{\sum Ax}{\sum A} = \frac{78000}{3000} = 26 \text{ mm}$$

Moment of inertia about the axis 2 – 2

$$= I_{2-2} = \sum I_{self} + \sum Ax^2$$

$$\therefore I_{2-2} = 420000 + 3180000 = 3600000$$

But $I_{2-2} = I_{yy} + (\sum A)(\bar{x}^2)$

$$3600000 = I_{yy} + 3000 \times 26^2$$

$$I_{yy} = 1572000 \text{ mm}^4$$

Problem 4.6 Calculate the moment of inertia about the horizontal and vertical gravity axes (I_{xx} and I_{yy}) of the section shown in Fig. 4.15.

Solution: The given section is split up into two components. The areas of the individual components, their centroidal distances from the axis 1 – 1 through the top edge and the Moments of Inertia of the individual components about their centroidal axes parallel to the axis 1 – 1 are tabulated below:

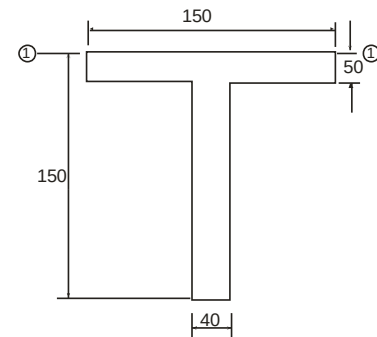


Fig. 4.15

Component	Area A mm ²	Centroidal distance y from 1 – 1 mm	Ay mm ³	Ay ² mm ⁴	I _{self} mm ⁴
Top Flange	7500	25	187500	4687500	$\frac{150 \times 50^3}{12} = 1562500$
Web	4000	100	400000	40000000	$\frac{40 \times 100^3}{12} = 3333333.3$
TOTAL	11500		587500	44687500	4895833.3

Distance of the centroidal axis XX from 1 – 1

$$= \bar{y} = \frac{\sum Ay}{\sum A} = \frac{587500}{11500} = 51.09 \text{ mm}$$

$$I_{1-1} = \sum I_{self} + \sum Ay^2 = 4895833.3 + 44687500 = 49583333.3$$

But $I_{1-1} = I_{xx} + (\sum A)\bar{y}^2$

$$49583333.3 = I_{xx} + 11500 \times 51.09^2$$

∴

$$I_{xx} = \frac{50 \times 150^3}{12} + \frac{100 \times 40^3}{12}$$

$$I_{yy} = \frac{50 \times 150^3}{12} + \frac{100 \times 40^3}{12} = 14595833.33 \text{ mm}^4$$

Problem 4.7 Find the moment of inertia about the centroidal axis XX for the section shown in

Fig. 4.16. The dimensions shown in the Fig. 4.16 are in cm.

Solution: The given section will be split up into three rectangular components, i.e., the top flange, the web and the bottom flange.

The properties of the components are tabulated below:

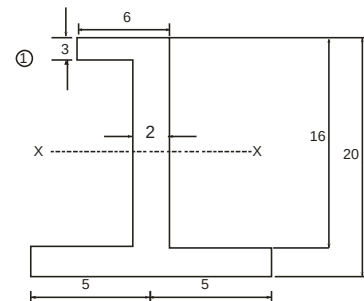


Fig. 4.16

Component	Area of (cm ²)	Centroidal distance y from 1 – 1 (cm) bottom edge	Ax (cm ³)	Ax ² (cm ⁴)	I _{self} (cm ⁴)
Top Flange	18	18.5	333	6160.50	$\frac{6 \times 3^3}{12} = 13.5$
Web	26	10.5	273	2866.50	$\frac{2 \times 16^3}{12} = 366.167$
Bottom flange	40	2	80	160	$\frac{10 \times 4^3}{12} = 53.333$
TOTAL	84		686	9187	433

Distance of centroidal axis XX from the bottom edge 1 – 1

$$= \bar{y} = \frac{\sum Ay}{\sum A} = \frac{686}{84} = 8.17 \text{ cm}$$

Moment of inertia about the axis 1

$$= I_{1-1} = \sum I_{self} + \sum Ay^2 = 433 + 9187 = 9620 \text{ cm}^4$$

But

$$I_{1-1} = I_{xx} + A \bar{y}^2$$

$$\therefore 9620 = I_{xx} + 84 \times 8.172$$

$$\therefore I_{xx} = 4015.38 \text{ cm}^4$$

Exercise

4.1 Define centre of gravity and centroid.

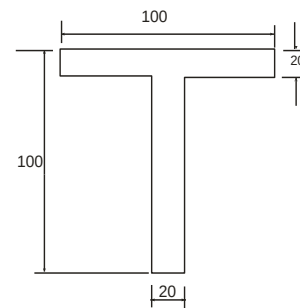
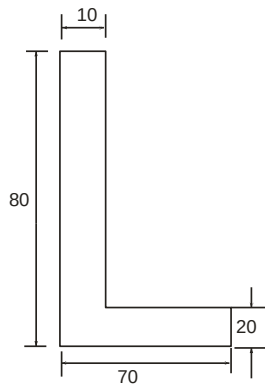
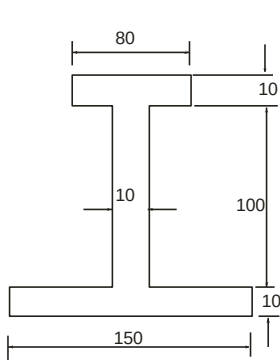
4.2 Define radius of gyration and moment of inertia.

4.3 Find the expression for moment of inertia of a rectangular section about the axis passing through its base.

4.4 State the parallel axis theorem and perpendicular axis theorem.

4.5 Find the expression for moment of inertia of a hollow rectangular section about the axis passing through its base.

4.6 Find the centre of gravity and moment of inertia of the following sections about their centroidal axis:



5. BENDING STRESSES IN BEAMS

5.1 Introduction

Due to external loads and moments bending moments and shearing forces are set up at all the sections of a beam. A beam section has to resist the action of bending moment and shearing force so induced and the beam deforms. The process of bending will stop when it has set up full resistance to the bending moment and shearing force.

The stress produced at the section to resist the bending moment is known as bending stress or direct stress or flexural stress and that to resist the action of shear force is known as shearing stress.

5.2 Theory of simple bending

Assumptions: Before developing elastic flexural formula it will be worthwhile to outline the assumptions upon which such relationship depends:

- (1) The material of the beam remains elastic during loading i.e. elastic limit is not exceeded.
- (2) The material of the beam is perfectly homogenous and isotropic and modulus of elasticity in tension is equal to that in compression.
- (3) Before loading takes place the beam is straight and each cross section is symmetric about plane of bending.
- (4) Plane cross section remains plane before and after bending.
- (5) There is no resultant force perpendicular to any cross section.
- (6) Every layer of material is free to expand or contract longitudinally and laterally under stress and does not exert pressure upon each other.

Pure Bending or Simple Bending:

Fig. 5.1 shows a beam AB with supports at C&D, L units apart. Let the overhangs AC = BD = a. Let a point load be applied at each end of the beam. It can be observed that there is a constant bending moment and no shear force (zero shear force) between C&D. The stresses will be set up in that length of the

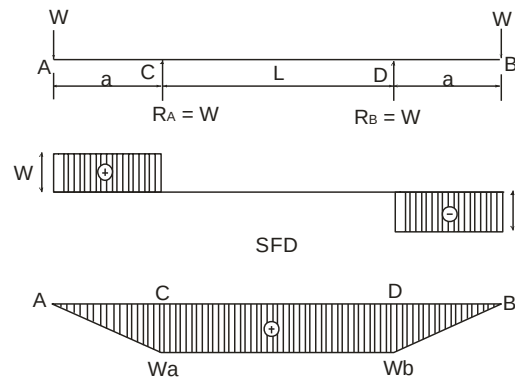


Fig. 5.1

beam due to bending moment only and that length of the beam is said to be in pure bending or simple bending.

Fig. 5.2 Shows the longitudinal section of a beam through plane of symmetry before and after it has been bent by two equal and apposite moments applied at its ends. The axes of coordinates are also shown in the same figure. The bent form of the longitudinal beam section shown in Fig. 5.2 (b) is highly exaggerated but it represents the essential features.

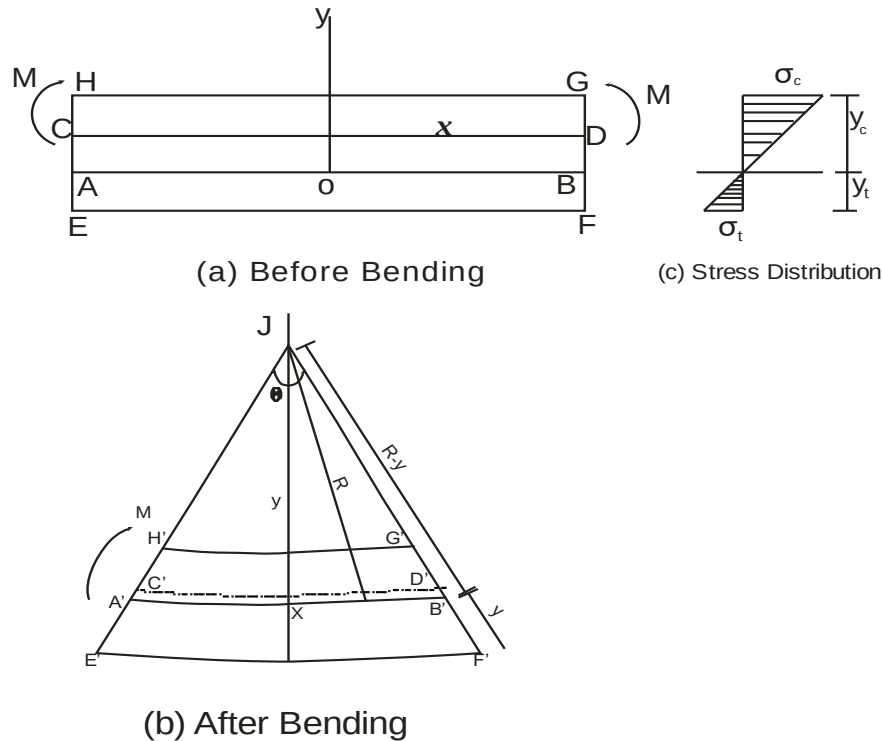


Fig. 5.2

Since the bending moment at any section is same, the section of the beam under consideration will bend uniformly or the curvature of any surface will be uniform as in Fig. 5.2 b. Thus the sides of longitudinal section GH and EF will bend into concentric arcs of circles, $G'H'$ and $E'F'$. All surfaces between and parallel to GH and EF will bend into contracted arcs of circles with centre at J . From Fig. 5.2(b) it can be seen that while $G'H'$ has contracted from its original length GH , the lower fiber $E'F'$ has extended from its original length EF . If we go down from $G'H'$ to $E'F'$, the contraction reduces becomes zero at some layer and then the layer begins to extend.

The layer of beam at which the extension or contraction does not take place, i.e. the layer which remains in its original length after bending is termed as or neutral layer. Line

AB in Fig. 5.2(a) and arc A'B' in fig. 5.2(b) represent neutral layer. The line of intersection of neutral layer and beam's cross section is defined as neutral axis.

The Fig. 5.2(b) shows the position of beam after bending. Let the radius of the neutral layer after and before bending be R so that the radius of layer C'D' is 'R-y'. It may be noted that AB = CD before bending and AB = A'B' after bending being neutral layer. C'D' has shortened from its original length CD. If the angle subtended by concentric arcs at common centre J is θ then

$$AB = CD = A'B' = R \theta$$

and $C'D' = (R - y) \theta$

strain of fiber CD $e = \frac{\text{Change in length}}{\text{Original Length}} = \frac{C'D' - A'B'}{A'B'}$

$$e = \frac{(R - y)\theta - R\theta}{R\theta} = \frac{-y}{R} \quad (\text{compressive}) \quad \dots(5.1)$$

The negative sign indicates that strain is compressive. If another layer at a distance y below the neutral plane is considered, e will be positive or tensile.

Equation $e = \frac{-y}{R}$ shows strain varies directly as the distance from neutral axis

because R is constant for a given loading.

Since stress = strain x elasticity

$$\sigma = e \times E = \frac{y}{R} E$$

$$\frac{\sigma}{y} = \frac{E}{R} \quad \dots\dots (5.2)$$

or $\sigma \propto y$

$\sigma_t \propto y_t$

$\sigma_c \propto y_c$

Hence the stress intensity in any fiber is proportional to the distance of the fiber from the neutral layer. For the case explained above all fibers below the neutral layer are subjected to tensile stresses while those above the neutral layer are subjected to compressive stresses. The stress distribution diagram is shown in Fig. 5.2(c).

Moment of Resistance:

Let us now determine the magnitude of stress from the equation of equilibrium, since there is no external force, the sum of all the compressive forces above the neutral layer must be equal to the sum of the tensile forces below the neutral layer.

Let us take a strip δy thick at a distance y from NA in Fig 5.3 and let its area be δa . Since the stress is equal to ' σ ' there the thrust exerted by this small area will be ' $\sigma \cdot \delta a$ '.

To find M_r let us again consider the elementary area δa in Fig. 5.3. The force set up by the area $\sigma \delta a$ and the moment of force about NA is

$$\delta M = \sigma \cdot \delta a \cdot y = \frac{E}{R} (\delta a) \cdot y^2$$

The total moment of resistance set up by such elementary area will be

$$\sum_{y=y_t}^{y=y_c} \frac{E}{R} \delta a \cdot y^2 = \frac{E}{R} \int_{y_t}^{y_c} y^2 \delta a$$

The expression in the integral is known as second moment of area or moment of inertia about NA and is shown as 'I'

Hence $M_r = \frac{E}{R} I$

for equilibrium $M = M_r = \frac{E}{R} I$

$$\frac{M}{I} = \frac{E}{R} \quad \text{.....(5.4)}$$

But we know $\frac{E}{R} = \frac{\sigma}{y}$

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y} \quad \text{.....(5.5)}$$

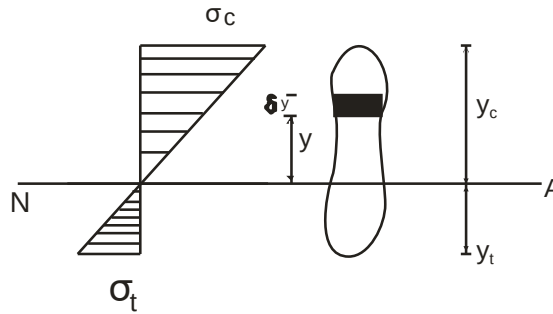


Fig. 5.3 Stress Distribution

$$\frac{M}{I} = \frac{\sigma}{y}$$

For tensile side use σ_t and y_t

$$\frac{M}{I} = \frac{\sigma}{y}$$

For compressive side use σ_c and y_c

'EI' is termed as *flexural rigidity* and may be specified as a single constant for a given beam.

Eq. 5.5 is applicable to a member which is subjected to a constant bending moment and the member is absolutely free from shear force but in actual practice, a member is subjected to such loading that the B.M. varies from section to section and also the shear force is not zero. But shear force is zero at a section where bending moment is maximum. Hence the condition of simple bending may be assumed to be satisfied at such a section.

5.3 Design Criterion And Section Modulus

The bending equation is quite significant while designing machine components. To ensure strength in a beam it is necessary that the greatest tensile and compressive bending stress at the critical section i.e. where bending moment is maximum should not exceed the corresponding allowable stress for the material.

$$\sigma_{tmax} = \frac{M}{I} y_t ;$$

$$\sigma_{cmax} = \frac{M}{I} y_c$$

Condition of strength or of design

$$\sigma_{tmax} \leq \sigma_t \text{ allowable and } \sigma_{cmax} \leq \sigma_c \text{ allowable}$$

for a given value of allowable stress

$$\sigma = \frac{M}{I} y$$

$$M = \sigma \frac{I}{y} = \sigma Z \quad [\text{Compare with } F = \sigma.A \text{ for tensile loading}] \dots\dots(5.6)$$

Where $Z = \frac{I}{y}$ and is known as *section modulus*. Greater the value of Z stronger will be the section.

5.4 Section Modulus For Various Shapes Of Beam Section :

5.4.1 Rectangular Section

Moment of inertia of a rectangular section (Fig 5.4) about an axis through its C.G. (or through NA) is given by,

$$I = \frac{bd^3}{12}$$

Distance of outer most layers from N.A. is given by,

$$y_{\max} = \frac{d}{2}$$

∴ Section modulus is given by

$$Z = \frac{I}{y_{\max}} = \frac{\frac{bd^3}{12}}{\left(\frac{d}{2}\right)}$$

$$= \frac{bd^3}{12} \times \frac{2}{d} = \frac{bd^2}{6}$$

.....(5.7)

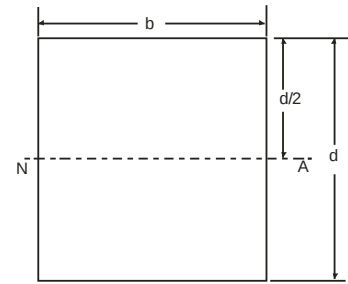


Fig. 5.4

5.4.2 Hollow Rectangular Section

Moment of inertia of a hollow rectangular section (Fig 5.5) about an axis through its C.G. (or through NA) is given by,

Here $I = \frac{BD^3}{12} - \frac{bd^3}{12}$

$$= \frac{1}{12} [BD^3 - bd^3]$$

$$y_{\max} = \frac{D}{2}$$

$$\therefore Z = \frac{I}{y_{\max}} = \frac{\frac{1}{12} [BD^3 - bd^3]}{\left(\frac{D}{2}\right)}$$

$$= \frac{1}{6D} [BD^3 - bd^3]$$

.....(5.8)

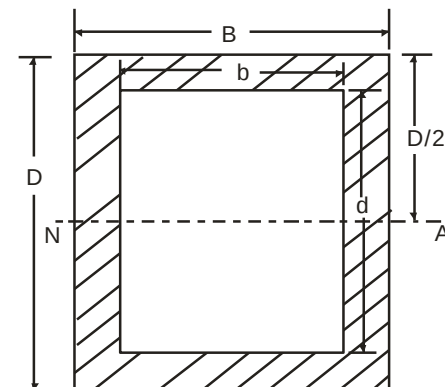


Fig. 5.5

5.4.3 Circular Section

Moment of inertia of a circular section (Fig 5.6) about an axis through its C.G. (or through NA) is given by,

$$I = \frac{\pi}{64} d^4$$

and

$$y_{\max} = \frac{d}{2}$$

$$\frac{I}{y_{\max}} = \frac{\frac{\pi}{64} d^4}{\left(\frac{d}{2}\right)}$$

∴

$$Z =$$

$$= \frac{\pi}{32} d^3$$

.....(5.9)

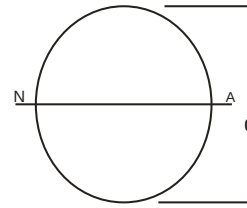


Fig. 5.6

5.4.4 Hollow Circular Section

Moment of inertia of a hollow circular section (Fig 5.7) about an axis through its C.G. (or through NA) is by,

Here $I = \frac{\pi}{64} [D^4 - d^4]$

and

$$y_{\max} = \frac{D}{2}$$

$$\begin{aligned} \therefore Z &= \frac{\frac{\pi}{64} [D^4 - d^4]}{\left(\frac{D}{2}\right)} \\ &= \end{aligned}$$

$$= \frac{\pi}{32D} [D^4 - d^4]$$

.....(5.10)

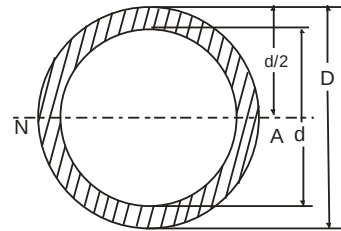


Fig. 5.7

5.7)
given

Problem 5.1 Copper plate is bent into a circular arc of radius 10 meters. If the plate section be 120 mm wide and 20 mm thick, find the maximum stress induced and the bending moment which can produce this stress. Take $E = 1 \times 10^5 \text{ N/mm}^2$.

Solution: Moment of inertia of the section about the neutral axis.

$$I = \frac{12 \times 20^3}{12} = 80000 \text{ mm}^4$$

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{E}{R} \cdot y$$

$$\therefore \sigma_{\max} = \frac{1 \times 10^5}{10 \times 10^3} \times \left(\frac{20}{2} \right) \text{ N/mm}^2 = 100 \text{ N/mm}^2$$

$$M = \frac{E}{R} \cdot I = \frac{1 \times 10^5}{10 \times 10^3} \times 8 \times 10^4 = 8 \times 10^5 \text{ N mm}$$

$$= 800 \text{ Nm}$$

Problem 5.2 A timber joist of 200 mm width and 400 mm depth is used as a cantilever beam of 3 meters length and loaded with a concentrated load at its free end in addition to its self-weight, so that the bending stress does not exceed 7 N/mm^2 . Find out the maximum value of the applied load. Unit weight of timber is 5 kN/m^3 .

Solution: Let W (N) be the concentrated load that can be placed safely at the free end of the cantilever under the given conditions of the problem. W = self weight of the joint per meter run.

$$= 0.2 \times 0.4 \times 1 \times 5 = 0.4 \text{ kN/m} = 400 \text{ N/m}$$

I_{xx} = Moment of inertia of the section about N.A.

$$= \frac{12 \times 200 \times 400^3}{12} = 32 \times 10^8 \text{ mm}^4$$

M = Maximum bending moment which will occur at the fixed end.

$$= WL + \frac{WL^2}{2}$$

$$= W \times 3 + \frac{100 \times 3 \times 3}{2} \text{ Nm} = (3W + 450) \times \frac{10^3}{400} \text{ N mm}$$

$$y = \text{Distance of extreme fiber from N.A.} = \frac{400}{2} = 200 \text{ mm.}$$

$$\sigma = \text{Bending stress} = 7 \text{ N/mm}^2$$

Using the relation $\frac{M}{I} = \frac{\sigma}{y}$ and substituting the values

$$\frac{(3W + 450) \times 10^3 \times 3}{32 \times 10^8} = \frac{7}{200}$$

or

$$3W + 450 = 37333.33$$

$\therefore$

$$W = 12294.44 \text{ N}$$

Problem 5.3 A cast iron T-beam has the following dimensions:

Overall depth	= 160 mm
Width of flange	= 150 mm
Flange thickness	= 40 mm
Web thickness	= 50 mm

The beam is simply supported over a span of 2.5 meters placed in the inverted T-position (i.e. flange at the bottom). If the maximum allowable tensile stress in the flange and compressive stress in the web are limited to 25 N/mm² and 80 N/mm² respectively, find the maximum central load that the beam can safely carry.

Solution: As the Fig. 5.8 is symmetrical about Y-Y axis, the centre of gravity will lie on this axis. Let $\bar{y}$ be the distance of centre of gravity from the bottom of flange. Divide the figure into two rectangles as shown in Fig. 5.8.

Rectangle 1

$$a_1 = 150 \times 40 = 6000 \text{ mm}^2, y_1 = \frac{40}{2} = 20$$

mm

Rectangle 2

$$a_2 = 120 \times 50 = 6000 \text{ mm}^2, y_2 = 40 + \frac{120}{2}$$

= 100 mm

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2}{a_1 + a_2} = \frac{6000 \times 20 + 6000 \times 100}{6000 + 6000}$$

= 60 mm

Moment of inertia of the section about NA

$$= \frac{1}{12} \times 150 \times 40^3 + 150 \times 40 \times (60 - 20)^2 + \frac{1}{12} \times 50 \times 120^3 + 50 \times 120 \times (100 - 60)^2$$

$$= 272 \times 10^5 \text{ mm}^4$$

σ_t = Permissible tensile stress at the bottom most face of flange = 25 N/mm²,
at a distance of $y_t = 60$ mm from neutral axis.

σ_c = Permissible compressive stress at the top most face of web = 80 N/mm²,
at a distance of $y_c = 100$ mm from neutral axis.

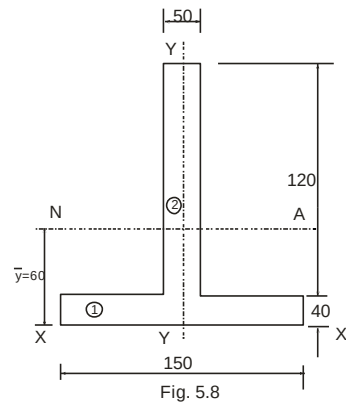
$$\frac{\sigma_t}{y_t} = \frac{25}{60} = 0.42, \quad \frac{\sigma_c}{y_c} = \frac{80}{100} = 0.80$$

$\frac{\sigma_t}{y_t}$

y_t being the least, is to be taken into consideration. Let W N be the concentrated load

that can be safely carried under the given condition of the problem. Maximum bending

moment at the centre, $M = \frac{Wl}{4} = \frac{4 \times 2.5}{4} \text{ Nm}$.



$$\frac{M}{I} = \frac{\sigma_t}{y_t} \text{ and substituting the values}$$

$$\frac{W \times 2.5 \times 1000}{4 \times 272 \times 10^5} = 0.42$$

$$\therefore w = 18278.4 \text{ N}$$

Problem 5.4 A 4 meters long beam with rectangular section of 100 mm width and 200 mm depth is simply supported at the ends. If it is loaded with a uniformly distributed load of 4 kN/m throughout the span and a concentrated load $P = 2 \text{ kN}$ placed at a distance of 1.5 m from one end, determine the maximum bending stress in the beam.

Solution: Let R_A and R_B be the reactions at A and B respectively. Taking moment of all the forces about A:

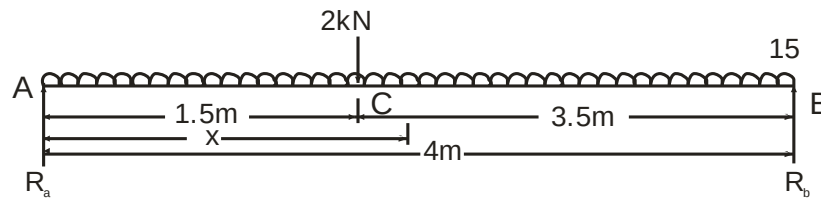


Fig. 5.9

$$R_B \times 4 - 4 \times 4 \times \frac{4}{2} - 2 \times 1.5 = 0 \quad \therefore R_B = 8.75 \text{ kN.}$$

$$\text{But } R_A + R_B = 4 \times 4 + 2 = 18$$

$$\therefore R_A = 18 - 8.75 = 9.25 \text{ kN.}$$

Let us consider a section at a distance x from A.

$$M_x = 9.25x - 2(x - 1.5) - 4 \cdot x \cdot \frac{x}{2} = 7.25x + 3 - 2x^2$$

M_x is maximum when

$$\frac{d}{dx} (7.25x + 3 - 2x^2) = 0$$

$$\text{or } 7.25 - 4x = 0 \quad \therefore x = 1.81 \text{ m}$$

$$\text{Moment bending of the section, } M = \frac{7.25 \times 1.81 + 3 - 2 \times (1.81)^2}{100 \times (200)} = 9.5703 \text{ kN.m}$$

$$\text{Moment of inertia of the section, } I = \frac{12}{100 \times (200)^3} = 6666.666 \times 10^4 \text{ mm}^4$$

$$\therefore \text{Maximum bending stress, } \sigma = \frac{M}{I} \times y = \frac{9.5703 \times 10^3 \times 10^3}{6666.66 \times 10^4} \times 100$$

$$= 14.35 \text{ N/mm}^2$$

Problem 5.5 A floor has to carry a load of 10000 N/sq.m. It is supported by joists 150 mm $\times$ 300 mm over a span of 5 m. How far apart may the joists be placed so that the bending stress does not exceed 8 N/mm²?

Solution: Let x be the centre-to-centre spacing of the joists. The area of the floor between any two joists = $5 \times x \text{ m}^2$.

The total load supported by one interior joist = $5 \times x \times 10000 = 50000x \text{ N}$.

$$\text{Maximum bending moment, } M = \frac{wl^2}{8} = \frac{50000x \times 5^2}{8} \text{ Nm.}$$

$$\text{Moment of inertia of the joist, } I = \frac{bd^3}{12} = \frac{0.15 \times (0.30)^3}{12} = 3.375 \times 10^{-4} \text{ m}^4$$

$$\frac{I}{y} \times \sigma = \frac{M}{y} \quad \text{We know, } M = \frac{I \times \sigma}{y}$$

$$= 18000 \text{ Nm}$$

Equating this to the maximum bending moment

$$\frac{50000x \times 5^2}{8} = 18000$$

$$\therefore x = 0.576 \text{ m}$$

Problem 5.6 Compare the weight of two equally strong beams of circular sections made of the same material, one being of solid section and the other of hollow section with inside diameter $\frac{2}{5}$ of the external diameter.

Solution: For solid let the diameter be d

For hollow section: Let d_1 = Outside diameter

d_2 = Inside diameter

$$\therefore \frac{d_1}{d_2} = \frac{5}{2}$$

$$\text{For solid section, } Z_s = \frac{\pi d^3}{32}$$

$$\frac{\pi(d_1^4 - d_2^4)}{32d_1} = \frac{\pi d_1^3}{32} \left[1 - \left(\frac{2}{5} \right)^4 \right] = \frac{609\pi d_1^3}{32 \times 625}$$

$$\text{For hollow section, } Z_H =$$

$$\text{As both the beams are equally strong, } Z_s = Z_H \quad \frac{d_1^3}{d^3} = \frac{625}{609}$$

$$\therefore \frac{d_1}{d} = 1.00868$$

Both the beams are made of same material. Therefore $\rho_s = \rho_H = \rho$

$$\text{Let } W_s = \text{weight of solid shaft} = \text{volume} \times \text{density}$$

$$= \frac{\pi}{4} d^2 \times L \times \rho$$

$$W_H = \text{Weight of hollow shaft} = \text{volume} \times \text{density}$$

$$= \frac{\pi}{4} \left[d_1^2 - \left(\frac{2}{5} d_1 \right)^2 \right] \times L \times \rho$$

$$\therefore \frac{W_s}{W_H} = \frac{\frac{\pi}{4} d^2 \times L \times \rho}{\frac{\pi}{4} \left[d_1^2 - \left(\frac{2}{5} d_1 \right)^2 \right] \times L \times \rho} = \frac{d^2}{d_1^2 - \frac{4}{25} d_1^2}$$

$$\frac{W_s}{W_H} = \frac{25d^2}{21d_1^2} = \frac{25}{21} \times \frac{1}{(1.00868)^2} = 1.17$$

Therefore the weight of solid shaft is 1.17 times the weight of the hollow shaft.

Problem: 5.7 Fig. 5.10 shows an I-section proposed to be used for a beam of span 8 meters. The beam carries a uniformly distributed load of 100 kN/m over the whole span. Find the extreme bending stress for the section and state if the stress is permissible. If not, find the width of a 15 mm thick cover plate for each flange for the section to be safe. Find also the length of this cover plate. Safe stress in bending is 160 N/mm².

Solution: Moment of inertia of the I-section³

$$= I = \frac{250 \times 640^3}{12} - \frac{235 \times 600^3}{12} \text{ mm}^4$$

$$= 1.2313 \times 10^9 \text{ mm}^4$$

Maximum bending moment for the beam

$$= M = \frac{wl}{8} = \frac{100 \times 8}{8} = 800 \text{ kNm}$$

Extreme stress due to bending moment

$$= f = \frac{M}{I} y = \frac{800 \times 10^6}{1.2313 \times 10^9} \times 320 \text{ N/mm}^2$$

$$= 208 \text{ N/mm}^2$$

This is greater than the permissible stress of 160 N/mm². Suppose a cover plate of 15 mm thickness of provided for each flange, Increase in moment of inertia

$$= 2 \left[\frac{b \times 15^3}{12} + b \times 15 \times 327.5^2 \right]$$

$$= 3218250 b \text{ mm}^4$$

Moment of inertia of the modified section

$$= I' = 1.2313 \times 10^9 + 3218250 b \text{ mm}^4$$

Limiting the extreme stress to the permissible value of 160 N/mm²

$$f = 160 = \frac{M}{I'} y'$$

$$\frac{800 \times 10^6 \times 327.5}{I'}$$

$$\therefore I' = \frac{160}{160} = 1.63752 \times 10^9 \text{ mm}^4$$

$$\therefore 1.2313 \times 10^9 + 3218250 b = 1.63752 \times 10^9$$

$$\therefore b = 126.2 \text{ mm say } 130 \text{ mm}$$

Moment of resistance of the I section alone

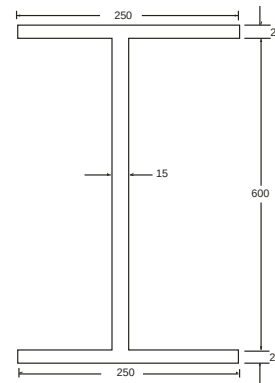


Fig. 5.10

$$= \frac{I}{y} \times f = \frac{1.2313 \times 10^9}{320} \times 160 = 6.1565 \times 10^8 \text{ Nmm} = 615.65 \text{ kNm}$$

Let the bending moment for the girder reach 615.65 kNm at a distance of x meter from the support

$$\therefore \frac{100 \times 10}{2} x - 100 \times \frac{x^2}{2} = 615.65$$

$$\therefore 50x^2 - 500x + 615.65 = 0$$

$$\therefore x^2 - 10x + 12.31 = 0 \quad \therefore 1.44 \text{ m from each end.}$$

Hence length of cover plate required

$$\therefore = 10 - 1 \times 1.44 = 7.124 \text{ m}$$

Problem 5.8 A timber beam is freely supported on supports 6 meters apart. It carried a uniformly distributed load of 12 kN per meter run and a concentrated load of 9 kN at 2.5 meters from the left support. If the stress in timber is not to exceed 10 N/mm², design a suitable section making the depth twice the width.

Solution: Fig. 5.11 shows the beam carrying the loading mentioned in the problem. Let R_a and R_b be the reactions at the left and right supports.

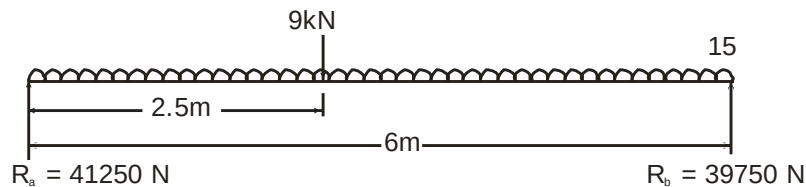


Fig. 5.11

Taking moments about the left support A, we have

$$R_b \times 6 = 12000 \times 6 \times 3 + 9000 \times 2.5$$

$$\therefore R_b = 39750 \text{ N}$$

$$\therefore R_a = 12000 \times 6 + 9000 - 39750 \text{ N} = 41250 \text{ N}$$

Let the shear force be zero at x meter from B

$$\therefore 39750 - 12000x = 0$$

$$\therefore x = \frac{39750}{12000} = 3.3125 \text{ m}$$

$\therefore$ Maximum bending moment occurs at 3.3125 meters from B

$$\therefore \text{Maximum bending moment} = 39750 \times 3.3125 - \frac{12000 \times (3.3125)^2}{2} \text{ Nm}$$

$$= 65836 \text{ Nm} = 65836 \times 1000 \text{ Nmm}$$

But the moment of resistance of the section

$$= \frac{1}{6} \sigma_b b d^2$$

Since $\sigma_b = 10 \text{ N/mm}^2$ and $b = \frac{d}{2}$, we have,

$$\frac{1}{6} \times 10 \times \frac{d}{2} \times d^2 = \frac{65836 \times 1000}{10}$$

$$d^3 = \frac{65836 \times 1000 \times 12}{10}$$

$$\begin{aligned} \therefore d &= 369.64 \text{ mm} \cong 370 \text{ mm} \\ \therefore b &= \frac{d}{2} = \frac{370}{2} = 184.8 \text{ mm} \cong 185 \text{ mm} \\ \therefore \text{The beam section is } 231 \text{ mm} \times 462 \text{ mm} \end{aligned}$$

Problem 5.9 Find the width x of the flange of a cast iron beam having the section shown in Fig. 5.12 so that the maximum compressive stress is three times the maximum tensile stress the member being in pure bending subjected to sagging moment. The depth h of the beam is 80 mm, the thickness t of the web and flange is 20 mm.

Solution: Let the distance of the centroidal neutral axis from the bottom edge be $\bar{y}$. Let σ_c and σ_t be the maximum compressive and maximum tensile stress

$$\therefore \frac{\sigma_c}{\sigma_t} = \frac{\bar{h} - \bar{y}}{\bar{y}} = 3$$

$$\therefore \frac{h - \bar{y}}{\bar{y}} = 3$$

$$\therefore \bar{y} = \frac{h}{4}$$

But from the geometry of the section,

$$\bar{y} = \frac{th \cdot \frac{h}{2} + (x-t)t \cdot \frac{t}{2}}{th + (x-t)t} = \frac{h}{4}$$

$$\therefore x = t + \frac{h^2}{h - 2t}$$

Putting, $t = 20 \text{ mm}$, and $h = 80 \text{ mm}$

$$\therefore x = 20 + \frac{80^2}{80 - 2 \times 20} \text{ mm}$$

$$\therefore x = 180 \text{ mm}$$

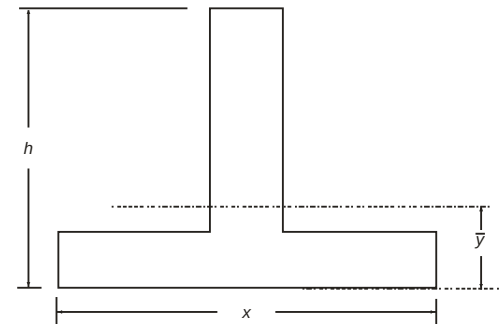


Fig. 5.12

Problem 5.10 A cast iron bracket subject to bending has cross-section of I-form with unequal flanges. The depth of the section is 280 mm and the metal is 40 mm thick throughout. The top flange is 200 mm wide and the bottom flange is 100 mm wide. Find the position of the neutral axis and the moment of inertia of the section about the neutral axis and determine the maximum bending moment that should be imposed on this section if the tensile stress in the top flange is not to exceed 20 N/mm^2 . What is then the value of the maximum compressive stress in the bottom flange?

Solution: Fig. 5.13 shows the section of the bracket. The section may be conveniently divided into three rectangular components. The areas of the individual components, their centroidal distance from the top edge and their moments about the top edge and their moments of inertia about their centroidal axes are tabulated below:

Component	Area a mm^2	Centroidal distance y from top edge y (mm)	ay (mm) ³	ay^2 (mm) ⁴	I_{self} (mm) ⁴
Top Flange	8000	20	16×10^4	320×10^4	$\frac{200 \times 40^3}{12} = 106.67 \times 10^4$
Web	8000	140	112×10^4	15680×10^4	$\frac{40 \times 200^3}{12} = 2666.67 \times 10^4$
Bottom Flange	4000	260	104×10^4	27040×10^4	$\frac{100 \times 40^3}{12} = 53.33 \times 10^4$
TOTAL	20000		232×10^4	43040×10^4	2826.67×10^4

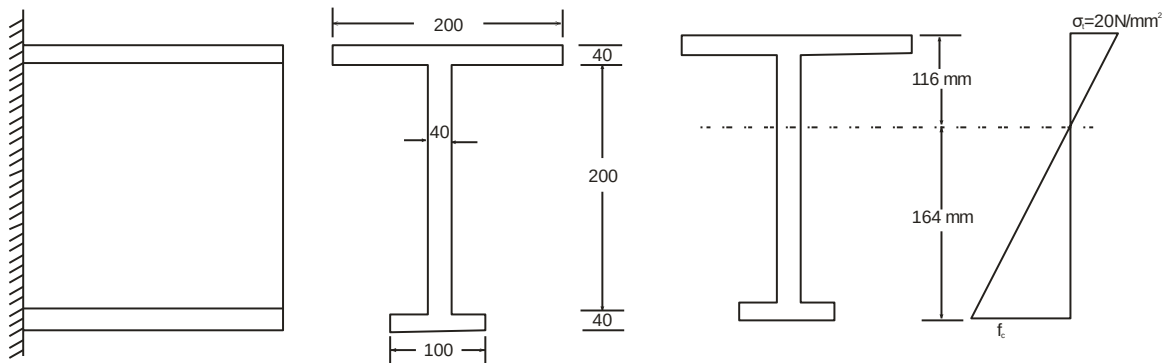


Fig. 5.13

$$\therefore \text{Distance of the neutral axis from the upper edge}$$

$$= \bar{y} = \frac{\sum ay}{\sum a} = \frac{232 \times 10^4}{20000} = 116 \text{ mm}$$

Moment of inertia about the upper edge

$$= I_{uu} = \sum I_{self} + \sum ay^2$$

$$2826.67 \times 10^4 + 43040 \times 10^4 = 45866.67 \times 10^4 \text{ mm}^4$$

But

$$I_{uu} = I_{xx} + (\sum a)\bar{y}^2$$

$$45866.67 \times 10^4 = I_{xx} + 20000 \times 116^2$$

$$\therefore I_{xx} = (45866.67 - 26912) \times 10^4 = 18954.67 \times 10^4$$

Let the maximum bending moment be in Nmm.

$$\frac{M}{I} = \frac{\sigma}{y_t}$$

$$\frac{\sigma_t}{y_t} \times I$$

$\therefore$

$$M = \frac{\sigma_t}{y_t} \times I \quad (\text{But } \sigma_t = 20 \text{ N/mm}^2)$$

$$= \frac{20}{116} \times 18954.67 \times 10^4 \text{ Nmm}$$

$$= 3268 \times 10^4 \text{ Nmm} = 32680 \text{ Nm}$$

Let the maximum compressive stress be $\sigma_c = \frac{164}{116} \times 20 \text{ N/mm}^2$

$$\therefore \sigma_c = 28.28 \text{ N/mm}^2$$

$$\therefore \sigma_c = 28.28 \text{ N/mm}^2$$

Problem 5.11 The following three beams have the same length and are of the same material and have same weight.

- (i) I-section 400 mm × 180mm with 20 mm thick flanges and 12.5 mm thick web.
- (ii) Rectangular section of depth twice its width.
- (iii) Solid circular section.

Compare the strength of the I-section with the rectangular and circular sections.

Solution: Obviously all the three beams have the same sectional sections area.

Properties of the I-section

$$\text{Area} = 2(180 \times 20) + (360 \times 12.5) = 11700 \text{ mm}^2$$

$$= 3.0876 \times 10^8 \text{ mm}^3$$

$$\text{Section modulus } Z_1 = \frac{3.0876 \times 10^8}{200}$$

$$= 1543800 \text{ mm}^3$$

Properties of the rectangular section

$$\text{Area} = 11700 \text{ mm}^2, \text{ Width} = b, \text{ depth} = d = 2b$$

$$\therefore \text{Area} = 2b^2 = 11700$$

$$\therefore b = 76.485 \text{ mm}$$

$$\text{and } d = 2 \times 76.485 = 152.97 \text{ mm}$$

$$\text{section modulus } Z_2 = \frac{bd^3}{6} = \frac{76.485 \times 152.97^3}{6} = 298289.22 \text{ mm}^3$$

Properties of circular section

$$\text{Area} = 11700 \text{ mm}^2, \text{ Diameter} = d$$

$$\therefore \text{Area} = \frac{\pi d^2}{4} = 11700 \quad \therefore d = 122.053 \text{ mm}$$

$$\text{Section modulus } Z_3 = \frac{\pi d^3}{32} = \frac{\pi \times 122.053^3}{32} = 178502.9 \text{ mm}^3$$

$$\text{Comparison of strength} = \frac{\text{Strength of I-section}}{\text{Strength of rectangular section}} = \frac{Z_1}{Z_2} = \frac{1543800}{298289.22} = 5.176$$

$$\frac{\text{Strength of I-section}}{\text{Strength of circular section}} = \frac{Z_1}{Z_3} = \frac{1543800}{178502.9} = 8.649$$

Exercise

5.1 Derive the expression for bending stress for a layer in a beam.

5.2 What do you mean by section modulus? Find an expression for section modulus for rectangular, circular and hollow circular section.

5.3 What is the meaning of 'strength of a section'?

5.4 What do you mean by 'simple bending' or 'pure bending'? What are the assumptions made in theory of simple bending?

5.5 A timber beam is 120 mm wide and 200 mm deep and is used on a span of 4m. the beam carries a udl of 2.8 kN/m run over the entire length. Find the maximum bending stress induced.

(Ans. 7 N/mm²)

5.6 A rectangular beam 10 m long is simply supported at its ends and is acted upon by a 10 kN/m udl throughout its length. Find the size of the beam if the thickness is twice that of its width. Take the allowable bending stress as 40 N/mm².

(Ans. $b = 167.3$ mm, $d = 334.6$ mm)

5.7 A rectangular beam 300mm deep is simply supported over a span of 4m. what udl per meter the beam may carry if the bending stress is not to exceed 120 N/mm². Take $I = 8 \times 10^6$ mm⁴.

(Ans. 3.2 kN/m)

5.8 For a given stress compare the moments of resistance of a beam of square section when placed (a) with its two sides horizontal (ii) with its diagonal horizontal.

(Ans. 1.414)

5.9 A 10m long C.I. pipe of 500mm internal diameter and 20 mm thickness is simply supported at its ends. Find the maximum intensity of bending stress induced in the metal when the pipe is running full with water. Take unit weights of C.I. and water as 70 kN/mm³ and 10 N/mm³.

(Ans. 13.17 N/mm²)

5.10 A beam of I-section 400mm deep has flanges 200 mm wide and 30 mm thick and web 24 mm thick. Compare the flexural strength with that of a beam rectangular section of the same weight, depth being twice the breadth.

(Ans. 3.27)

Problem 12.1 A horizontal cantilever of uniform section of length l carries two point loads, W at the free end and $2W$ at a distance of a from the free end. Find the maximum deflection due to this loading. If the cantilever is a steel tube of circular section 100 mm external diameter and 6 mm thick and $l=1.50$ meters and $a = 0.60$ meters determine the value of W so that the maximum bending stress is 140 N/mm^2 and calculate the maximum deflection for the loading. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution: Deflection at the free end due to the load W alone

$$= \frac{Wl^3}{3EI}$$

Deflection at the free end due to the load $2W$ alone

$$= \frac{2W(l-a)^3}{3EI} + \frac{2W(l-a)a^2}{2EI}$$

$$\therefore \text{Total deflection at the free end } y = \frac{Wl^3}{3EI} + \frac{2W(l-a)^3}{3EI} + \frac{W(l-a)^2 a}{EI}$$

$$y = \frac{3EI}{W} [l^3 + 2(l-a)^3 + 3(l-a)^2 a]$$

$$= \frac{3EI}{W} [l^3 + 2l^3 - 6l^2a + 6la^2 - 2a^3 + 3l^2a - 6la^2 + 3a^3]$$

$$= \frac{3EI}{W} [3l^3 - 3l^2a + a^3]$$

$$\therefore y = \frac{3EI}{W} [3l^2(l-a) + a^3]$$

For the numerical

$$\text{Max B.M. } M = Wl + 2W(l-a)$$

$$W(3l-2a) = W(3 \times 1.5 - 2 \times 0.6) \text{ Nm} = 3.3 W \text{ Nm.}$$

$$I = \frac{\pi}{64} (100^4 - 88^4) \text{ mm}^4 = 1.965 \times 10^6 \text{ mm}^4$$

$$\text{Stress due to max. B.M. } = \sigma = \frac{M}{I} y = 140 \text{ N/mm}^2$$

$$\therefore \frac{3.3W \times 1000}{1.965 \times 10^6} \times 50 = 140$$

$$\therefore W = \frac{140 \times 1.965 \times 10^6}{3.3 \times 1000 \times 50} \text{ N} = 1667 \text{ N.}$$

$$\text{Max deflection } y = \frac{3EI}{1667} [3l^2(l-a) + a^3]$$

$$= \frac{3 \times 2.1 \times 10^5 \times 1.965 \times 10^6}{1667} [3 \times 2.25 \times 0.9 + 0.216] \times 1000^3 \text{ mm}$$

$$= 8.47 \text{ mm.}$$

Problem 12.2 A cantilever of length 2 meters carries a uniformly distributed load of 5000 N per meter for a length of 1.25 meter from the fixed end and a point load of 1000 N at the free end. If the section is rectangular 120 mm wide and 240 mm deep, find the deflection at the free end. Take $E = 10000 \text{ N/mm}^2$.

Solution: Moment of inertia of the section

$$= I = \frac{bd^3}{12} = \frac{120 \times 240^3}{12}$$

$$= 1.3824 \times 10^8 \text{ mm}^4$$

Deflection at C due to uniformly distributed load on AB

$$= \frac{wd^4}{8EI} + \frac{wd^4}{6EI}(l-a)$$

$$= \frac{5000 \times 1.25^4}{8EI} + \frac{5000 \times 1.25^3 \times 0.75}{6EI} = \frac{2146}{EI}$$

Deflection at C due to point load at C

$$= \frac{wl^3}{3EI} = \frac{2000 \times 2^3}{3EI} = \frac{5334}{EI}$$

$$\therefore \text{Net deflection at C} = \frac{2146}{EI} + \frac{5334}{EI} = \frac{8080}{EI} = \frac{8080(1000)^3}{10^4 \times 1.3824 \times 10^8} \text{ mm}$$

$$= 5.844 \text{ mm}$$

Problem: 12.3 A cantilever beam 120 mm wide and 20 mm deep is 2.5 m long. What uniformly distributed load should the beam carry to produce a deflection of 5 mm at the free end and what concentrated load at free end would produce the same deflection? give the slope at the free end in both cases. Ignore self-weight of the beam. Take $E = 2 \times 10^5 \text{ N/mm}^2$.

Solution: Let I = moment of inertia of the rectangular section about centroidal axis =

$$12 \times 120 \times 200^3 = 8000 \times 10^4 \text{ mm}^4$$

Maximum deflection at the free end of the cantilever

$$\frac{wl^4}{8EI} = 5$$

$$\frac{w \times (2500)^4}{8 \times 2 \times 10^5 \times 8 \times 10^7} = 5$$

$$\therefore w = 16.384 \text{ N/mm}$$

$$\text{Slope at the free end} = \frac{wL^3}{6EI} = \frac{16.384 \times 2500^3}{6 \times 2 \times 10^5 \times 8 \times 10^7} = 2.666 \times 10^{-3} \text{ rad.}$$

Let W Newton be the concentrated load that can be placed under the given conditions of the problem.

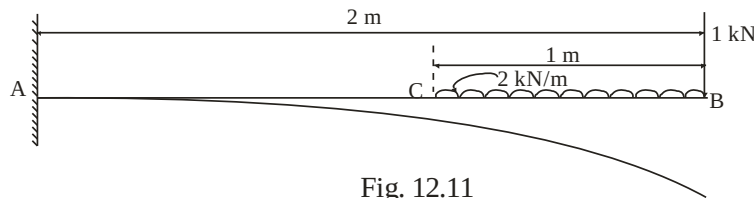
$$\text{Maximum deflection at the free end} = \frac{wL^3}{3EI} = 5$$

$$\text{or } \frac{W \times 2500^3}{3 \times 2 \times 10^5 \times 8 \times 10^7} = 5$$

$$\text{Slope at the free end} = \frac{wL^2}{2EI} = \frac{W}{2 \times 2 \times 10^5 \times 8 \times 10^7} = 0.003 \text{ rad.}$$

Problem 12.4 A cantilever 100 mm wide and 200 mm deep projects 2 m from a wall into which it is cast. The cantilever carries a u.d.l. of 2 kN/m over a length of 1 m from the free end and a point load of 1 kN at the free end. Find the slope and deflection at the free end. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution: Here, as in Fig. 12.11



$$L = AB = 2\text{m}$$

$$L_1 = AC = 1\text{m}$$

$$W = 1 \text{ kN}, w = 2 \text{ kN/m}$$

Moment of inertia of the beam section,

$$I = \frac{1}{12} \times 100 \times (200)^3 = \frac{2}{3} \times 10^8 \text{ mm}^4.$$

Slope at the free end B is given by

$$\begin{aligned} i_B &= \frac{WL^2}{2EI} + \left[\frac{wL^3}{6EI} - \frac{w(L-L_1)^3}{6EI} \right] = \frac{1}{EI} \left[\frac{WL^2}{2} + \frac{wL^3}{6} - \frac{w(L-L_1)^3}{6} \right] \\ &= \frac{1}{2.1 \times 10^5 \times \frac{2}{3} \times 10^8} \times \left[\frac{2 \times 10^3 \times 2000^2}{2} + \frac{2 \times 2000^3}{6} - \frac{2 \times (2-1)^3 \times 1000^3}{6} \right] \end{aligned}$$

$$= 0.000309 \text{ radian}$$

$$W = 10^3 \text{ N}$$

$$L = 2000 \text{ mm} \quad L_1 = 1000 \text{ mm}$$

$$w = 2 \text{ N/mm}$$

Deflection at the free end B is given by

$$y_B = \frac{WL^3}{3EI} + \frac{wL^4}{8EI} - \left[\frac{w(L-L_1)^3}{8EI} + \frac{w(L-L_1)^4}{6EI} \right]$$

$$= \frac{10^3 \times 2^3 \times 3 \times 10^4}{3 \times 210 \times 10^9 \times 2} + \frac{2 \times 10^3 \times 2^4 \times 3 \times 10^4}{8 \times 210 \times 10^9 \times 2}$$

$$- \left[\frac{2 \times 10^3 \times (2-1)^4 \times 3 \times 10^4}{8 \times 210 \times 10^9 \times 2} + \frac{2 \times 10^3 \times (2-1)^3 \times 3 \times 10^4 \times 1}{6 \times 210 \times 10^9 \times 2} \right]$$

$$= 0.00045 \text{ m} = 0.45 \text{ mm},.$$

Problem 12.5 A simply supported beam is loaded as shown in Fig 12.12. Determine the deflection at points C and D in the beam. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 2 \times 10^8 \text{ mm}^4$.

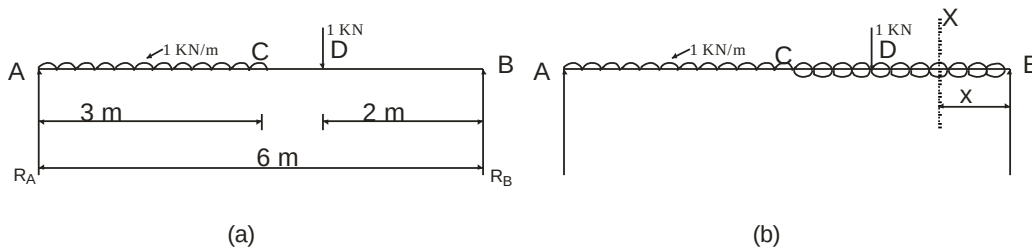


Fig.12.12

Solution: Taking moment about B

$$R_A \times 6 = 1000 \times 3 \times 4.5 + 1000 \times 2$$

$$\therefore R_A = \frac{15500}{6} \text{ N}$$

On portion BC a u.d.l. of 1 kN/m is introduced both above and below so that these loads neutralise each other and the net effect remains unchanged. Consider a section X-X at a distance x from left end A.

$$M = \frac{15500x}{6} - \frac{1000x^2}{2} - 1000(x-4) + \frac{1000}{2}(x-3)^2$$

$$EI \frac{d^2 y}{dx^2} = -M = -\frac{15500x}{6} + \frac{1000x^2}{2} + 1000(x-4) + \frac{1000}{2}(x-3)^2$$

Integrating the above equation for slope, we get

$$EI \frac{dy}{dx} = -\frac{15500x^2}{12} + \frac{1000x^3}{6} + \frac{1000(x-4)^2}{2} + \frac{1000(x-3)^3}{6} + C_1$$

Integrating again for deflection, we have

$$EI y = -\frac{15500x^3}{36} + \frac{1000x^4}{24} + \frac{1000(x-4)^3}{6} + \frac{1000(x-3)^4}{24} + C_1 x + C_2$$

At $x = 0$, deflection is zero, i.e. $y = 0 \quad \therefore C_2 = 0$

At $x = 6$, deflection is zero, i.e. $y = 0 \quad \therefore C_1 = 6.84 \times 10^3$

Therefore the final equation for slope is

$$EI \frac{dy}{dx} = -\frac{15500x^2}{12} + \frac{1000x^3}{6} + \frac{1000(x-4)^2}{6} + \frac{1000(x-3)^4}{24} + 6.84 \times 10^3$$

The final equation for deflection is

$$EI y = -\frac{15500x^3}{36} + \frac{1000x^4}{24} + \frac{1000(x-4)^3}{6} + \frac{1000(x-3)^4}{24} + 6.84 \times 10^3 x$$

For deflection at C, put $x = 3$

$$y_c = \frac{1}{EI} \left[-\frac{15500 \times 3^3}{36} + \frac{1000 \times 3^4}{24} + 6.84 \times 10^3 \times 3 \right] = \frac{12.27 \times 10^3}{EI}$$

$$= \frac{12.27 \times 10^3}{2 \times 10^5 \times 2 \times 10^8} = 0.3 \text{ mm}$$

For deflection at D, put $x = 4$

$$y_D = \frac{1}{EI} \left[-\frac{15500 \times 4^3}{36} + \frac{1000 \times 4^4}{24} - \frac{1000}{24} + 6.84 \times 10^3 \times 4 \right]$$

$$= \frac{10.429 \times 10^3}{EI} = \frac{10.429 \times 10^3}{2 \times 10^5 \times 2 \times 10^8} = 0.26 \text{ mm.}$$

Problem 12.6 A rolled steel beam having a span of 6 meters carries a point load of 80 kN at 4 meters from the left support. Find the deflection under the load and the position and amount of maximum deflection.

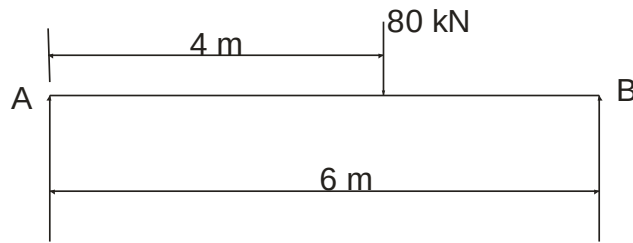


Fig.12.13

I_{xx} for the section = $7.33 \times 10^7 \text{ mm}^4$. Take $E = 200 \text{ kN/mm}^2$.

Solution: Fig. 12.13 shows the beam carrying the given loading.

Bigger segment = $AC = a = 4 \text{ m}$.

Smaller segment = $BC = b = 2 \text{ m}$.

$$\begin{aligned} \text{Deflection under the load} &= y_c = \frac{W a^2 b^2}{3EI l} \\ &= \frac{80 \times 4^2 \times 2^2}{3 \times 200 \times 7.33 \times 10^7 \times 6 (1000)^3} \text{ mm} = 19.4 \text{ mm} \end{aligned}$$

Maximum deflection occurs at $\sqrt{\frac{l^2 - b^2}{3}} = \sqrt{\frac{36 - 4}{3}} = 3.27 \text{ meters from the left end.}$

$$\begin{aligned} \text{Max. deflection} &= y_{\max} = \frac{Wb(l^2 - b^2)^{3/2}}{9\sqrt{3} EI l} \\ &= \frac{80 \times 2(32)^{3/2} (1000)^3}{9\sqrt{3} (200 \times 7.33 \times 10^7 \times 6)} = 21.12 \text{ mm} \end{aligned}$$

Problem 12.7 A horizontal girder of steel having uniform section is 14 meters long and is simply supported at its ends. It carries concentrated loads of 120 kN and 80 kN at two points 3 meters and 4.5 meters from the two ends respectively. I for the section of the girder is $32 \times 10^8 \text{ mm}^4$, and $E_s = 210 \text{ kN/mm}^2$. Calculate the deflections of

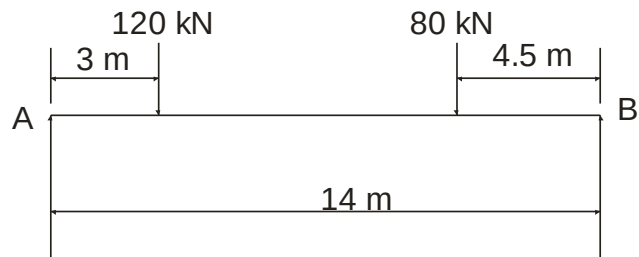


Fig.12.14

the girder at points under the two loads.

Find also the maximum deflection.

Solution: Let R_a and R_b be the vertical reactions at the supports A and B.

Taking moments about A, we have

$$R_b \times 14 = 120 \times 3 + 80 \times 9.5$$

$$\therefore R_b = 80 \text{ kN} \quad R_a = 200 - 80 = 120 \text{ kN}$$

The B.M. at any section distance x from A is given by,

$$\frac{d^2 y}{EI dx^2} = 120x \Big| -120(x-3) \Big| -80(x-9.5)$$

Integrating, we get,

$$\frac{dy}{EI dx} = 60x^2 + C_1 \Big| -60(x-3)^2 \Big| -40(x-9.5)^2$$

$$\text{Integrating again, we get} \quad EIy = 20x^3 + C_1x + C_2 \Big| -20(x-3)^3 \Big| \frac{40}{3} (x-9.5)^3$$

$$\text{At } x = 0$$

$$y = 0$$

$$\therefore C_2 = 0$$

$$\text{At } x = 14,$$

$$y = 0$$

$$\therefore 0 = 20(14)^3 + 14 C_1 - 20(14-3)^3 - \frac{40}{3} (14-9.5)^3$$

$$\therefore 14 C_1 = -27045 \quad \therefore C_1 = -1931.78$$

Hence the deflection at any section is given by

$$EIy = 20x^3 - 1931.78x \Big| -20(x-3)^3 \Big| \frac{40}{3} (x-9.5)^3$$

To find the deflection at C, put $x = 3\text{m}$ in the deflection equation

$$EIy_c = 20 \times 3^3 - 1931.78 \times 3 = -5255.34$$

$$y_c = -\frac{5255.34}{210 \times 32 \times 10^8 (1000)^3} = -7.82 \text{ mm}$$

$$y_c = 7.82 \text{ mm (downward equation)}$$

to find the deflection at D, put $x = 9.5 \text{ m}$ in the deflection equation

$$EIy_d = 20(9.5)^3 - 1931.78 \times 9.5 - 20(9.5-3)^3$$

$$EIy_d = - \frac{6696.91}{6696.91}$$

$$y_d = - \frac{210 \times 32 \times 10^8}{(1000)^3} \text{ mm} = - 9.965 \text{ mm}$$

$$y_d = 9.965 \text{ (downward deflection)}$$

Maximum deflection: Let us assume that the deflection will be maximum at a section between C and D.

Equating the slope at the section to zero, we have,

$$EI \frac{dy}{dx} = 60x^2 - 1931.78 - 60(x - 3)^2 = 0$$

Solving, we get $x = 6.87 \text{ m.}$

Substituting in the deflection equation, we get,

$$EIy_{\max} = 20(6.87)^3 - 1931.78 \times 6.87 - 20(6.87 - 3)^3$$

$$EIy_{\max} = - \frac{7945.7}{7945.7}$$

$$y_{\max} = - \frac{210 \times 32 \times 10^8}{(1000)^3} \text{ mm} = - 23.6 \text{ mm}$$

i.e. $y_{\max} = 11.8 \text{ mm (downward deflection)}$

Problem 12.8 For the loaded beam shown in Fig. 12.15 find (i) the slopes at A, C and D (ii) the deflection at C and D and (iii) the maximum deflection. Take $E = 200 \text{ kN/mm}^2$ and $I = 9.25 \times 10^7 \text{ mm}^4$.

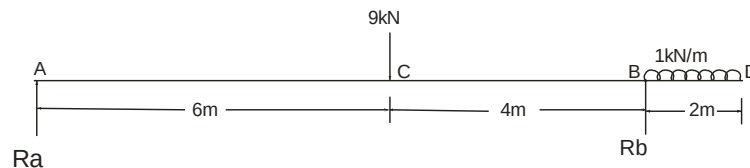


Fig. 12.15

Solution: Taking moments about A, $R_b \times 10 = (9 \times 6) + (1 \times 2 \times 11) \quad \therefore R_b = 7.60 \text{ kN}$

$$\therefore R_a = 11 - 7.60 = 3.40 \text{ kN}$$

The equation of the deflected shape of the beam is given by,

$$EI \frac{d^2 y}{dx^2} = \text{B.M.}$$

$$EI \frac{d^2 y}{dx^2} = 3.40x | - 9 (x - 6) | + 7.60(x - 10) - \frac{1}{2} (x - 10)^2$$

$$\frac{1.70}{3}$$

Integrating again, $EIy = \frac{1.70}{3} x^3 + C_1x + C_2 | - 1.50(x - 6)^3 | + (x - 10)^3 - \frac{1}{24} (x - 10)^4$

At $x = 0, y = 0 \quad \therefore C_2 = 0$

At $x = 10, y = 0$

$\therefore 0 = \frac{1.70}{3} \times 10^3 + 10C_1 - 1.50 \times 4^3 \quad \therefore C_1 = -47.067$

Hence, the slope and deflection equations are,

$$EI \frac{dy}{dx} = 1.70 x^2 - 47.067x | - 4.50 (x - 6)^2 | + 3.80 (x - 10)^2 - \frac{1}{6} (x - 10)^3 \dots \text{(slope equation)}$$

$$EIy = \frac{1.70}{3} x^3 - 47.067x | - 1.50 (x - 6)^3 | + \frac{3.80}{3} (x - 10)^3 - \frac{1}{24} (x - 10)^4 \dots \text{(deflection equation)}$$

Slope at A: Put $x = 0$ in the slope equation,

$$EIi_a = -47.067 \quad i_a = \frac{-47.067}{200 \times 9.25 \times 10^7 \times 1000^2} = -0.002544 \text{ radian}$$

Slope at C: Put $x = 6$ m in the slope equation,

$$EIi_c = 1.70 \times 6^2 - 47.067 = 14.133$$

$$i_c = \frac{14.133}{200 \times 9.25 \times 10^7 \times 1000^2} = 0.0007639 \text{ radian}$$

Slope at D: Put $x = 12$ m in the slope equation,

$$EIi_d = 1.70 \times 12^2 - 47.067 - 4.50 \times 6^2 + 3.80 \times 2^2 - \frac{1}{6} \times 2^3 = 49.60$$

$$i_d = \frac{49.60}{200 \times 9.25 \times 10^7 \times 1000^2} = 0.002681 \text{ radian}$$

Deflection at C: Put $x = 6$ m in the deflection equation,

$$EIy_c = \frac{1.70}{3} \times 6^3 - 47.067 \times 6 - 160.002 \quad \therefore y_c = - \frac{160.002}{200 \times 9.25 \times 10^7 \times 1000^3} = -8.65 \text{ mm}$$

Deflection at D: Put $x = 12$ m in the deflection equation.

$$\therefore y_d = \frac{99.863}{200 \times 9.25 \times 10^7 \times 1000^3} = 5.4 \text{ mm}$$

Maximum deflection

Assuming the deflection to be maximum at a section between general expressions for slope to zero,

$$0 = 1.70 x^2 - 47.067 \quad \therefore x^2 = \frac{47.067}{1.7}$$

$$\therefore x = 5.262 \text{ m}$$

(Since this distance is less than 6m, our assumption about the maximum deflection is correct). Putting $x = 5.262 \text{ m}$ in the deflection.

$$EIy_{\max} = \frac{1.70}{3} \times 5.262^3 - 47.067 \times 5.26$$

$$\therefore y_{\max} = - \frac{165.105}{200 \times 9.25 \times 10^7 \times 1000^3} = -8.925 \text{ mm}$$

Problem 12.9 Fig. 12.16 (a) shows a simply supported beam of uniform section whose moment of inertia is $4.3 \times 10^8 \text{ mm}^4$. For the loading shown, find the position and magnitude of the maximum deflection. Take $E = 210 \text{ kN/mm}^2$.

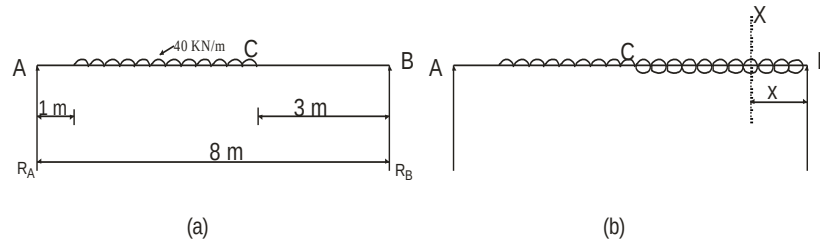


Fig.12.16

Solution: Taking moments about the end A, we have

$$R_b \times 8 = 40 \times 4 \times 3$$

$$R_b = 60 \text{ kN}$$

$$R_a = 40 \times 4 - 60 \text{ kN} = 100 \text{ kN}$$

In order the general expression for the bending moment at any section at any section may be expressed in the form suitable for application of Macaulay's Method the loading on the beam is arranged as shown in Fig. 12.16 (b).

Now, following Macaulay's method, the B.M. at any section distant x from A is given by,

$$EI \frac{d^2 y}{dx^2} = 100x \left| -\frac{40(x-1)^2}{2} \right| + \frac{40(x-5)^2}{2}$$

$$EI \frac{d^2 y}{dx^2} = 100x \left| -20(x-1)^2 \right| + 20(x-5)^2$$

Integrating, we get, $EI \frac{dy}{dx} = 50x^2 + C_1 \left| -\frac{20}{3}(x-1)^3 \right| + \frac{20}{3}(x-5)^3$

Integrating again, we get $EI y = \frac{50}{3}x^3 + C_1 x + C_2 \left| -\frac{5}{3}(x-1)^4 \right| + \frac{5}{2}(x-5)^4$

At A, the deflection is zero

At $x = 0, y = 0$

$\therefore C_2 = 0$

At B also the deflection is zero,

$\therefore$ At $x = 8, y = 0$

$$0 = \frac{50 \times 8^3}{3} + 8C_1 - \frac{5 \times 7^4}{3} + \frac{5 \times 3^4}{3}$$

$\therefore C_1 = -583.33$

Position of maximum deflection

Let us assume that the deflection will be maximum between C and D.

Equating the slope to zero, we have,

$$50x - 583.33 - \frac{20}{3}(x-1)^3 = 0$$

or $15x^2 - 175 = 2(x-1)^3$

solving the above equation by trial and error we get $x = 3.82$ meters.

The value of x obtained confirm that our assumption about the position of maximum deflection is correct.

Substituting $x = 3.82$ meters in the deflection equation, we get

$$EI y_{\max} = \frac{50}{3} \times 3.82^3 - 583.33 \times 3.82 - \frac{5}{3} \times 2.82^4 = -1404.67$$

$\therefore y_{\max} = -\frac{1404.67}{210 \times 4.3 \times 10^6} \times (1000)^3 \text{ mm} = -15.55 \text{ mm}$

Problem 12.10 A beam has supports l apart with equal overhangs $\frac{l}{3}$ over each support. It carries a load W at each end and a load $2W$ at the centre. Find the slopes at the free end and at the support. Find also the deflections at the free end and at the center.

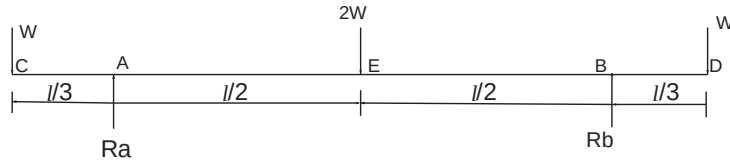


Fig. 12.17

Solution: Fig. 12.17 shows the beam subjected to the given load system.

By symmetry, $R_a = R_b = 2W$

We will consider one half of the beam length.

At any section distance x from the free end C, the bending moment is given by,

$$EI \frac{d^2 y}{dx^2} = -Wx \left| + 2W \left(x - \frac{l}{3} \right) \right.$$

Integrating, $EI \frac{dy}{dx} = -\frac{Wx^2}{2} + C_1 \left| + W \left(x - \frac{l}{3} \right)^2 \right.$

$$\text{At } x = \frac{5l}{6}, \frac{dy}{dx} = 0$$

$$\therefore 0 = -\frac{25}{72}Wl^2 + C_1 + \frac{Wl^2}{4} \quad \therefore C_1 = \frac{7}{72}Wl^2$$

$$\therefore EI \frac{dy}{dx} = -\frac{Wx^2}{2} + \frac{7}{72}Wl^2 \left| + W \left(x - \frac{l}{3} \right)^2 \right. \quad \dots (\text{Slope equation})$$

Integrating again, $EI y = -\frac{Wx^3}{6} + \frac{7}{72}Wl^2 x + C_2 \left| + \frac{W}{3} \left(x - \frac{l}{3} \right)^3 \right.$

$$\text{At } x = \frac{l}{3}, y = 0$$

$$\therefore 0 = -\frac{Wl^3}{162} + \frac{7}{216}Wl^3 + C_2 \quad \therefore C_2 = -\frac{17}{648}Wl^3$$

$$\therefore EI y = - \left[\frac{Wx^3}{6} + \frac{7}{72} Wl^2 x - \frac{17}{648} Wl^3 \right] + \frac{W}{3} \left(x - \frac{l}{3} \right)^3 \quad (\text{deflection equation})$$

Slope at C: Putting $x = 0$ in the slope equation,

$$EI i_c = \frac{7}{72} Wl^2 \quad \therefore i_c = \frac{7}{72} \frac{Wl^2}{EI}$$

Slope at A: Putting $x = \frac{l}{3}$ in the slope equation,

$$EI i_a = - \left[\frac{Wl^2}{18} + \frac{7}{72} Wl^2 \right] = - \frac{Wl^2}{24}$$

$$\therefore i_a = - \frac{Wl^2}{24EI}$$

Deflection at C: Putting $x = 0$ in the deflection equation,

$$EI y_c = - \frac{17}{648} Wl^3 \quad \therefore y_c = - \frac{17}{648} \frac{Wl^3}{EI}$$

Deflection at E: Putting $x = \frac{5l}{6}$ in the deflection equation,

$$EI y_e = - \left[\frac{W}{6} \left(\frac{5l}{6} \right)^3 + \frac{7}{72} Wl^2 \frac{5l}{6} - \frac{17}{648} Wl^3 + \frac{W}{3} \frac{l^3}{8} \right]$$

$$= \left[-\frac{125}{1296} + \frac{35}{432} - \frac{17}{468} + \frac{1}{25} \right] Wl^3 = 0$$

$$\therefore y_e = 0$$

Alternative Solution:

The above problem can also be solved easily by the principal of super-position.

First load system : Loads W each at the ends C and D only

Second load system ; Load $2W$ at the centre only.

Analysis due to the first load system

In this case it may be noted that the part of the beam AB between the supports is subjected to a constant bending moment $M = \frac{Wl}{3}$ Hence due to this load system,

$$\text{Slope at A} = \frac{ML}{2EI} = \frac{Wl}{3} \cdot \frac{l}{2EI} = \frac{Wl^2}{6EI} \quad (\text{anticlockwise})$$

Slope at C = Slope at A + Slope at C consideration AC as a cantilever fixed at A

$$\frac{Wl^2}{6EI} + W\left(\frac{l}{3}\right)^2 \frac{1}{2EI} = \frac{Wl^2}{6EI} + \frac{Wl^2}{18EI} = \frac{2}{9} \cdot \frac{Wl^2}{EI} \quad (\text{anticlockwise})$$

Deflection at C = Deflection at C due to slope at A + Deflection at C considering AC as a cantilever fixed at A

$$= \frac{Wl^2}{6EI} \cdot \frac{l}{3} + \frac{W}{3EI} \left(\frac{l}{3}\right)^3 = \frac{Wl^3}{18EI} + \frac{Wl^3}{81EI} = \frac{11}{162} \frac{Wl^3}{EI} \downarrow$$

$$\text{Deflection at the center} = \frac{Wl^2}{8EI} = \frac{Wl}{3} \cdot \frac{l^2}{8EI} = \frac{Wl^3}{24EI} \uparrow$$

Analysis due to the second load system

In this case,

$$\text{Slope at A} = \text{Slope at C} = \frac{2W \cdot l^2}{16EI} = \frac{Wl^2}{8EI} \quad (\text{clockwise})$$

$$\text{Deflection at C} = \frac{Wl^2}{8EI} \times \frac{l}{3} = \frac{Wl^3}{24EI} \uparrow$$

$$\text{Deflection at the center} = \frac{2Wl^3}{48EI} = \frac{Wl^3}{24EI} \downarrow$$

Hence,

$$\text{Net slope at A} = \frac{Wl^2}{6EI} - \frac{Wl^2}{8EI} = \frac{Wl^2}{24EI} \quad (\text{anticlockwise})$$

$$\text{Net slope at C} = \frac{2}{9} \frac{Wl^2}{EI} - \frac{Wl^2}{8EI} = \frac{7}{72} \frac{Wl^2}{EI} \quad (\text{anticlockwise})$$

$$\text{Net deflection at the centre} = \frac{Wl^3}{24EI} - \frac{Wl^3}{24EI} = 0$$

Exercise

12.1 Derive expression for the slope and deflection of a beam subjected to uniform bending moment.

12.2 What is Macaulay's method? Where is it used? Find an expression for the deflection of a simply supported beam which is subjected to eccentric point load using this method.

12.3 What are the different methods of finding slope and deflection of a cantilever?

12.4 A wooden beam 4 m long simply supported at its ends carries a point load of 7.25 kN at its centre. The cross section of the beam is 140 mm wide and 240 mm deep. If E for the beam is $6 \times 10^3 \text{ N/mm}^2$, find the deflection at the centre.

(Ans. 10 mm)

12.5 A beam of length 10 m is simply supported at its ends carries two point loads of 100 kN and 60 kN at a distance of 2 m and 5 m respectively from the left support. Calculate the deflection under each load. Find also the maximum deflection. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 18 \times 10^8 \text{ mm}^4$.

(Ans. -4.35, -6.76, -6.78 mm)

12.6 A beam of length 20 m is simply supported at its ends carries two point loads of 4 kN and 10 kN at a distance of 8 m and 12 m respectively from the left support. Calculate the deflection under each load. Find also the maximum deflection. Take $E = 2 \times 10^6 \text{ N/mm}^2$ and $I = 1 \times 10^9 \text{ mm}^4$.

(Ans. -10.3, -10.6, -11 mm)

12.7 A cantilever beam of length 3 m carries a point load of 60 kN at a distance of 2 m from the fixed end. Calculate the slope and deflection at the free end. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 1 \times 10^8 \text{ mm}^4$.

(Ans. 0.006 radian, 14 mm)

12.8 A cantilever beam of length 3 m carries a udl over the entire length. If the slope at the free end is 0.01777 radians calculate the deflection at the free end.

(Ans. 39.99 mm)

12.9 A cantilever beam of length 4 m carries a udl 12 kN/m over a length of 3 m from the fixed end. Calculate the slope and deflection at the free end. Take $EI = 2 \times 10^{13} \text{ Nmm}^2$.

(Ans. 0.0027 radian, 8.775 mm)

12.10 A beam of length 4.8 m and of rectangular section is simply supported at its ends carries a udl of 9.375 kN/m over the entire length. Calculate the width and depth of the beam if permissible bending stress is 7 N/mm^2 and the maximum deflection should exceed 9.5 mm. Take $E = 1.05 \times 10^4 \text{ N/mm}^2$.

(Ans. 240, 336.8 mm)

10. COLUMNS AND STRUTS

10.1 Introduction:

In many incidents we see the members subject to compressive loads such as truss, building structures etc. A structural member subjected to axial compressive force is called a strut. A strut may be horizontal or inclined. If the member is vertical & subjected to axial compressive force is termed as column, pillar or stanchion.

10.2 Axially Loaded Compressive Members: Elastic Stability:

When a column or strut is subjected to some compressive force, then the compressive stress induced

$$\sigma = \frac{P}{A} \quad \dots(10.1)$$

Where P = Compressive Force

A = Cross – Sectional area of the column.

If the column is short it fails by compressive stress, but if the column is long and subjected to a compressive load it fails under bending or buckling. The load at which the column just buckles is called buckling load, critical load or crippling load and the column is said to have an elastic instability. Equilibrium of an absolutely original body may be stable, neutral and unstable.

If a long column is subjected to an axial force P (less than a certain critical load P_{cr}) it will compress a little and will remain straight. The column is in a condition of stable equilibrium.

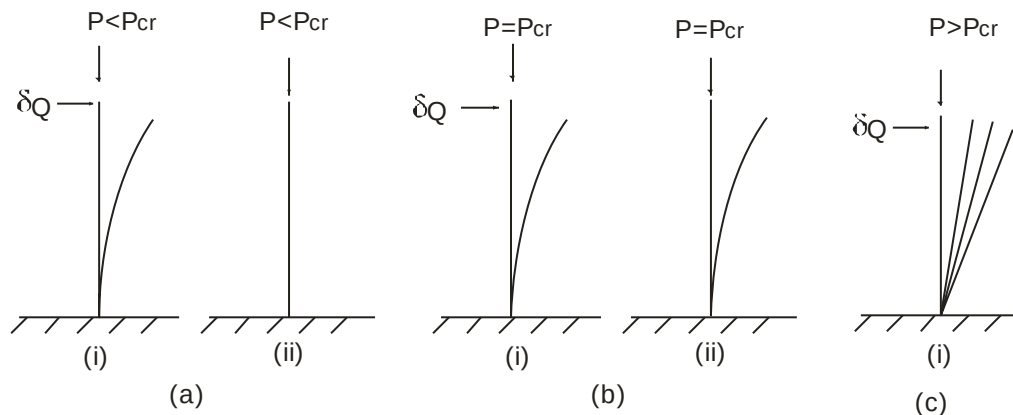


Fig. 10.1

Fig. 10.1(a) If it is deflected slightly by some transverse load δQ and then the load is removed the column become straight again and will assume the original condition. The Column is said to be in stable condition. For the case Fig. 10.1(b) the applied load is equal to the critical load. If the lateral load δQ is applied and removed after deflection, the column remains in deformed position after removal of δQ . This instantaneous condition is condition of Neutral Equilibrium. Finally if the load is increased beyond the critical load value, the straight configuration of equilibrium will become unstable and the column will continue to deform after the removal of the lateral load Fig. 10.1(c).

This instability is to be avoided in the design and hence knowledge of the critical load for the member becomes necessary.

10.3 Classification Of Columns:

Depending upon the ratio between the length and diameter column and slenderness ratio (ratio of length of the column to the minimum radius of gyration of the cross sectional area of the column) can be divided into three classes:

(a) Short Column:

Columns which have length less than 8 times their diameter or slenderness ratio is less than 32 are called short column or stocky struts. In short column buckling is negligible and hence short columns are always subjected to direct compressive stresses only.

(b) Medium Size Column:

Columns which have length diameter l/d ratio 8 to 30 or slenderness ratio 32 to 120 are called medium size columns. In this case buckling and direct stresses both are to be taken into account while designing.

(c) Long Columns:

Column which have length diameter ratio l/d more than 30 or slenderness ratio less than 12 are called long columns. In this case direct compressive stress is much less than buckling stress.

10.4 Euler's Theory - Long Columns:

Assumptions:

1. The column is very long in proportions to its cross sectional dimensions.
2. The section of column is uniform.

3. The column is assumed to be perfectly straight.
4. Column material is homogeneous in quality, isotropic and perfectly elastic.
5. The direct stress is very small as compared to bending stress.
6. The compressive load is applied perfectly axially.
7. The self weight of column is ignorable.
8. The column will fail by buckling alone.

Buckling Load: When a column is loaded and the load is steadily increased a certain limiting value of the load will be reached when the column will just tend to deflect or have a lateral displacement. At this position the internal forces which tend to straighten the column are just equal to the applied load. The minimum limiting load at which column ends to have lateral displacement or tends to buckle is known as buckling load, crippling load or critical load. Buckling takes place about the axis having the least moment of inertia.

The strength to resist buckling is greatly affected by the condition of the ends. The following four cases arise:

- (1) Both ends are hinged.
- (2) One end is fixed and other is free.
- (3) Both ends are fixed
- (4) One end is fixed and other is hinged.

10.4.1 Both Ends Hinged:

Let us consider a column AB with ends free to rotate around frictionless pins i.e. ends are hinged carrying a critical load at B. The buckled shape is shown in Fig. 10.2 at the critical load. The figure is possible only at critical load and prior to this load column remains straight.

Let us consider a section at a distance x from A and let y be its deflection from the center line. We have therefore, the bending moment at the section is

$$EI \frac{d^2 y}{dx^2} = M_x = -Py$$

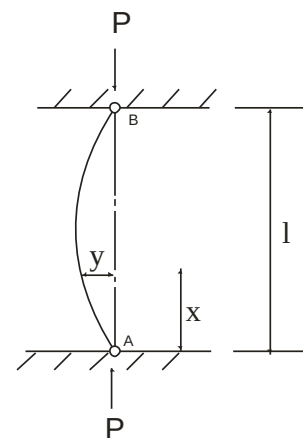


Fig.10.2

[-ve sign has been taken as bend is concave to original line]

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = 0$$

This is a standard differential equation, the solution is

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}} \quad \dots(10.2)$$

where C_1 and C_2 are constant and can be found by applying boundary conditions.

At A deflection is zero $x = 0$ & $y = 0 \therefore C_1 = 0$

At B also deflection is zero $x = l$ & $y = 0$

$$\therefore C_2 \sin l \sqrt{\frac{P}{EI}} = 0$$

Either $C_2 = 0$ (i.e. there is no bend in column hence C_2 cannot be 0)

$$\text{Hence } \sin l \sqrt{\frac{P}{EI}} = 0$$

$$\therefore l \sqrt{\frac{P}{EI}} = n\pi, \quad n = 0, 1, 2, 3, \dots \quad \dots(10.3)$$

Taking least significant value

$$l \sqrt{\frac{P}{EI}} = \pi \quad \therefore P = \frac{\pi^2 EI}{l^2} \quad \dots(10.4)$$

Hence the least critical load, buckling load or Euler load

$$P_E = \frac{\pi^2 EI}{l^2}$$

The equation (10.2) becomes

$$y = C_2 \sin x \sqrt{\frac{P}{EI}}$$

This is the characteristic or eigen function of this problem and since $\sqrt{\frac{P}{EI}} = \frac{n\pi}{L}$, n can assume any integer value, there is infinite

no. of such functions. In this linearized solution amplitude C_2 of buckling mode remains indeterminate. For fundamental case $n = 1$, the

elastic curve is a half wave sine curve. Shape and modes corresponding to $n = 2, 3$ are

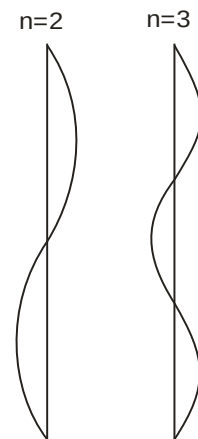


Fig.10.3

shown in Fig.10.3. The higher modes have no physical significance in buckling problems since the least critical buckling load occurs at $n = 1$.

10.4.2 One End Is Fixed Other End Is Free :

Fig. 10.4 shows a column AB of length l whose one end is fixed and other is free. The shape after applying buckling load P is shown in Fig. 10.4.

Let 'a' be the deflection of the free end and let y be the deflection at any distance x from free end A.

The bending moment at the section is given by

$$= P(a - y)$$

$$\text{Hence } EI \frac{d^2 y}{dx^2} = P(a - y)$$

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{Pa}{EI}$$

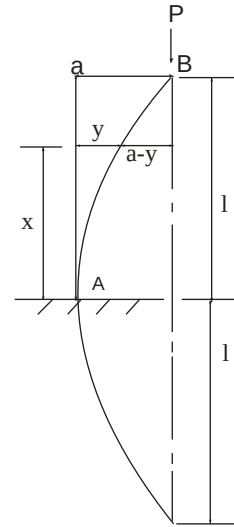


Fig. 10.4

the solution of the differential equation is given by

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}} + a \quad \dots(10.5)$$

C_1 and C_2 are constants of integration and are found out by applying boundary conditions.

At A deflection is zero $x = 0$; $y = 0$ $\therefore C_1 = -a$

At A slope is zero $x = 0$ $\frac{dy}{dx} = 0$

$$\therefore \frac{dy}{dx} = -C_1 \sqrt{\frac{P}{EI}} \sin x \sqrt{\frac{P}{EI}} + C_2 \sqrt{\frac{P}{EI}} \cos x \sqrt{\frac{P}{EI}}$$

$$\text{At A } 0 = -a \sqrt{\frac{P}{EI}} + C_2 \sqrt{\frac{P}{EI}} \cdot 1$$

$$C_2 \sqrt{\frac{P}{EI}} = 0$$

Either $P = 0$ (Not applicable)

or $C_2 = 0$

in equation (10.5) $C_1 = -a$; $C_2 = 0$

$$y = -a \cos x \sqrt{\frac{P}{EI}} + a$$

$$y = a \left(1 - \cos x \sqrt{\frac{P}{EI}} \right)$$

$$\text{at } x = l, y = a \quad a = a \left(1 - \cos l \sqrt{\frac{P}{EI}} \right)$$

$$1 = 1 - \cos l \sqrt{\frac{P}{EI}}$$

$$\cos l \sqrt{\frac{P}{EI}} = 0$$

$$l \sqrt{\frac{P}{EI}} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

Taking the least significant value

$$l \sqrt{\frac{P}{EI}} = \frac{\pi}{2}$$

Hence the least critical load, buckling load or Euler load

$$P = P_E = \frac{\pi^2 EI}{4l^2} \quad \dots\dots(10.6)$$

10.4.3 Both Ends Are Fixed:

Fig. 10.5 shows a column AB of length l whose both ends are fixed. There will be a restraint moment say M_0 at each end. The deformed shape under the application of buckling load P has been shown in Fig. 10.5. The column is convex for a length of $\frac{l}{4}$

from either end and concave for the middle half.

Considering at any distance x from A we have

$$EI \frac{d^2 y}{dx^2} = M_0 - P y$$

Where M_0 is the fixed moment at the ends as in the case of a fixed beam

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{PM_0}{EI P}$$

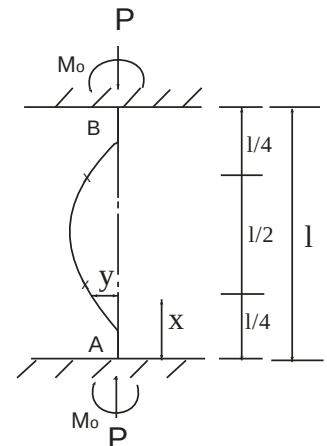


Fig. 10.5

The solution of the above differential equation is given by

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}} + \frac{M_o}{P} \quad \dots(10.7)$$

$$\frac{dy}{dx} = -C_1 \sqrt{\frac{P}{EI}} \sin x \sqrt{\frac{P}{EI}} + C_2 \sqrt{\frac{P}{EI}} \cos x \sqrt{\frac{P}{EI}} \quad \dots(10.8)$$

Where C_1 and C_2 are constants and found by applying boundary conditions

At A Deflection is zero $x = 0$; $y = 0$ $\therefore C_1 = -\frac{M_o}{P}$

At A slope is zero $x = 0$ $\frac{dy}{dx} = 0$ $\therefore C_2 = 0 (P \neq 0)$

In equation (10.7) substituting $C_1 = -\frac{M_o}{P}$; $C_2 = 0$

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} - C_1 = C_1 (\cos x \sqrt{\frac{P}{EI}} - 1)$$

at B deflection is zero $x = l$; $y = 0$; $0 = C_1 (\cos l \sqrt{\frac{P}{EI}} - 1)$

either $C_1 = 0$ in which case column is not bent at all

or $\cos l \sqrt{\frac{P}{EI}} = 1$

$$l \sqrt{\frac{P}{EI}} = 0, 2\pi, 4\pi \quad \dots(10.9)$$

also at B slope is zero $x = l$, $\frac{dy}{dx} = 0$

from equation (10.8)

$$0 = -C_1 \sqrt{\frac{P}{EI}} \sin l \sqrt{\frac{P}{EI}}$$

$$\sin l \sqrt{\frac{P}{EI}} = 0$$

$$l \sqrt{\frac{P}{EI}} = 0, \pi, 2\pi, 3\pi \dots \quad \dots(10.10)$$

The minimum significant value of $l \sqrt{\frac{P}{EI}}$ consistent both with (10.9) & (10.10) is 2π .

$$\therefore l \sqrt{\frac{P}{EI}} = 2\pi$$

Hence the least critical load, buckling load or Euler load

$$P = P_E = \frac{4\pi^2 EI}{l^2} \quad \dots(10.11)$$

10.4.4 One End Is Fixed, The Other Is Hinged:

Fig. 10.6 shows the deformed shape of a column which is fixed at one end A and hinged at B carrying a buckling load P. Since A is fixed a bending moment M_0 is induced at A and this involves the presence of a force R at right angle to AB (i.e. horizontal) to maintain equilibrium, without which no BM can occur at A. Let us consider any section at a distance x from A the bending moment at section is given by

$$EI \frac{d^2 y}{dx^2} = -P_y + R(l-x)$$

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{R}{EI} (l-x)$$

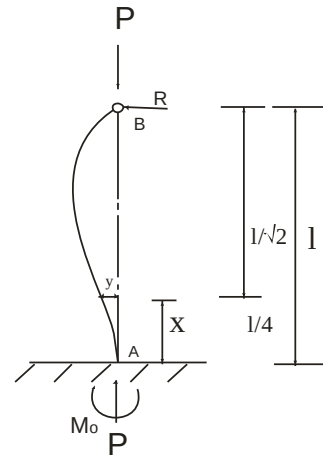


Fig. 10.6

The solution of the above differential equation is given by

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}} + \frac{R(l-x)}{P} \quad \dots(10.12)$$

$$\frac{dy}{dx} = -C_1 \sqrt{\frac{P}{EI}} \sin x \sqrt{\frac{P}{EI}} + C_2 \sqrt{\frac{P}{EI}} \cos x \sqrt{\frac{P}{EI}} - \frac{R}{P} \quad \dots(10.13)$$

At A deflection is zero $x = 0$ $y = 0$ $\therefore C_1 = - \frac{Rl}{P}$

At A slope is zero $x = 0$ $\frac{dy}{dx} = 0$ $\therefore 0 = C_2 \sqrt{\frac{P}{EI}} - \frac{R}{P}$

$$\therefore C_2 = \frac{R}{P} \sqrt{\frac{EI}{P}}$$

In equation (10.12)

$$y = -\frac{Rl}{P} \cos x \sqrt{\frac{P}{EI}} + \frac{R}{P} \sqrt{\frac{EI}{P}} \sin x \sqrt{\frac{P}{EI}} + \frac{R}{P} (l - x)$$

at B deflection is zero

$$0 = -\frac{Rl}{P} \cos l \sqrt{\frac{P}{EI}} + \frac{R}{P} \sqrt{\frac{EI}{P}} \sin l \sqrt{\frac{P}{EI}} + 0$$

$$\tan l \sqrt{\frac{P}{EI}} = l \sqrt{\frac{P}{EI}} \quad \{\text{such that tangent is equal to itself}\}$$

The smallest root of above equation

$$l \sqrt{\frac{P}{EI}} = 4.49 \text{ radian} \quad \therefore \frac{l^2 P}{EI} = 20 = 2\pi^2 \text{ (approximately)}$$

Hence the least critical load, buckling load or Euler load

$$P = P_E = \frac{2\pi^2 EI}{l^2} \quad \dots(10.14)$$

10.5 Equivalent Length Of Columns :

In general we can write the Euler load $P_E = \frac{\pi^2 EI}{Cl^2}$ with given end conditions

equivalent length is the length of an equivalent column of the same material and section with hinged ends having the value of buckling load equal to that of the given column.

Where C is a constant which takes into account of the end conditions. If

- a) Column is hinged at both ends , $C = 1$
- b) Column is fixed at one end free at other , $C = 4$
- c) Both ends are fixed , $C = 1/4$
- d) One end is fixed and other is hinged , $C = 1/2$

In the place of using actual length of the column we can use equivalent length L_E

$$P_E = \frac{\pi^2 EI}{l_e^2} \quad \dots (10.15)$$

$$l_E = l\sqrt{C} \text{ known as equivalent length}$$

Table 1. Equivalent Length L_e Of Column Corresponding To Different End Conditions.

End Cond.	Both end hinged	One fix other free	Both fix	One fix other hinged
L_e	L	$2l$	$\frac{l}{2}$	$\frac{l}{\sqrt{2}}$

I is the moment of inertia of section about the axis of least resistance. Putting $I = Ak^2$. k is least radius of gyration.

$$P_E = \frac{\pi^2 E Ak^2}{l_e^2}$$

$$\frac{P_E}{A} = \frac{\pi^2 E}{(l/K)^2} \quad \dots\dots\dots(10.16)$$

The ratio (l/k) is the slenderness ratio for the column Euler formula is applicable only if l/k is greater than 80.4 for steel column hinged at both ends.

10.6 Limitation of Euler's Formula:

$$P_{CR} = \frac{\pi^2 EI}{L_e^2} = \frac{\pi^2 EAK^2}{l_e^2}$$

$$\text{Crippling stress } \sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 E}{(l_e/k)^2}$$

Where l/k is slenderness ratio see figure 10.6. If the (l/k) is small the crippling stress (or stress at failure) will be high but for the column material the σ_{cr} can not be greater than crushing stress

(Proportional limit). Hence when the slenderness ratio is less than a certain limit, Euler formula gives a value greater than the crushing stress.

In the limiting case we can find the value of (l/k) for which crippling stress is equal to crushing stress. For mild steel column with both ends hinged.

$$\text{Crushing stress} = 330 \text{ N/mm}^2$$

$$\frac{\pi^2 EI}{(l/k)^2} = 330 \quad E = 2 \times 10^5 \text{ N/mm}^2$$

$$\frac{l}{k} = \sqrt{\frac{\pi^2 EI}{330}} = \sqrt{6282} = 79.27 \approx 80$$

Hence for $(l/k) < 80$ for mild steel Euler's formula is not valid.

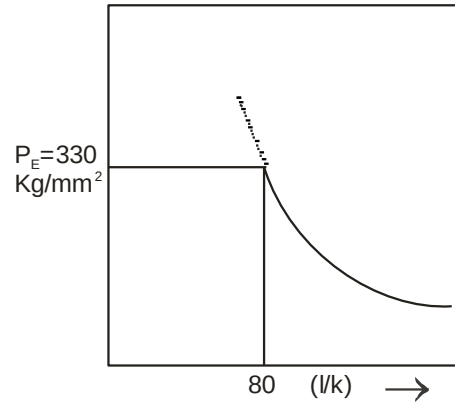


Fig. 10.7

10.7 Empirical Relations :

Euler formula does not take into account the direct compressive stress and hence give correct results only for long columns. For medium struts or columns Euler's formula does not give correct results. The following empirical formulas are important from the subject point of view.

10.7.1 Rankine's Formula For Columns :

Rankine proposed an empirical formula for columns which covers all cases of columns ranging from very short to long struts. He proposed the relation

$$\frac{1}{P} = \frac{1}{P_C} + \frac{1}{P_E} \quad \dots(10.17)$$

Where $P_C = \sigma_c A$ (crushing load)

$$P_E = \frac{\pi^2 EI}{l^2} \quad (\text{Euler crippling load}).$$

In equation (10.17) $\frac{1}{P_C}$ is constant for a material. For short column P_E is large and hence

$\frac{1}{P_E}$ is small as compared to $\frac{1}{P_C}$.

$$\text{Hence } \frac{1}{P} \cong \frac{1}{P_C} \text{ or } P \cong P_C.$$

Similarly for long column P_E is small and $\frac{1}{P_E}$ is much large as compared to $\frac{1}{P_C}$.

$$\text{Hence } \frac{1}{P} \cong \frac{1}{P_E}, \quad P \cong P_E.$$

Thus value of p obtained from above relation covers extreme cases of long as well as short columns. Hence Rankine formula is taken to be valid for length of columns.

$$\text{Hence } P = \frac{P_C P_E}{P_E + P_C} = \frac{1 + \frac{P_C}{P_E}}{P_E} = \frac{1 + \frac{\sigma_c A}{\pi^2 EI}}{\pi^2 EI}$$

Putting $I = AK^2$ where K is least radius of gyration of column section.

$$P = \frac{\sigma_c A}{1 + \frac{\sigma_c}{\pi^2 E} \left(\frac{l}{K} \right)^2} = \frac{\sigma_{Cr} A}{1 + a \left(\frac{l}{K} \right)^2} \quad \dots(10.18)$$

σ_c – Crushing stress for the metal

a – Rankine constant for metal = $\frac{\sigma_{Cr}}{\pi^2 E}$

Value of σ_c and a are shown in table 2

Table 2

No.	Material	σ_c (N/mm ²)	a (for ends hinged)
1.	Wrought Iron	250	1/9000

2.	CI	550	1/1600
3.	MS	320	1/7500
4.	Strong Timber	50	1/750

For all condition of column

$$P = \frac{\sigma_c A}{1 + a \left(\frac{L_e}{K} \right)^2} = \frac{\text{Crushing load}}{1 + a \left(\frac{L_e}{K} \right)^2}$$

$1 + a \left(\frac{L_e}{K} \right)^2$ is the factor takes into account the buckling effect.

10.7.2 Johnson's Formula For Column :

Euler's formulae and Rankine formula are not exact compared with actual result obtained by experiments because

- Effect of direct compression is neglected in Euler's Formula.
- The loading is not exactly applied as desired.
- The hinge joints are not perfectly frictionless.
- The fixed ends are not absolutely fixed.

10.7.2.1 Johnson's Straight Line Formula For Columns

$$P = A \left[\sigma_c - n \left(\frac{L}{K} \right) \right] \quad \dots(10.19)$$

Where P = Safe load on the column

n = constant whose value depends upon column material (Table 3)

Table 3

S.no.	Material	n	r
1.	M.S.	0.0053	0.000057
2.	Wrought Iron	0.0053	0.000039
3.	CI	0.008	0.000016

10.7.2.2 : Johnson's Parabolic Formula For Columns

$$P = A \left[\sigma_c - r \left(\frac{L}{K} \right) \right] \quad \dots(10.20)$$

$$\frac{P}{A} = \frac{\sigma_{CA}}{1 + a \left(\frac{Le}{k} \right)^2} \quad \text{Constant given in Table 3} \quad \dots(10.21)$$

This formula involves factor of safety. σ_{CA} is allowable crushing stress.

10.8 Long Columns Subjected To Eccentric Loading :

In actual practice load is eccentric. Direct compression stress

$$\sigma_C = \frac{P}{A}$$

Let P = load on column

A – Area of cross section

e – Eccentricity

z – Section modulus

k – Least radius of gyration

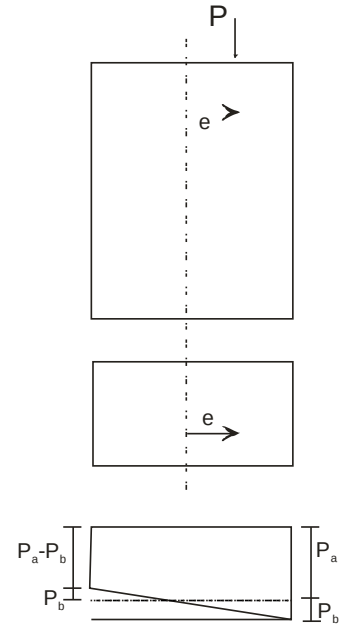


Fig. : 10.8

Bending couple will cause longitudinal tensile and

compressive stresses. The fiber stress σ_b at any distance y_c from neutral axis NA is given

by

$$\sigma_b = \frac{My_c}{I} = \frac{P_e y_c}{I_x}$$

Where y_e – distance of extreme fiber (on comp. side) from the axis of the column.

Maximum stress at any section is given by

$$\sigma_{\max} = \frac{P}{A} + \frac{P_e y_c}{I_x} = \frac{P}{A} + \frac{M}{Z} \quad \dots(10.22)$$

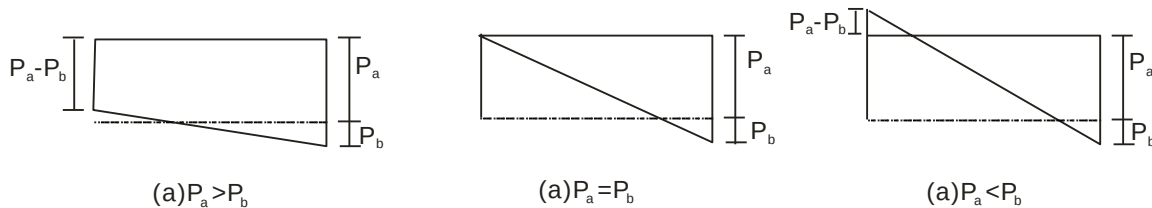


Fig. : 10.9

$$\sigma_{\max} = \frac{P}{A} + \frac{P_e y_c}{AK^2} \quad \left(z = \frac{I}{y_e} = \frac{AK^2}{y_e} \right)$$

$$\sigma_{\max} = \frac{P}{A} \left(1 + \frac{e y_c}{K^2} \right)$$

σ_c = permissible stress for the metal.

$$P = \frac{\sigma_c A}{\left(1 + \frac{e y_c}{K^2} \right)} \quad \dots(10.23)$$

$1 + \frac{e y_c}{K^2}$ is the reduction factor which take into account the eccentricity of loading. If the effect of buckling is also to be taken into account the safe axial load with eccentricity is given by.

$$P = \frac{A}{\left(1 + \frac{e y_c}{K^2} \right) \left(1 + a \frac{l^2}{K^2} \right)} \quad \dots(10.24)$$

10.9 The Euler Secant Formula :

Let us consider a long column AB of length l and both ends hinged loaded with force P . At any section from distance x , y is the deflection with reference to the line of action of force P , which is acting at an eccentricity e at the ends.

The BM due to load is $-Py$.

Hence $EI \frac{d^2 y}{dx^2} = -Py$

$$\text{or } \frac{d^2 y}{dx^2} + \frac{P}{EI} y = 0$$

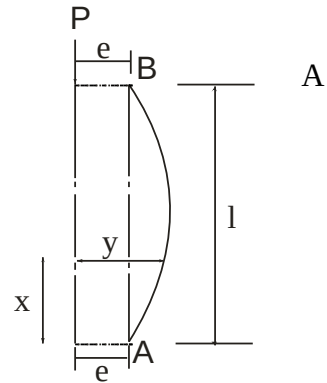


Fig.10.10

Solution of the above differential equation is given by

$$y = C_2 \cos x \sqrt{\frac{P}{EI}} + C_1 \sin x \sqrt{\frac{P}{EI}} \quad \dots(10.25)$$

C_1 and C_2 are constants found by applying boundary conditions.

$$\text{At } x = 0, y = e \quad \therefore C_2 = e$$

$$\frac{dy}{dx} = -C_2 \sqrt{\frac{P}{EI}} \sin x \sqrt{\frac{P}{EI}} + C_1 \sqrt{\frac{P}{EI}} \cos x \sqrt{\frac{P}{EI}}$$

$$\text{at } x = \frac{l}{2} \text{ (mid of column)}$$

$$\text{slope } \frac{dy}{dx} = 0$$

$$0 = -e \sqrt{\frac{P}{EI}} \sin \frac{l}{2} \sqrt{\frac{P}{EI}} + C_1 \sqrt{\frac{P}{EI}} \cos \frac{l}{2} \sqrt{\frac{P}{EI}}$$

$$C_1 = e \left[\frac{\sin \frac{l}{2} \sqrt{\frac{P}{EI}}}{\cos \frac{l}{2} \sqrt{\frac{P}{EI}}} \right]$$

$$\dots\dots(10.26)$$

$$x = l/2 \left[\frac{\sin^2 \frac{l}{2} \sqrt{\frac{P}{EI}}}{\cos \frac{l}{2} \sqrt{\frac{P}{EI}}} + \cos \frac{l}{2} \sqrt{\frac{P}{EI}} \right] = e \sec \frac{l}{2} \sqrt{\frac{P}{EI}}$$

$$y_{\max} = e \left[\dots\dots(10.27) \right]$$

$$\text{Maximum bending moment} = P y_{\max} = P e \sec \frac{l}{2} \sqrt{\frac{P}{EI}}$$

Maximum compressive stress

$$\sigma_{\max} = \frac{P}{A} + \frac{M}{Z}$$

$$= \frac{P}{A} + \frac{P_e \sec \frac{l}{2} \sqrt{\frac{P}{EI}}}{Z} \quad \left(Z = \frac{AK^2}{y_c} \right)$$

$$= \frac{P}{A} \left[1 + \frac{e y_c}{K^2} \sec \frac{l}{2} \sqrt{\frac{P}{EI}} \right] \dots\dots(10.28)$$

This known s Euler Secant formula for columns for all columns for all cases

$$\sigma = \frac{P}{A} + \frac{P_e \sec \frac{LE}{Z} \sqrt{\frac{P}{EI}}}{Z} \dots\dots(10.29)$$

It can be easily proved that if one end is fixed and the other end is free, the maximum bending moment is $= P e \sec l \sqrt{\frac{P}{EI}}$

Similarly if both ends are fixed maximum bending moment $= P.e . \sec \frac{l}{4} \sqrt{\frac{P}{EI}}$.

8. Torsion of Circular Shafts

8.1 Introduction

When equal and opposite forces are applied tangentially to the ends of a shaft it is subject to a twisting moment which is equal to the product of force applied and radius of shaft. This causes shaft to rotate with constant angular velocity. A shaft is said to be in pure torsion when it is subjected to equal and opposite torques at its two ends. It results in shearing off of shaft at every cross section perpendicular to its longitudinal axis. While a beam bends as an effect of bending moment, a shaft twists as an effect of torsion.

8.2 Theory of pure torsion

Following assumption are made, while finding out shear stress in a circular shaft subjected to torsion:

1. No distortion of parallel planes normal to axis takes place.
2. Torsion is uniform along the rod i.e. all the normal cross sections which are at the same axial distance suffer equal relative rotation.
3. All the stresses are within the elastic limit.
4. The material is homogenous and isotropic.
5. The shaft is circular in section remains circular after twisting.

Let one end of the shaft is fixed and a moment is applied at the other end. This is valid because whether it rotates at uniform speed to transmit the power or is at rest, the stress and strains due to equal and opposite couples at its ends will remain the same.

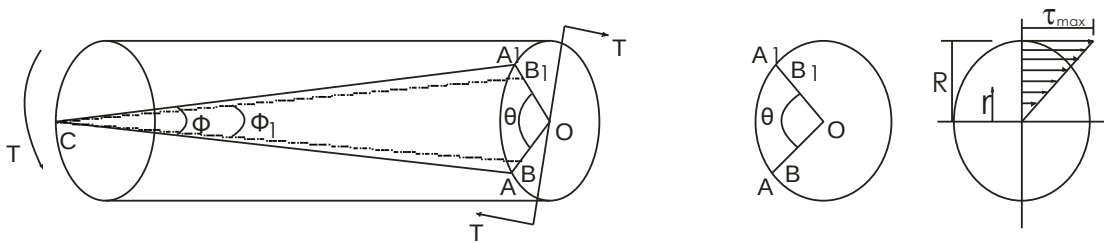


Fig. 8.1

Let us consider a torque T applied at the free end of the shaft as in Fig 8.1. A balancing torque of equal magnitude and opposite in direction will be induced at the fixed end. Let the cross section be distorted through an angle θ , shifting the radii OA to OA_1 . On the surface of the shaft any line such as CA will be distorted to CA_1 , through an

angle ϕ . The angle $\angle ACA' = \phi$ is called the shear strain and $\angle BOB' = \theta$ is called the angle of twist.

We know that shear strain = deformation per unit length = $\frac{AA_1}{l} = \tan \phi = \phi$

(ϕ being small $\therefore \tan \phi = \phi$)

If τ be the intensity of shear stress at the surface of the shaft and C be the modulus of rigidity or shear modulus:

$$\text{Shear strain } \phi = \frac{\tau}{C} \quad [\because \text{shear stress} \propto \text{shear strain}] \quad \dots\dots(8.1)$$

$$\text{Also } \phi = \frac{AA'}{CA} = \frac{OA \cdot \theta}{CA} = \frac{R \theta}{l} \quad \dots\dots(8.2)$$

$$\therefore \phi = \frac{\tau}{C} = \frac{R \theta}{l} \quad \dots\dots(8.3)$$

$$\frac{\tau}{R} = \frac{C \theta}{l} \quad \dots\dots (8.4)$$

Similarly a layer such as DB distant r from the axis of the shaft will be distorted to a new position DB_1 through an angle ϕ_1 and we have

$$\frac{q}{C} = \phi_1 = \frac{BB_1}{DB_1} = \frac{r \theta}{l}$$

$$\frac{q}{r} = \frac{C \theta}{l} \quad \dots\dots(8.5)$$

$$\text{Hence } \frac{\tau}{R} = \frac{q}{r} = \frac{C \theta}{l} \quad \dots\dots (8.6)$$

Intensity of shear stress at any point in the cross section of a shaft subject to pure torsion is proportional to its distance from the centre. This shear intensity is zero at axis and increases linearly to maximum of τ at surface as shown in Fig. 8.1.

Resisting Torque:

To calculate the torque, a shaft can withstand or transmit, let us consider an elementary ring of radius r and thickness δr . Let the shear intensity of stress at this radius be q. we have

$$q = \frac{\tau}{R} r \quad (\text{from equation 8.6})$$

the resistance couple set up by the ring

$$\delta T_r = q \cdot \delta a \cdot r$$

$$(\delta a = \text{area of the ring} = 2\pi r \delta r)$$

$$\delta T_r = \frac{\tau}{R} \cdot r \cdot \delta a \cdot r$$

Resistance by the whole shaft

$$T_r = \sum_0^R \delta T_r = \int_0^R \frac{\tau}{R} \delta a r^2 = \frac{\tau}{R} \int_0^R \delta a r^2$$

$$\int_0^R \delta a r^2 \text{ represents the polar moment of inertia } J$$

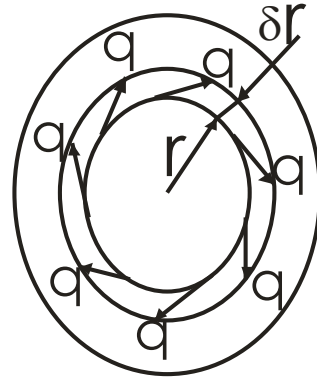


Fig. 8.2

about the longitudinal axis of the solid shaft. The moment of inertia of a plane area w.r.t. an axis perpendicular to the plane of the figure is called polar moment of inertia or w.r.t. the point where the axis intersects the plane.

$$\text{Hence } T_r = \frac{\tau}{R} J$$

$$\frac{T_r}{J} = \frac{\tau}{R}$$

.....(8.7)

Combining equations (8.6) & (8.7) we have torsion equation

$$\frac{q}{r} = \frac{\tau}{R} = \frac{T_r}{J} = \frac{C \theta}{l}$$

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

Note: for bending

In a beam the bending moment produces a bend or deflection, in the same manner a torque produces a twist in a shaft. The expression CJ corresponds to a similar quantity EI in expression for deflections of beams. The quantity CJ is called torsional rigidity while EI is called flexural rigidity.

8.3 Power Transmitted By A Shaft

Let a shaft turning at N rpm transmits P KW. Let the mean torque to which the shaft is subjected to be T Nm.

Power transmitted P = mean torque x angle turned / sec.

$$P = \frac{T \times \frac{N}{60} \times 2\pi}{W} \text{ W}$$

$$P = \frac{2\pi NT}{60} W \quad \dots(8.8)$$

8.6 Composite Circular Shafts

8.6.1 Shafts in Series:

Consider a shaft made of two material connected in series as shown in Fig. 8.4. The shaft is subjected to a torque T. As both the shafts are subjected to the same torque

$$\therefore T_1 = T_2 \quad \dots(8.13)$$

$$\text{Now } \frac{T_1}{J_1} = \frac{\tau_{s1}}{\frac{D_1}{2}} = \frac{C_1 \theta_1}{L_1}$$

and

$$\frac{T_2}{J_2} = \frac{\tau_{s2}}{\frac{D_2}{2}} = \frac{C_2 \theta_2}{L_2}$$

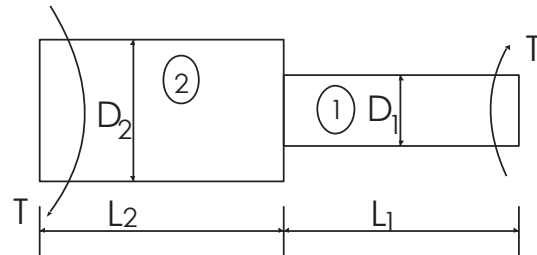


Fig. 8.4

Total angle of twist, $\theta = \theta_1 + \theta_2$

8.6.2 Shafts in Parallel:

Now consider a shaft made of two material connected in parallel as shown in Fig. 8.5. The torque applied in this case is divided between the two shafts but the angle of twist is the same for each shaft. Let T be the applied torque.

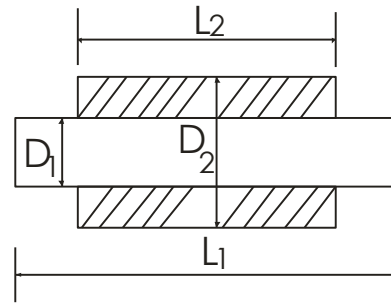


Fig. 8.5

Then $T = T_1 + T_2$ and $\theta_1 = \theta_2$.

8.5 Comparison Between Hollow And Solid Shaft

Let	D_1	=	Diameter of solid shaft
	L	=	length of shafts
	ρ	=	density of metal
	D	=	Outer diameter of hollow shaft
	d	=	inner diameter of hollow shaft
	n	=	D/d

$$\text{Polar moment of inertia of solid shaft} = \frac{\pi D_1^4}{32}$$

$$\text{Polar moment of inertia of hollow shaft} = \frac{\pi (D^4 - d^4)}{32}$$

Comparison can be done on two conditions:

- (a) **By Strength:** In this case we assume both solid and hollow shafts to be of same length, weight and material and we find the ratio of torque transmitted by both the shafts:

$$\begin{aligned} \text{Using torsion equation } \frac{T}{J} &= \frac{\tau}{R} \\ \frac{T_{\text{hollow}}}{T_{\text{solid}}} &= \frac{J_{\text{hollow}} R_{\text{solid}}}{R_{\text{hollow}} J_{\text{solid}}} = \frac{D^4 - d^4}{D_1^4} \times \frac{D_1}{D} = \frac{d^3 (n^4 - 1)}{n D_1^3} \end{aligned} \quad \dots(8.9)$$

Since weight of both the shafts are same

$$\text{Hence } \frac{\pi}{4} D_1^4 l \rho = \frac{\pi}{4} (D^2 - d^2) l \rho$$

$$\text{or } D_1 = \sqrt{D^2 - d^2} = d \sqrt{n^2 - 1}$$

$$\text{or } D_1^3 = d^3 (n^2 - 1) \sqrt{n^2 - 1} \quad \dots(8.10)$$

$$\text{putting this into equation (8.9)} \quad \frac{T_{\text{hollow}}}{T_{\text{solid}}} = \frac{n^4 - 1}{n \sqrt{n^2 - 1} (n^2 - 1)} = \frac{n^2 + 1}{n \sqrt{n^2 - 1}} \quad \dots(8.11)$$

Since $n > 1$ always hence hollow shaft is stronger than the solid shaft for same metal and weight.

$$\frac{T_{\text{hollow}}}{T_{\text{solid}}} = \frac{\text{If } n = \frac{2}{1}}{2\sqrt{4-1}} = 1.44$$

- (b) **Comparison By Weight :** In this case for the same torque transmitted the weight

$$\text{ratio of hollow and solid shafts are found out} \\ \frac{W_{\text{hollow}}}{W_{\text{solid}}} = \frac{\pi (D^2 - d^2) R \times \text{density}}{\pi D_1^2 r l \times \text{density}} = \frac{D^2 - d^2}{D_1^2}$$

$$= \frac{d^2(n^2-1)}{D_1^2} \quad (D = nd)$$

Since the Torque transmitted are same

$$\begin{aligned} \frac{\tau}{R} \times \frac{\pi}{32} T_{\text{hollow}} &= T_{\text{solid}} \\ \frac{\tau}{R} \times \frac{\pi}{32} (D^4 - d^4) &= \tau_s \times \frac{\pi}{32} D_1^3 \\ \frac{D^4 - d^4}{D} &= D_1^3 \\ \frac{d^4(n^4-1)}{D} &= D_1^3 \\ \frac{d^3(n^4-1)}{n} &= D_1^3 \\ D_1^2 &= \frac{d^3(n^4-1)}{n} \\ \frac{W_{\text{hollow}}}{W_{\text{solid}}} &= \left(\frac{n^4-1}{n} \right)^{2/3} = \frac{n^{2/3}(n^2-1)}{(n^4-1)^{2/3}} \end{aligned} \quad \dots(8.12)$$

$$\begin{aligned} \text{If } n &= 2 \\ \frac{W_{\text{hollow}}}{W_{\text{solid}}} &= 0.7829 \end{aligned}$$

For a given torque weight of the hollow shaft will be lesser than solid shaft.

Problem 2.1 A steel bar 50mm wide, 12mm thick and 300 mm long is subjected to an axial pull of 96 kN. Find the changes in the length, width, thickness and the volume of the bar. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio $= 0.32$.

Solution: Longitudinal stress $= \sigma = \frac{P}{A} = \frac{96000}{50 \times 12} = 160 \text{ N/mm}^2$

$\therefore$ Longitudinal strain $= e = \frac{\sigma}{E} = \frac{160}{2 \times 10^5} = 0.0008$

Lateral strain = Poisson's ratio $\times$ Longitudinal strain

$\therefore$ Increase in length of the bar $= \Delta l = e l = 0.0008 \times 300 = 0.24 \text{ mm}$

Decrease in width $= \Delta b = \text{Lateral strain} \times \text{Original width}$
 $= 0.000256 \times 12 = 0.0031 \text{ mm}$

Volumetric strain $= e_v = e_l + e_b + e_d$
 $= 0.0008 - 2 \times 0.000256 = 0.000288 \text{ (+ve)}$

$\therefore$ Increase in volume $= e_v \cdot V$
 $= 0.000288 \times 300 \times 50 \times 12 \text{ mm}^3 = 51.84 \text{ mm}^3$

Problem 2.2 A bar uniform rectangular section A is subjected to an axial tensile load P.

Show that the volumetric strain is given by $e_v = \frac{P}{AE} \left(1 - \frac{2}{m} \right)$ where E is the Young's

Modulus and $\frac{1}{m}$ is the Poisson's ratio.

Solution: Strain of the length $= e_l = \frac{P}{AE}$

$\therefore$ Lateral strain $= \text{Strain of width} = \text{Strain of depth}$
 $= -\frac{1}{m} e_l = -\frac{P}{mAE}$

$\therefore$ Volumetric strain $= \text{Strain of length} + \text{Strain of width} + \text{Strain of depth}$

$\therefore e_v = \frac{P}{AE} - \frac{P}{mAE} - \frac{P}{mAE}$

$\therefore e_v = \frac{P}{AE} \left(1 - \frac{2}{m} \right)$

Problem 2.3 A steel rod 5 meters long and 20 mm diameter subjected to an axial tensile load of 45 kN. Find the change in length, diameter and the volume of the rod. Take E_{steel}

$= 2 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio $= \frac{1}{4}$.

Solution : Area of the rod = $A = \frac{\pi}{4} \times 20^2 = 314.2 \text{ mm}^2$

$\therefore$ Tensile stress $\sigma = \frac{P}{A} = \frac{45000}{314.2} = 143.2 \text{ N/mm}^2$

$\therefore$ Strain of length = $\frac{\sigma}{E} = \frac{143.2}{2 \times 10^5} = 0.000716 \text{ (+ve)}$

$\therefore$ Increase in length = $0.000716 \times 5000 = 3.580 \text{ mm (+ve)}$

Lateral strain = Strain of diameter = $\frac{\sigma}{mE} = \frac{1}{4} \times 0.000716 = 0.000179 \text{ (-ve)}$

$\therefore$ Decrease in diameter = $0.000179 \times 20 = 0.000358 \text{ mm}$

Volumetric strain = $\frac{\sigma}{E} - \frac{2\sigma}{mE} = 0.000716 - 2 \times 0.000179 = 0.000358 \text{ (+ve)}$

$\therefore$ Increase in volume = $0.000358 \times \frac{\pi}{4} \times 20^2 \times 5000 = 562.06 \text{ mm}^3$

Problem 2.4 A material has a Young's Modulus of $1.30 \times 10^5 \text{ N/mm}^2$ and a Poisson's ratio of 0.25. Calculate the modulus of Rigidity and the Bulk Modulus.

Solution: $E = 1.30 \times 10^5 \text{ N/mm}^2$
 $\frac{1}{m} = 0.25$

C = Modulus of Rigidity

K = Bulk Modulus

We know, $E = 2C \left(1 + \frac{1}{m} \right) = 2C (1 + 0.25) = 2.5 C$

$\therefore C = \frac{1.30 \times 10^5}{2.5} = 0.52 \times 10^5 \text{ N/mm}^2$

We know, $E = 3K \left(1 - \frac{2}{m} \right) = 3K (1 - 2 \times 0.25) = 1.5K$

$\therefore K = \frac{1.30 \times 10^5}{1.5} = 0.867 \times 10^5 \text{ N/mm}^2$

Problem 2.5 In a tensile test, a test piece 30 mm in diameter , 250 mm gauge length stretched 0.106 mm under a pull of 65 kN. In a torsion test, the same rod twisted 0.017 radian over a length of 250 mm, when a torque of 450 Nm was applied. Find the three elastic moduli and the Poisson's ratio for the material of the test piece.

Solution: $d = 30 \text{ mm}$; $l = 250 \text{ mm}$; pull = 65 kN = 65000 N

$$\text{Tensile stress} = \sigma = \frac{\frac{\pi}{4}(30)^2}{65000} = 91.95 \text{ N/mm}^2$$

$$\text{Tensile strain} = e = \frac{0.106}{250} = 4.24 \times 10^{-4}$$

$$\text{Modulus of Elasticity} E = \frac{\sigma}{e} = \frac{91.95}{4.24 \times 10^{-4}} = 2.169 \times 10^5 \text{ N/mm}^2$$

$$\text{Modulus of Rigidity} C = \frac{E}{2(1 + \mu)} = \frac{2.169 \times 10^5}{2(1 + 0.303)} = 0.832 \times 10^5 \text{ N/mm}^2$$

$$\text{We know, } E = 2C \left(1 + \frac{1}{m} \right)$$

$$\frac{1}{1 + m} = \frac{E}{2C} = \frac{2.169 \times 10^5}{2 \times 0.832 \times 10^5} = 1.303$$

$$\therefore \text{Poisson's Ratio} = \frac{1}{m} = 0.303$$

$$\text{We know } E = 3K \left(1 - \frac{2}{m} \right)$$

$$K = \frac{E}{3 \left(1 - \frac{2}{m} \right)} = \frac{2.169 \times 10^5}{3(1 - 2 \times 0.303)} = 1.835 \times 10^5 \text{ N/mm}^2$$

Problem 2.6 A rectangular block 250mm × 100mm × 80mm. is subjected to axial loads as follows:

500 kN tensile in the direction of its length

900 kN tensile on the 250 mm × 80 mm faces

1000 kN compressive on the 250mm × 100 mm faces

Assuming Poisson's ratio as 0.25, find in terms of the modulus of elasticity E of the material the strains in the direction of each force.

If $E = 2 \times 10^5 \text{ N/mm}^2$, find the value of the modulus of rigidity and bulk modulus for the material of the block. Also calculate the change in the volume of the block.

Solution: Fig 2.7 shows the block with its 250 mm, 100mm and 80 mm dimension parallel to X, Y and Z axes.

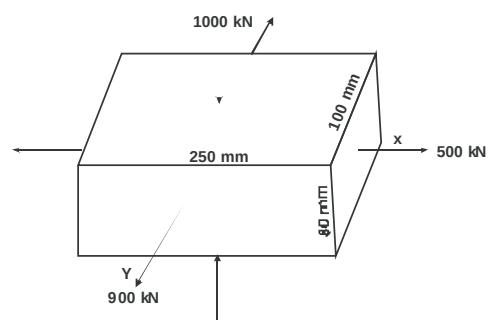


Fig. 2.7

The stresses in the direction of these axes are ,

$$\sigma_x = \frac{500 \times 1000}{100 \times 80} = 62.5 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_y = \frac{900 \times 1000}{250 \times 80} = 45 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_z = \frac{1000 \times 1000}{250 \times 100} = 40 \text{ N/mm}^2 \text{ (Compressive)}$$

The strains along the three principal directions are,

$$e_x = \frac{62.5}{E} - \frac{45}{mE} + \frac{40}{mE} = \frac{1}{E} \left(62.5 - \frac{45}{m} + \frac{40}{m} \right)$$

$$= \frac{1}{E} \left(62.5 - \frac{5}{m} \right) = \frac{1}{E} (62.5 - 5 \times 0.25) = + \frac{61.25}{E}$$

$$e_y = \frac{45}{E} - \frac{62.5}{mE} + \frac{40}{mE}$$

$$= \frac{1}{E} \left(45 - \frac{22.5}{m} \right) = \frac{1}{E} (45 - 22.5 \times 0.25) = + \frac{39.375}{E}$$

$$e_z = - \frac{40}{E} - \frac{62.5}{mE} - \frac{45}{mE} = - \frac{1}{E} \left(40 + \frac{107.5}{m} \right)$$

$$= - \frac{1}{E} (40 + 107.5 \times 0.25) = - \frac{66.875}{E}$$

$$\text{Volumetric strain} = e_v = e_x + e_y + e_z$$

$$= \frac{1}{E} (61.25 + 39.375 - 66.875) = + \frac{33.75}{E} = \frac{33.75}{2 \times 10^5}$$

$$\therefore \text{Increase in volume} = e_v \times V = \frac{33.75}{2 \times 10^5} \times (250 \times 100 \times 80) \text{ mm}^3 = 337.5 \text{ mm}^3$$

We know the following relation between the Young's Modulus E and the modulus of rigidity C and the bulk modulus K.

$$E = 2C \left(1 + \frac{1}{m} \right)$$

And

$$E = 3K \left(1 - \frac{2}{m} \right)$$

$$\therefore 2 \times 10^5 = 2C(1 + 0.25) = 3K(1 - 2 \times 0.25)$$

∴

$$C = 0.8 \times 10^5 \text{ N/mm}^2$$

And

$$K = 1.33 \times 10^5 \text{ N/mm}^2$$

Problem 2.7 A square prism of wood 50mm × 50mm in cross-section and 300 mm long is subjected to a tensile stress of 40 N/mm² along its longitudinal axis and lateral compressive stresses of 20 N/mm² on one set of lateral face and a lateral tensile stress of 10 N/mm² on the other set of lateral faces. Find the changed dimensions of the prism. Take Poisson's Ratio = 0.4 and Modulus of Elasticity = 2 × 10⁴ N/mm².

Solution: Fig. 2.8 shows the prism subjected to the given stresses.

The strain along the length, breadth and depth are given by

$$e_x = \frac{40}{E} - \frac{10}{mE} + \frac{20}{mE}$$

$$= \frac{1}{E} (40 - 10 \times 0.4 + 20 \times 0.4)$$

$$= \frac{44}{E} = \frac{44}{2 \times 10^4} = 2.2 \times 10^{-3}$$

$$e_y = \frac{10}{E} - \frac{40}{mE} + \frac{20}{mE}$$

$$= \frac{1}{E} (10 - 40 \times 0.4 + 20 \times 0.4)$$

$$= \frac{2}{E} = \frac{2}{2 \times 10^4} = 1 \times 10^{-4}$$

$$e_z = -\frac{20}{E} - \frac{40}{mE} - \frac{10}{mE} = -\frac{1}{E} (20 + 40 \times 0.4 + 10 \times 0.4)$$

$$= -\frac{40}{E} = -\frac{40}{2 \times 10^4} = 2 \times 10^{-3}$$

∴

$$e_v = e_x + e_y + e_z = 2.2 \times 10^{-3} + 1 \times 10^{-4} - 2 \times 10^{-3}$$

$$= 0.3 \times 10^{-3}$$

$$\text{Increase in length} = 2.2 \times 10^{-3} \times 300 = 0.66 \text{ mm}$$

$$\text{Increase in width} = 1 \times 10^{-4} \times 50 = 0.0050 \text{ mm}$$

$$\text{Decrease in depth} = 2 \times 10^{-3} \times 50 = 0.1 \text{ mm}$$

$$\text{Increase in volume} = 0.3 \times 10^{-3} \times 50 \times 50 \times 300 = 225 \text{ mm}^3$$

Problem 2.8 A bar of copper is 40 mm × 40 mm in section and is 120 mm long. It is subjected to a tensile load of 200 kN along the longitudinal axis and tensile loads of 500 kN and 400 kN on the lateral faces.

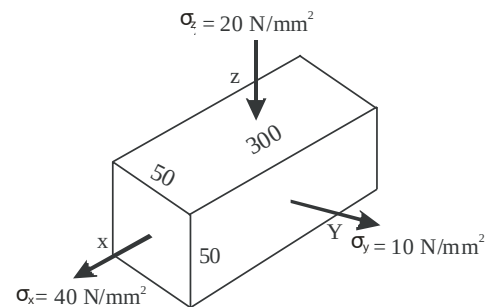


Fig. 2.8

- (a) Find the change in the dimensions of the bar and the change in volume.
 (b) Find also what axial longitudinal tensile load acting alone can produce the same longitudinal strain as in (a)

Take $E = 1 \times 10^5 \text{ N/mm}^2$ and $m = 0.3$

Solution: Fig. 2.9 shows the bar subjected to the given load system.

Let reference axes X, Y and Z be chosen as shown in the Fig. 2.9

Then stresses along these axes are

$$\sigma_x = \frac{200 \times 1000}{40 \times 40} = 125 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_y = \frac{500 \times 1000}{40 \times 120} = 104.17 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_z = \frac{400 \times 1000}{40 \times 120} = 83.33 \text{ N/mm}^2 \text{ (tensile)}$$

Strain along the X-axis

$$= e_x = \frac{\sigma_x}{E} - \frac{\sigma_y}{mE} - \frac{\sigma_z}{mE}$$

$$= \frac{1}{1 \times 10^5} (125 - 0.3 \times 104.17 - 0.3 \times 83.33)$$

$$= 0.0006875$$

$$\therefore \text{Increase in length} = 0.0006875 \times 120 \text{ mm}$$

$$= 0.0825 \text{ mm}$$

Strain along the Y-axis

$$= e_y = \frac{\sigma_y}{E} - \frac{\sigma_x}{mE} - \frac{\sigma_z}{mE}$$

$$= \frac{1}{1 \times 10^5} \{104.17 - 0.3(125 + 83.3)\} = 0.0004166$$

$$\therefore \text{Increase in width}$$

$$= 0.0004166 \times 40 \text{ mm} = 0.0167 \text{ mm}$$

$$= e_z = \frac{\sigma_z}{E} - \frac{\sigma_x}{mE} - \frac{\sigma_y}{mE} = \frac{1}{E} \left(\sigma_z - \frac{\sigma_x + \sigma_y}{m} \right)$$

Strain along the Z-axis

$$= \frac{1}{1 \times 10^5} \{83.33 - 0.3(125 + 104.17)\} = 0.0001458$$

$$\therefore \text{Increase in depth}$$

$$= 0.0001458 \times 40 \text{ mm} = 0.005832 \text{ mm}$$

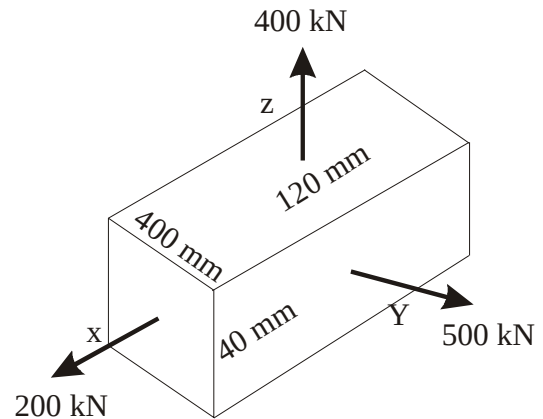


Fig. 2.9

Let P be the axial stress acting alone longitudinally i.e., along the X-axis to produce the same longitudinal strain .

$$\therefore \frac{P}{E} = 0.0006875$$

$$\therefore P = 0.0006875 \times 1 \times 10^5 = 68.75 \text{ N/mm}^2$$

$$\therefore \text{Longitudinal load} = 68.75 \times 40 \times 40 = 11000 \text{ N} = 110 \text{ kN}$$

Exercise

2.1 Define and explain lateral strain, longitudinal strain and Poisson's ratio.

2.2 Explain the 'state of simple shear'.

2.3 A bar of steel has rectangular cross section 30 x 20mm. Find the percentage decrease of area of cross section, when it is subjected to a tensile force of 120 kN in the direction of its length. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$.

(Ans. 0.06%)

2.4 A bar of steel has rectangular cross section 30 x 30mm and 300 mm long. Find the change in volume, when it is subjected to a tensile force of 150 kN in the direction of its length. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$.

(Ans. 90 mm³)

2.5 A bar of steel has rectangular cross section 40 x 40mm and 200 mm long. When it is subjected to a tensile force of 320 kN, 0.2 mm extension and 0.012 change in thickness were observed. Calculate the Poisson's ratio and the value of Young's modulus.

(Ans. 0.3, $2 \times 10^5 \text{ N/mm}^2$)

2.6 A steel cube of 50 mm side is subjected to a force of 10 kN (tension), 12.5 kN (compression) and 7.5 kN (tension) along x, y and z directions respectively. Determine the change in the volume of the block. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$.

(Ans. 0.5 mm³)

2.7 Calculate the modulus of rigidity and bulk modulus of a cylindrical bar of diameter 25 mm and of length 1.2 m if the longitudinal strain in a bar during a tensile test is four times the lateral strain. Find the change in the volume when the bar is subjected to pressure of 120 N/mm^2 . Take $E = 1.2 \times 10^5 \text{ N/mm}^2$.

(Ans. $\mu=0.25$, $C=0.48 \times 10^5 \text{ N/mm}^2$, $K=0.8 \times 10^5 \text{ N/mm}^2$, change in volume=884 mm³)

2.8 For a material Young's modulus is given as $1.4 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio 0.28. Calculate the Bulk modulus.

(Ans. $1.06 \times 10^5 \text{ N/mm}^2$)

2.9 For a material Young's modulus is given as $1.2 \times 10^5 \text{ N/mm}^2$ and modulus of rigidity is $4.5 \times 10^4 \text{ N/mm}^2$. Calculate the Bulk modulus and Poisson's ratio.

(Ans. $1.2 \times 10^5 \text{ N/mm}^2$, 0.33)

2.10 A steel bar of 320mm long, 40 mm wide and 30 mm thick is subjected to a pull of 250 kN in the direction of its length. Determine the change in volume. Take $E = 1.2 \times 10^5$ N/mm² and Poisson's ratio 0.25

(Ans. 200 mm²)

2.11 A bar of 25 mm diameter is subjected to a pull of 40 kN. The measured extension on length of 200 mm is 0.085 mm and the change in diameter is 0.003 mm. Calculate the Poisson's ratio and the value of Young's modulus.

(Ans. 0.282, 1.917×10^5 N/mm²)